Diamond iO: A Straightforward Construction of Indistinguishability Obfuscation from Lattices

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Abstract. Indistinguishability obfuscation (iO) has seen remarkable theoretical progress, yet it remains impractical due to its high complexity and inefficiency. A common bottleneck in recent iO schemes is the reliance on bootstrapping techniques from functional encryption (FE) into iO, which requires recursively invoking the FE encryption algorithm for each input bit—creating a significant barrier to practical iO schemes.
In this work, we propose diamond iO, a new lattice-based iO construction that replaces the costly recursive encryption process with lightweight matrix operations. Our construction is proven secure under the learning with errors (LWE) and evasive LWE assumptions, as well as our new assumption—all-product LWE—in the pseudorandom oracle model. By leveraging the FE scheme for pseudorandom functionalities introduced by Agrawal et al. (ePrint'24) in a non-black-box manner, we remove the reliance on prior FE-to-iO bootstrapping techniques and thereby significantly reduce complexity. We further show that a variant of the all-product LWE assumption reduces to standard LWE, and we argue that known attacks on evasive LWE do not threaten our construction.

1 Introduction

Program obfuscation aims to make programs unintelligible without altering their functionality. A general-purpose obfuscator that offers an ideal security guarantee converts any program into an obfuscated version that can be distributed safely, revealing no information beyond its input-output behavior.

Unfortunately, Barak et al. [9] proved that such an obfuscator, defined as a virtual black-box obfuscator, for arbitrary programs is impossible to achieve.[4] They proposed a weaker security notation of the obfuscator called indistinguishability obfuscation (iO), where the obfuscated versions of two arbitrary programs

[4] However, the work of [38] showed a construction of an ideal obfuscation for arbitrary programs in the pseudorandom oracle model. In this model, a non-black-use of the random oracle is formally allowed.

with the same size and functionality should be computationally indistinguishable. Despite its limited security guarantee, iO has been seen as a "central hub [48]" for its role in building numerous powerful cryptographic primitives [48, 30, 11, 10, 23, 15, 5].

The first candidate of iO in [23] kick-started a quest for constructing iO from reasonable cryptographic assumptions. Following a series of studies in this context [23, 45, 27, 42, 8, 6, 1, 39, 24], Jain et al. [40] have succeeded in building iO relying solely on four standard assumptions. Furthermore, subsequent works [41, 46] reduced the number of the necessary assumptions to just three.

However, most studies on iO have not focused on evaluating its concrete efficiency or implementing the proposed schemes, and thus iO remains more of a theoretical cryptographic primitive than a practical one. In recently proposed iO schemes [42, 6, 16, 17, 40, 41, 46, 25, 24, 21], one common bottleneck is the reliance on bootstrapping techniques from functional encryption (FE) to iO [7, 12]. We note that FE is a variant of public-key encryption in which a special secret key associated with a **public** function f—called functional secret key sk_f—allows its holder to decrypt an encryption of an input x and learn only the output $f(x)$ [14]. These bootstrapping techniques are costly because they recursively invoke the FE encryption algorithm for each input bit—specifically, they run the FE encryption algorithm as f during decryption to produce other FE ciphertexts. As a result, the size and complexity of the FE encryption algorithm becomes a lower bound on the size of the functions that must be evaluated by the underlying FE scheme, creating a significant barrier to practical iO constructions.

Our contributions. To address this issue, we propose diamond iO: a straightforward construction of iO using lattice techniques. This replaces the costly FE recursive encryption process with simple matrix operations. The security of our construction relies on the learning with errors (LWE) and evasive LWE assumptions, as well as a new lattice assumption we introduce, called all-product LWE, in the pseudorandom oracle model (PROM) [38]. A non-black-box use of the FE scheme for pseudorandom functionalities proposed in [2] enables us to remove the reliance on the existing FE-to-iO bootstrapping techniques.

This paper is organized as follows. The rest of Section 1 reviews related works and provides a technical overview of our iO construction. Section 3 introduces the notions and preliminaries necessarily to describe our construction. Section 4 presents our iO construction along with its correctness and security proofs. Section 5 proves that a variant of the all-product LWE assumption can be reduced to the standard LWE assumption. Additionally, this section provides a detailed analysis of attacks against the evasive LWE assumption, demonstrating that our iO construction is secure against all known attacks. Finally, Section 6 concludes the paper.

1.1 Related Works

Bootstrapping from FE to iO. We review the bootstrapping technique from FE to iO, which is a common bottleneck in recent iO schemes. Goldwasser et al. [30] showed how to build iO from secret-key multi-input functional encryption (MiFE). Subsequently, Ananth and Jain [7] proved that this secret-key MIFE scheme can be constructed from a public-key single-key FE scheme that is compact, i.e., the running time of the encryption algorithm must be independent of the output size. Furthermore, Bitansky et al. [12] presented a similar bootstrapping technique from FE to iO but with a weaker requirement on the FE scheme: that is, the encryption running time only must be sublinear in the output size. Taken together, these results imply that the single-key FE whose encryption algorithm runs in sublinear time in the output size suffices for building iO.

Specifically, the technique proposed in [7] constructs a secret-key MiFE scheme for any function with n distinct inputs, denoted by MiFE_i, by recursively increasing the arity of MiFE_{i-1} by one bit. As an initial ingredient for this recursion, the public-key compact FE, referred to as PKFE, is employed. To obfuscate a circuit f, the obfuscator provides a functional secret key sk_f with respect to a function $U(\mathsf{Sym.Enc}(\cdot, f), \cdot)$: it takes as input a key of the symmetric key encryption (SKE) scheme and an input $\mathbf{x}_L := (x_1, \ldots, x_n)$, decrypts the hardcoded ciphertext of f—denoted by $\mathsf{Sym.Enc}(\cdot, f)$—with the provided SKE key, and returns $f(\mathbf{x}_L)$ by evaluating a universal circuit U. Additionally, the obfuscator provides the evaluator with a pair of ciphertexts for each of the n input bits. For $i = 1$, these two ciphertexts are, respectively, the encryption of $(0, sk)$ and $(1, sk)$ under MiFE_2. For $i \in \{2, \ldots, L-1\}$, the pair of ciphertexts are functional secret keys with respect to MiFE_i for a function that takes the $(i-1)$-bit input vector $\mathbf{x}_{i-1} := (x_1, \ldots, x_{i-1})$, concatenates it with either 0 or 1, and returns its re-encryption under MiFE_{i+1}. Lastly, for $i = L$, the two ciphertexts are functional secret keys with respect to MiFE_L for a function that takes the $(L-1)$-bit input vector $\mathbf{x}_{L-1}$, concatenates it with either 0 or 1, and returns its re-encryption with respect to PKFE.

To evaluate the obfuscated circuit on an input $\mathbf{x}_L$, the evaluator picks the ciphertext corresponding to their first input bit and recursively decrypts it with the functional secret key corresponding to the next input bit. By iterating this process for each input bit, the evaluator eventually obtains an encryption of $(\mathbf{x}_L, K)$ under FE, which can be decrypted with sk_f to recover $f(\mathbf{x}_n)$. In this way, the recursive FE encryption lets the evaluator obtain a single ciphertext that contains both the obfuscator's private date, namely the symmetric key K, and the evaluator's L-bit input $\mathbf{x}_L$.

The technique proposed in [12] is similar to the one in [7]. However, instead of building a MiFE scheme, it constructs a variant of the public-key single-input FE scheme called puncturable FE (PFE). While it is impossible to have a functional secret key hide the function in the public-key setting [12], that of PFE guarantees a weaker nortion of the function-hiding property. The authors first showed that a public-key single-input FE scheme with sublinear encryption efficiency can be extended to a PFE scheme. They then constructed iO by recursively invoking the PFE encryption algorithm for each input bit. In this manner, even though multiple studies have addressed FE-to-iO bootstrapping techniques, to the best of our knowledge, they all rely on the recursive FE encryption process.

Lattice-based iO. Existing iO schemes from standard assumptions [40, 41, 46] rely on bilinear maps and are therefore not post-quantum secure. To achieve (plausibly) post-quantum iO, a parallel line of research [17, 21, 25, 53, 16, 3, 19] focuses on lattice-based iO constructions.

As the first study in this direction, Brakerski et al. [16] introduced a new primitive called split-FHE and extended it to exponential iO (XiO), which implies iO [43]. Their proposed split-FHE construction employs a standard LWE-based FHE scheme, a linearly-homomorphic encryption (LHE) scheme with short decryption hints, and a hash function that instantiates the random oracle.[5] Unfortunately, its security analysis remains heuristic due to partial leakage of smudging noises.

Subsequent works [17, 21, 25, 53] focused on improving the security analysis by providing constructions that rely on different flavors of the circular security assumption. However, Hopkins et al. [32], followed by Jain et al. [37], demonstrated the fragility of these constructions by providing polynomial-time attacks against some of their new assumptions.

Unlike the above works, Agrawal et al. [3] and Branco et al. [19] independently propose constructions of iO for pseudorandom functionalities, proven secure under commonly used assumptions, namely LWE and evasive LWE assumptions [52]. Although the latter is still relatively new, it has already been employed in various cryptographic schemes such as broadcast encryption [52], attribute-based encryption [51, 35, 34], and witness encryption [49, 50]. Furthermore, Branco et al. [19] introduce a transformation from iO for pseudorandom functionalities into one for general circuits in PROM.

Our work is directly inspired by these two schemes that utilize LWE and evasive LWE assumptions. Specifically, we build on their approach to lattice-based iO by relying on the same two assumptions and the transformation in PROM, while also introducing a new assumption—all-product LWE—for our construction. Although similar methods to the aforementioned attacks have been applied to the evasive LWE assumption [33, 4], we confirm that our construction is secure against all known attacks as discussed in Subsection 5.2. Nevertheless, it is evident that further cryptanalysis is necessary.

2 Technical Overview

2.1 Review of the FE scheme introduced by Agrawal et al. (ePrint'24)

Before introducing our technical ideas, we recall the defintion of BGG+ encodings [13], which is a homomorphic encoding scheme using lattices, and the

[5] While the first construction in [17] still relies on the decisional composite residuosit (DCR) assumption for the LHE scheme, which is not post-quantum secure, the construction in Subsection 4.4 of [17] are based on LWE, which removes the DCR assumption.

construction of the FE scheme introduced by Agrawal et al. in [2]—hereafter denoted as AKY24.

BGG+ encodings. The BGG+ encoding of bits $\mathbf{x} \in \{0,1\}^L$ is defined by

$$\mathbf{c}_{\mathsf{att},\mathbf{x}}^T := \underline{\mathbf{s}^T(\mathbf{A}_{\mathsf{att}} - (1, \mathbf{x}^T) \otimes \mathbf{G}_{n+1})},$$

where $\mathbf{s} \in \mathbb{Z}_q^{n+1}$ is a secret key, $\mathbf{A}_{\mathsf{att}} \in \mathbb{Z}_q^{(n+1)\times Lm}$ is a public matrix with the column size $m = (n+1)\lceil \log_2 q \rceil$, and an underlined term contains LWE error. There are efficient determinsitic algortimhs [18] $\mathsf{EvalC}(\mathbf{A}_{\mathsf{att}}, C) \to \mathbf{A}_C$ and $\mathsf{EvalCX}(\mathbf{A}_{\mathsf{att}}, C, \mathbf{x}) \to \mathbf{H}_{C,x}$, where C is a matrix-valued circuit $C : \{0,1\}^L \to \mathbb{Z}_q^{n\times m'}$, such that the following homomorphism holds:

$$(\mathbf{A}_{\mathsf{att}} - (1, \mathbf{x}^T) \otimes \mathbf{G}_{n+1})\mathbf{H}_{C,x} = \mathbf{A}_C - C(\mathbf{x})$$

We highlight that the public matrix $\mathbf{A}_C$ is deterministic to the input public matrix $\mathbf{A}_{\mathsf{att}}$ and the circuit C, namely independent of the input $\mathbf{x}$. In other words, one can compute $\mathbf{A}_C$ without knowing $\mathbf{x}$.

Functional encryption. FE allows an untrusted decryptor to evaluate a specified circuit C over the encryption of $\mathbf{x}$ to obtain $C(\mathbf{x})$ without learning anything about $\mathbf{x}$ other than the output of f [14]. More specifically:

- A trusted key generator outputs a master public key mpk and a master secret key msk.
- An encryptor publishes an FE ciphertext $\mathsf{ct}_{\mathsf{FE},\mathbf{x}}$ for the input $\mathbf{x}$ using the master public key mpk.
- The key generator—holding msk—computes a functional secret key sk_f for a circuit f and shares it to the decryptor.
- Provided $\mathsf{ct}_{\mathsf{FE},\mathbf{x}}$ and sk_f, the decryptor can obtain $C(\mathbf{x})$ without interactions with any other parties.

In the AKY24 scheme, the key generator samples the input public matrix $\mathbf{A}_{\mathsf{att}}$ and generates a trapdoor pair $(\mathbf{B}, \mathbf{B}^{-1})$. The master public and private keys are defined by $\mathsf{mpk} := (\mathbf{A}_{\mathsf{att}}, \mathbf{B})$ and $\mathsf{msk} := \mathbf{B}^{-1}$, respectively.

The encryptor first generates an FHE key pair $\big(\mathsf{s}, \mathsf{pk}\big)$ and a PRF key K. They then encrypt the private input $\mathbf{x}$ together with K under pk, producing the FHE ciphertext $\mathsf{ct}_{\mathsf{FHE},\mathbf{x},K}$. They finally compute the encoding of $\xi^T := \mathsf{bits}(\mathsf{ct}_{\mathsf{FHE},\mathbf{x},K}) \in \{0,1\}^{s'}$—defined by $\mathbf{c}_{\mathsf{att},\xi} = \underline{\mathbf{s}^T(\mathbf{A}_{\mathsf{att}} - (1, \xi^T) \otimes \mathbf{G}_{n+1})}$—and the LWE instance $\mathbf{c}_{\mathbf{B}} = \underline{\mathbf{s}^T\mathbf{B}}$. The FE ciphertext is published as $\mathsf{ct}_{\mathsf{FE},\mathbf{x}} := (\mathsf{ct}_{\mathsf{FHE},\mathbf{x},K}, \mathbf{c}_{\xi}, \mathbf{c}_{\mathbf{B}})$. Notably, the secret key $\mathbf{s}$ is used both for the FHE ciphertext and the BGG+ encoding, based on the dual-use technique introduced in [18].

The key generator defines a circuit F that performs the homomorphic evaluation of a circuit that outputs $f(\mathbf{x}, K) := \frac{q}{2}C(\mathbf{x}) + \mathsf{PRF}(K, \mathbf{x})$ over the provided FHE ciphertext. The functional secret key is defined by a lattice preimage $\mathsf{sk}_f := \mathbf{K}_F \leftarrow\!\!\$\ \mathbf{B}^{-1}(\mathbf{A}_F)$. For any target matrix $\mathbf{A}$, a preimage $\mathbf{K} \leftarrow\!\!\$\ \mathbf{B}^{-1}(\mathbf{A})$ is a low-norm matrix that satisfies $\mathbf{BK} = \mathbf{A}$ [28, 44, 26]. The preimage can be sampled only by the person who knows the trapdoor $\mathbf{B}^{-1}$, namely the key generator.

The decryptor first evalautes the **public** circuit F over the encoding $\mathbf{c}_\xi$ to obtain $\mathbf{c}_\mathbf{Y}$, an encoding of the FHE encryption $\mathbf{Y}$ of the circuit output $\frac{q}{2}C(\mathbf{x}) + \mathsf{PRF}(K, \mathbf{x})$. Many lattice-based FHE schemes, such as GSW [29], have a linear decryption property such that the multiplication between the secret key $\mathbf{s}$ and the FHE ciphertext $\mathbf{Y}$ yields the decryption result with some LWE errors. Therefore, the dual-use of the secret key $\mathbf{s}$ automatically converts the second term of the encoding into the decryption result $\frac{q}{2}C(\mathbf{x}) + \mathsf{PRF}(K, \mathbf{x})$ (plus some error) as follows:

$$\begin{aligned}\mathbf{c}_\mathbf{Y}^T &:= \underline{\mathbf{s}^T(\mathbf{A}_F - \mathbf{Y})} \\ &= \underline{\mathbf{s}^T\mathbf{A}_F - \left(\frac{q}{2}C(\mathbf{x}) + \mathsf{PRF}(K, \mathbf{x})\right)}\end{aligned}$$

The functional secret key sk_f, together with the LWE instance $\mathbf{c}_\mathbf{B}$, allows the decryptor to recover the term $\underline{\mathbf{s}^T\mathbf{A}_F}$. This allows the decryptor to remove the first term in $\mathbf{c}_\mathbf{Y}^T$, obtaining $\underline{\frac{q}{2}C(\mathbf{x}) + \mathsf{PRF}(K, \mathbf{x})}$. Notably, the sum of the PRF output and the remaining errors is bounded by $\frac{q}{4}$; therefore, the decryptor can recover the desired output $C(\mathbf{x})$ by extracting the highest-order bits.

The AKY24 scheme is proven secure only for pseudorandom functionalities. However, the following works provide methods to transform such a FE scheme into iO for general functionalities:

- Agrawal et al. [3] constructs pseudorandom obfuscation (PRO) from any pseudorandom FE scheme assuming LWE and evasive LWE assumptions [52, 20], where PRO is a variant of iO that only supports pseudorandom functionalities.
- Branco et al. [19] provides a generic bootstrapping technique from PRO to iO in the pseudorandom oracle model (PROM).

While the second transformation remains simple because it only requires computing a hash function inside and outside the obfuscation, the first transformation still suffers from the costly recursive encryption process in similar to [7, 12]. Therefore, we focus on building a PRO scheme without the reliance on the existing bootstrapping techniques from (pseudorandom) FE to iO.

2.2 Our initial insight

The initial insight that guided our PRO construction–allowing us to avoid the existing FE bootstrapping techniques–was the following question: why must we execute the entire FE encryption algorithm again just to derive a fresh ciphertext from an existing one? In fact, if we can allow an evaluator to directly alter the input under the FE ciphertext provided by an obfuscator, but only in some restricted manner, we can straightforwardly construct PRO from FE as follows:

1. The obfuscator provides an FE encryption of a **private** circuit $C : \{0,1\}^L \to \{0,1\}^\ell$, defined by $\mathsf{ct}_{\mathsf{FE},C}$.

2. The obfuscator also provides a functional secret key sk_f for a circuit f that takes as input C and the evaluator's input $\mathbf{x}$ and outputs $U(C, \mathbf{x}) = C(\mathbf{x})$ and U is a universal circuit.
3. The evaluator with the input $\mathbf{x} \in \{0,1\}^L$ alters $\mathsf{ct}_{\mathsf{FE},C}$ into $\mathsf{ct}_{\mathsf{FE},C,\mathbf{x}}$.
4. The evaluator finally decrypts $\mathsf{ct}_{\mathsf{FE},C,\mathbf{x}}$ with sk_f, obtaining $C(\mathbf{x})$.

Therefore, the problem we should solve is how to concretely realize the third step. One naive solution is to generate functional secret keys for all 2^L input patterns, where L is the input bit size. Specifically, we define a circuit $U_{\mathbf{x}}(\cdot) := U(\cdot, \mathbf{x})$ for every input $\mathbf{x} \in \{0,1\}^L$ and include $\{\mathsf{sk}_{U_{\mathbf{x}}}\}_{\mathbf{x} \in \{0,1\}^L}$ in the obfuscated circuit. While it allows the evaluator to directly decrypt the FE encryption of C with arbitrary input, the size of the obfuscated circuit and the obfuscator's running time increase in the order $\mathcal{O}(2^L)$, which is infeasible.

2.3 Merging public matrices

To address the inefficiency of the above solution, we adopt an essence of the existing bootstrapping techniques from FE to iO. Specifically, we allow the evaluator to insert each input bit one-by-one into the encoding provided by the obfuscator without recursively invoking the FE encryption algorithm for each input bit. Hereafter, we utilize the structure of AKY24 in a non-black-box manner.

The obfuscator provides the initial BGG+ encoding

$$\mathbf{c}_{\mathsf{att},\epsilon}^T := \underline{\mathbf{s}^T(\mathbf{A}_{\mathsf{att},0} - (1, \xi^T, \mathbf{0}_{1\times L}^T) \otimes \mathbf{G}_{n+1})},$$

where $\xi^T := \mathsf{bits}(\mathsf{ct})$ and $\mathsf{ct} := \mathsf{FHE.Enc}(\mathsf{pk}, (C, K))$, and the sequence of zeros denoted by $\mathbf{0}_{1\times L}^T$ will be altered into the evaluator's input bits $\mathbf{x}_L$ later.

For every $i \in [L]$ and $b \in \{0,1\}$, we let $\mathsf{ADD}[i, b]$ be a circuit that adds b to the i-th bit of the input sequence. For the first input bit x_1, the evaluator can apply $\mathsf{ADD}[1, x_1]$ to the encoding $\mathbf{c}_{\mathsf{att},\epsilon}$ to obtain the encoding of $(1, \xi^T, x_1, \mathbf{0}_{1\times(L-1)}^T)$. The first term in this encoding is expressed as $\mathbf{s}^T\mathbf{A}_{\mathsf{att},\mathsf{ADD}[1,x_1]}$. Unlike the property of the original BGG+ encoding described above, the matrix $\mathbf{A}_{\mathsf{att},\mathsf{ADD}[1,x_1]}$ depends on the input bit x_1. When this process is repeated until the last input bit, even though it allows the evaluator to insert the input bits to the obfuscator's initial encoding, it still requires the obfuscator to perform $\mathcal{O}(2^L)$ amount of work to generate the functional secret keys for all possible input patterns.

Therefore, we consider merging $\mathsf{ADD}[i, 0]$ and $\mathsf{ADD}[i, 1]$, depending on x_i, into a common public matrix $\mathbf{A}_{\mathsf{att},i}$, which does not depend on x_i. As a result, when the circuit $\mathsf{ADD}[i+1, x_{i+1}]$ is applied to the modified encoding, the resulting public matrix depends only on x_{i+1} and no longer on the previously inserted bits $\mathbf{x}_i := (x_1, \ldots, x_i)$. In order for the evaluator to replace the input-bit dependent public matrices with a common public matrix, for each $i \in \{1, \ldots, L\}$, the obfuscator only needs to include two variants of the functional secret keys that can produce two masks: $\mathbf{s}^T(-\mathbf{A}_{\mathsf{att},\mathsf{ADD}[i,0]} + \mathbf{A}_{\mathsf{att},i+1})$ and $\mathbf{s}^T(-\mathbf{A}_{\mathsf{att},\mathsf{ADD}[i,1]} + \mathbf{A}_{\mathsf{att},i+1})$. By adding $\mathbf{s}^T(-\mathbf{A}_{\mathsf{att},\mathsf{ADD}[i,x_i]} + \mathbf{A}_{\mathsf{att},i+1})$ to the encoding after applying $\mathsf{ADD}[i, x_i]$, the evaluator can obtain the following encoding:

$$\mathbf{c}^T_{\mathsf{att},\mathbf{x}_i} := \underline{\mathbf{s}^T(\mathbf{A}_{\mathsf{att},i} - (1, \xi^T, \mathbf{x}_i^T, \mathbf{0}^T_{1\times(L-i)}))}.$$

In this manner, this replacement technique can maintain the size of the obfuscated circuit and the obfuscator's running time at $\mathcal{O}(L)$.

Unfortunately, the above construction is insecure. Since the same secret key $\mathbf{s}$ is reused for all encodings, an adversary can remove $\mathbf{s}^T\mathbf{A}_{\mathsf{att},i}$ from two encoding of two distinct encoded bits, allowing the adversary to recover $\mathbf{s}$ by solving an easy LWE problem $\underline{\mathbf{s}G}$ [52]. This suggests that distinct secret keys should be used for every combination of input bits $\mathbf{x}_L$. In other words, the secret key should depend on the trace of input bits that have been inserted.

2.4 Diamond iO

We now provide an overview of our straightforward PRO construction in three steps. After building PRO, we finally rely on the transformation introduced by [19] to obtain iO, involving only hash computations.

Step 1: encode evaluator's input bits. To solve for the security issue highlighted in the previous construction, we define a input-dependent secret key for every $i \in [L]$ and every trace of input bits $\mathbf{x}_i \in \{0,1\}^i$ by

$$\mathbf{s}^T_{\mathbf{x}_i} := (\mathbf{s}^T_\epsilon \prod_{j\in[i]} \mathbf{S}_{j,x_j}),$$

where the initial secret is defined as $\mathbf{s}^T_\epsilon := (\bar{\mathbf{s}}_\epsilon, -1)$ where $\bar{\mathbf{s}}_\epsilon \leftarrow\!\!\$\ \{0,1\}^n$ and the input-trace dependent part is $\mathbf{S}_{i,b} := \begin{pmatrix} \bar{\mathbf{S}}_{i,b} & \mathbf{0}_{n\times 1} \\ \mathbf{0}_{1\times n} & 1 \end{pmatrix}$ where $\bar{\mathbf{S}}_{i,b} \leftarrow\!\!\$\ \{0,1\}^{n\times n}$ for every $i \in [L]$ and $b \in \{0,1\}$.

We also define an intial helper vector $\mathbf{p}^T_\epsilon := \underline{((1, \xi^T, \mathbf{0}^T_{1\times L}, \mathbf{t}^T) \otimes \mathbf{s}^T_\epsilon)\mathbf{B}_0}$ instead of the initial BGG+ encoding defined above. Here, $\mathbf{t}^T$ denotes a n_t-sized vector acting as a secret key of the underlying FHE scheme with a linear decryption property [16], and the corresponding public key pk is used to encrypt the private circuit C and a PRF key K, procucing a ciphertext ct. $\mathbf{B}_0$ is a random matrix defined soon.

The helper vector, rather than a BGG+ encoding, is dynamically updated by the evaluator when inserting their input bits as follows:

$$\mathbf{p}^T_{\mathbf{x}_i} := \underline{\Big((1, \xi^T, \mathbf{x}_i^T, \mathbf{0}^T_{L-i}, \mathbf{t}^T) \otimes \mathbf{s}^T_{\mathbf{x}_i}\Big)\mathbf{B}_i}$$

We further define the size $L' := 1 + s' + L + n_t$. The data that allows the evaluator to obtain the encoding $\mathbf{c}^T_{\mathsf{att},\mathbf{x}_L}$—filled with their input bits—are provided by the obfuscator as follows:

1. For every $i \in [0, L]$ sample a random matrix $\mathbf{B}_i$ and its trapdoor $\mathbf{B}_i^{-1}$.
2. Compute $\mathbf{p}^T_\epsilon$.

3. For every $i \in [L]$ and $b \in \{0,1\}$, sample the preimage $\mathbf{K}_{i,b} \leftarrow\$ \mathbf{B}_{i-1}^{-1}((\mathbf{U}_{i,b} \otimes \mathbf{S}_{i,b})\mathbf{B}_i + \mathbf{E}_{K_{i,b}})$, where $\mathbf{U}_{i,b}$ is a matrix such that $(1, \xi^T, \mathbf{x}_{i-1}^T, \mathbf{0}_{L-i+1}^T, \mathbf{t}^T)\mathbf{U}_{i,b} = (1, \xi^T, \mathbf{x}_{i-1}^T, b, \mathbf{0}_{L-i}^T, \mathbf{t}^T)$ holds, and $\mathbf{E}_{K_{i,b}}$ is an error matrix.
4. Sample a public matrix $\mathbf{A}_{\mathsf{att}}$.
5. Compute $\mathbf{P}_{\mathsf{att}} := \mathbf{u}_{1,L'} \otimes \mathbf{A}_{\mathsf{att}} - \mathbf{I}_{L'} \otimes \mathbf{G}_{n+1}$, where $\mathbf{u}_{1,L'} \in \{0,1\}^{L'}$ is the first unit vector and $\mathbf{I}_{L'}$ is the squared identity matrix of size L'.
6. Sample the preimage $\mathbf{K}_{\mathsf{att}} \leftarrow\$ \mathbf{B}_L^{-1}(\mathbf{P}_{\mathsf{att}})$
7. Add $\mathbf{p}_\epsilon$, $\{\mathbf{K}_{i,b}\}_{i\in[L], b\in\{0,1\}}$, $\mathbf{A}_{\mathsf{att}}$ and $\mathbf{K}_{\mathsf{att}}$ to the obfuscated circuit.

Using the above additional data, the evaluator can progressively insert their input bits (and dynamically switch the secret key) inside the helper vector and eventually obtain the desired encoding as follows:

1. For every $i \in [L]$, compute $\mathbf{p}_{\mathbf{x}_i}^T := \mathbf{p}_{\mathbf{x}_{i-1}}^T \mathbf{K}_{i,x_i}$.
2. Compute the final encoding

$$\begin{aligned}\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T &:= \mathbf{p}_{\mathbf{x}_L}^T \mathbf{K}_{\mathsf{att}} \\ &= \underline{\mathbf{s}_{\mathbf{x}_L}^T (\mathbf{A}_{\mathsf{att}} - (1, \xi^T, \mathbf{x}_L^T, \mathbf{t}^T) \otimes \mathbf{G}_{n+1})}\end{aligned}$$

The diamond-shaped bit insertion operation is illustrated in Figure 1. Notably:

– The secret key $\mathbf{s}_{\mathbf{x}_L}^T$ used for the final encoding depends on the evaluator's all input bits $\mathbf{x}_L$.
– The public key $\mathbf{A}_{\mathsf{att}}$ used for the final encoding is independent from the evaluator's input bits $\mathbf{x}_L$.
– The i-th preimages $\mathbf{K}_{i,0}$ and $\mathbf{K}_{i,1}$ provided by the obfuscator are independent of the trace of the previous input bits $\mathbf{x}_{i-1}$, maintaining the size of the obfuscated circuit and the obfuscator's running time at $\mathcal{O}(L)$.
– Unlike the other encoded bits, the **private** FHE secret key $\mathbf{t}$ remains hidden from the evaluator[6]. We will later explain why this preserves the homomorphic property for the encoding of $\mathbf{t}$ required by our construction.

Since a different secret key $\mathbf{s}_{\mathbf{x}_L}$ is associated to each of the possible 2^L encodings, the above attack, which uses the encodings of different bits under the same secret key, seems to be prevented. However, to prove this formally, we need to introduce a new lattice assumption called all-product LWE, claiming

[6] Although the evaluator learns the BGG+ encoding of $\mathbf{t}$, its second term—depending on $\mathbf{t}$—still hides $\mathbf{t}$, because the first term of the encoding, together with the error, is pseudorandom under the all-product LWE assumption.

$$
\begin{tikzcd}[column sep=tiny]
& \mathbf{p}_{\epsilon}^T := \underline{((1, \xi^T, \mathbf{0}_{1\times L}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\epsilon}^T)\mathbf{B}_0} \arrow[dl] \arrow[dr] & \\
\mathbf{K}_{1,0} \leftarrow\!\!\!\$\; \mathbf{B}_0^{-1}((\mathbf{U}_{1,0} \otimes \mathbf{S}_{1,0})\mathbf{B}_1 + \mathbf{E}_{K_{1,0}}) \arrow[dr] & & \mathbf{K}_{1,1} \leftarrow\!\!\!\$\; \mathbf{B}_0^{-1}((\mathbf{U}_{1,1} \otimes \mathbf{S}_{1,1})\mathbf{B}_1 + \mathbf{E}_{K_{1,1}}) \arrow[dl] \\
& \mathbf{p}_{\mathbf{x}_1}^T := \underline{((1, \xi^T, x_1, \mathbf{0}_{1\times(L-1)}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_1}^T)\mathbf{B}_1} \arrow[d, dash, dotted] & \\
& \mathbf{p}_{\mathbf{x}_{L-1}}^T := \underline{((1, \xi^T, \mathbf{x}_{L-1}^T, 0, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_{L-1}}^T)\mathbf{B}_{L-1}} \arrow[dl] \arrow[dr] & \\
\mathbf{K}_{L,0} \leftarrow\!\!\!\$\; \mathbf{B}_{L-1}^{-1}((\mathbf{U}_{L,0} \otimes \mathbf{S}_{L,0})\mathbf{B}_L + \mathbf{E}_{K_{L,0}}) \arrow[dr] & & \mathbf{K}_{L,1} \leftarrow\!\!\!\$\; \mathbf{B}_{L-1}^{-1}((\mathbf{U}_{L,1} \otimes \mathbf{S}_{L,1})\mathbf{B}_L + \mathbf{E}_{K_{L,1}}) \arrow[dl] \\
& \mathbf{p}_{\mathbf{x}_L}^T := \underline{((1, \xi^T, \mathbf{x}_L^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_L}^T)\mathbf{B}_L} &
\end{tikzcd}
$$

$$
\begin{aligned}
\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T &:= \mathbf{p}_{\mathbf{x}_L}^T \mathbf{K}_{\mathsf{att}} = \underline{\mathbf{s}_{\mathbf{x}_L}^T (\mathbf{A}_{\mathsf{att}} - (1, \xi^T, \mathbf{x}_L^T, \mathbf{t}^T) \otimes \mathbf{G}_{n+1})} \\
\mathbf{v}_{F,\mathbf{x}_L}^T &:= \mathbf{p}_{\mathbf{x}_L}^T \mathbf{K}_F = \underline{\mathbf{s}_{\mathbf{x}_L}^T \mathbf{A}_F (\mathbf{I}_\ell \otimes \mathbf{G}_{n+1}^{-1}(\mathbf{u}_{n+1,n+1}))}
\end{aligned}
$$

Fig. 1. An illustration of the diamond-shaped key switching operation in our final construction.

the pseudorandomness of the following distribution:

$$\begin{pmatrix} \mathbf{A}, \{\mathbf{B}_i\}_{i\in[0,L]}, \\ \{\mathbf{c}_{A,\mathbf{x}_L} := \underline{\mathbf{s}_{\mathbf{x}_i}^T(\mathbf{A} - \mathbf{x}_L^T \otimes \mathbf{G}_{n+1})}\}_{\mathbf{x}_L\in\{0,1\}^L} \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \underline{\mathbf{s}_{\mathbf{x}_i}^T\mathbf{B}_i}\}_{\substack{i\in[0,L],\\ \mathbf{x}_i\in\{0,1\}^i}} \end{pmatrix}$$

$$\overset{c}{\approx} \begin{pmatrix} \mathbf{A}, \{\mathbf{B}_i\}_{i\in[0,L]}, \\ \{\mathbf{c}_{A,\mathbf{x}_L} \leftarrow\$ \mathbb{Z}_q^{(n+1)\lceil\log_2 q\rceil}\}_{\mathbf{x}_L\in\{0,1\}^L} \\ \{\mathbf{c}_{B,\mathbf{x}_i} \leftarrow\$ \mathbb{Z}_q^{m_{B,i}}\}_{\substack{i\in[0,L],\\ \mathbf{x}_i\in\{0,1\}^i}} \end{pmatrix},$$

where $\mathbf{A} \leftarrow\$ \mathbb{Z}_q^{(n+1)\times(n+1)\lceil\log_2 q\rceil}$ and $\mathbf{B}_i \leftarrow\$ \mathbb{Z}_q^{(n+1)\times m_{B,i}}$ with the column size $m_{B,i}$ given as the parameter. We can reduce a variant of this assumption from which the BGG+ encoding $\mathbf{c}_{A,\mathbf{x}_i}$ is removed to the standard LWE assumption as shown in Subsection 5.1. However, it remains an open problem to reduce this assumption as it is to any standard assumptions.

Step 2: evaluate and decrypt. Using the encoding $\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T$ the evaluator can homomorphically compute the encoding of $\mathsf{bits}(\mathbf{Y})$, which is the result of the homomorphic evaluation of the encrypted circuit ξ on the evaluator inserted input bits $\mathbf{x}_L$, defined as follows:

$$\mathbf{c}_{\mathbf{Y},\mathbf{x}_L}^T := \underline{\mathbf{s}_{\mathbf{x}_L}^T(\mathbf{A}_{\mathsf{att}'} - \mathsf{bits}(\mathbf{Y}) \otimes \mathbf{G}_{n+1})}$$

The evaluator next needs to decrypt it on the BGG+ encodings. However, since the secret key of the BGG+ encoding has dynamically changed based on the evaluator's input bits, we cannot rely on the dual-use technique to decrypt the output ciphertext.

Intuitively, our solution is to manually perform FHE decryption over the BGG+ encodings. More specifically, we rely on the linear decryption property of the chosen FHE scheme and allow the evaluator to perform an inner product between the resulting encoding of $\mathsf{bits}(\mathbf{Y})$ and that of $\mathbf{t}^T$ by a similar method introduced in [31] to produce the encoding $\mathbf{c}_{\mathbf{y},\mathbf{x}_L}^T$ of the decryption result with some error, namely $\mathbf{y}^T + \mathbf{e}_{\mathsf{fhe}}^T = \frac{q}{2}C(\mathbf{x}) + \mathsf{PRF}(K,\mathbf{x}) + \mathbf{e}_{\mathsf{fhe}}^T$. We recall that ℓ denotes the output size of the obfuscated circuit. For every $j \in [\ell]$, the encoding of the j-th output y_j is computed as below:

$$\begin{aligned} \mathbf{c}_{y_j,\mathbf{x}_L}^T :=\ & \Sigma_{k\in[\lceil\log_2 q\rceil]}\left(\Sigma_{i\in[n_t]}\left(\underline{\mathbf{c}_{y_{i,j,k},\mathbf{x}_L}^T}, \underline{\mathbf{c}_{t_i,\mathbf{x}_L}^T}\right)\begin{pmatrix}\mathbf{G}_n^{-1}(\mathbf{A}_{t,i}) \\ y_{i,j,k}\mathbf{I}_m\end{pmatrix}\right) \\ & \mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n) \\ =\ & \underline{\mathbf{s}_{\mathbf{x}_L}^T(\mathbf{A}_{y_j,\mathbf{x}_L} - (y_j + e_{\mathsf{fhe},j})\mathbf{G}_{n+1})}, \end{aligned} \tag{1}$$

where $\mathbf{c}_{y_{i,j,k},\mathbf{x}_L}^T$ and $\mathbf{c}_{t_i,\mathbf{x}_L}^T$ are the parts of the $\mathbf{c}_{\mathbf{Y},\mathbf{x}_L}^T$ and $\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T$ that correspond to the k-th bit of the entry (i,j) of $\mathbf{Y}$ and the i-th element of $\mathbf{t}$, respectively.

Notably, since the multiplication over BGG+ encodings only requires the left-hand side encoded bit to be public, the encoded attribute $\mathbf{t}^T$ can remain private without having to encrypt it under FHE.

In the second term of Equation 1, the decryption result y_j is still multiplied by the secret key $\mathbf{s}_{\mathbf{x}_L}$. We remove this by leveraging the fact that the last element of $\mathbf{s}_{\mathbf{x}_L}$ is fixed to (-1) as follows:

$$
\begin{aligned}
\mathbf{c}^T_{\mathbf{y},\mathbf{x}_L} &:= (\mathbf{c}^T_{y_1,\mathbf{x}_L}, \dots, \mathbf{c}^T_{y_\ell,\mathbf{x}_L})(\mathbf{I}_\ell \otimes \mathbf{G}^{-1}_{n+1}(\mathbf{u}_{n+1,n+1})) \\
&= \underline{\mathbf{s}^T_{\mathbf{x}_L}((\mathbf{A}_{y_1}, \dots, \mathbf{A}_{y_\ell}) - (\mathbf{y}^T + \mathbf{e}^T_{\mathsf{fhe}}) \otimes \mathbf{G}_{n+1})}(\mathbf{I}_\ell \otimes \mathbf{G}^{-1}_{n+1}(\mathbf{u}_{n+1,n+1})) \\
&= \underline{\mathbf{s}^T_{\mathbf{x}_L}\mathbf{A}_F - (\mathbf{y}^T + \mathbf{e}^T_{\mathsf{fhe}}) \otimes \mathbf{s}^T_{\mathbf{x}_L}\mathbf{u}_{n+1,n+1}} \\
&= \underline{\mathbf{s}^T_{\mathbf{x}_L}\mathbf{A}_F + (\mathbf{y}^T + \mathbf{e}^T_{\mathsf{fhe}})} \\
&= \underline{\mathbf{s}^T_{\mathbf{x}_L}\mathbf{A}_F + (\frac{q}{2}C(\mathbf{x}) + \mathsf{PRF}(K, \mathbf{x}) + \mathbf{e}^T_{\mathsf{fhe}})}
\end{aligned} \tag{2}
$$

where $\mathbf{A}_F := (\mathbf{A}_{y_1}, \dots, \mathbf{A}_{y_\ell})(\mathbf{I}_\ell \otimes \mathbf{G}^{-1}_{n+1}(\mathbf{u}_{n+1,n+1}))$.

The formal construction in Subsection 4.1 is slightly more complex than the above description. Specifically, it divides the homomorphic evaluation and decryption into two parts—denoted by low and high—and involves rounding operations to avoid recent attacks on the private-coin evasive LWE [2, 4, 33].

Step 3: unmask the output. The evaluator's final task is to recover the decryption result by removing the first term from Equation 2. We recall that the public matrix $\mathbf{A}_F$ in the first term of the BGG+ encoding after evaluating the circuit is independent of the input $\mathbf{x}_L$. Therefore, the obfuscator can additionally provide a preimage to allow the evaluator to obtain $\mathbf{s}^T_{\mathbf{x}_L}\mathbf{A}_F$ defined as follows:

$$\mathbf{K}_F \leftarrow\!\!\$\, \mathbf{B}_L^{-1}(\mathbf{P}_F),$$

where $\mathbf{P}_F := \mathbf{u}_{1,L'} \otimes \mathbf{A}_F$. This allows the evaluator to compute $\mathbf{v}^T_{F,\mathbf{x}_L} := \mathbf{p}^T_{\mathbf{x}_L}\mathbf{K}_F = \underline{\mathbf{s}^T_{\mathbf{x}_L}\mathbf{A}_F}$. Hence, the evaluator can remove the first term from $\mathbf{c}^T_{F,\mathbf{x}_L}$, obtaining the (noisy) plaintext output $\underline{\mathbf{y}^T} = \underline{\frac{q}{2}C(\mathbf{x}) + \mathsf{PRF}(K, \mathbf{x})}$. The parameters are chosen so that the accumulated error in $\underline{\mathbf{y}^T}$ are bounded by $B << \frac{q}{4}$, and the PRF output is within the range $[-\frac{q}{4} + B, \frac{q}{4} - B]^{1\times\ell}$, implying that the norm of the PRF output along with the error is bound by $\frac{q}{4}$. Therefore, the evaluator can recover the desired circuit output $C(x)$ by extracting the highest bits of $\underline{\mathbf{y}^T}$.

3 Preliminaries

3.1 Notations

For non-negative integers $i, j \in \mathbb{Z}$ such that $i \leq j$ holds, $[i, j]$ denotes the set $\{i, i+1, \dots, j\}$. For simplicity, $[n]$ and $[0]$ represent the sets $\{1, \dots, n\}$ and $\emptyset$, respectively.

A vector and a matrix are denoted by bold letters, such as $\mathbf{a}$ and $\mathbf{A}$. Unless otherwise specified, we define vectors as column vectors. For any vector $\mathbf{a}$, the size of $\mathbf{a}$ is denoted by $|\mathbf{a}|$, and the i-th element of $\mathbf{a}$ is denoted by a_i for every $i \in [|\mathbf{a}|]$. For any matrix $\mathbf{A}$ with n rows and m columns, let $a_{i,j}$ denote the (i, j)-th entry of $\mathbf{A}$, for every $i \in [n]$ and $j \in [m]$. The ℓ_2-norm of the n-sized vector $\mathbf{a}$ is defined by $\|a\|_2 := \sqrt{\Sigma_{i\in[|\mathbf{a}|]} a_i^2}$. For a matrix $\mathbf{A}$ with n rows and m columns, $\|A\|_\infty$ denotes an entry-wise infinity norm, defined by $\|A\|_\infty := \max_{i\in[n],j\in[m]}\{|a_{i,j}|\}$. For any multiplication between two matrices $\mathbf{A}$ and $\mathbf{B}$, where the column size of $\mathbf{A}$ and the row size of $\mathbf{B}$ are m, it holds that $\|AB\|_\infty \leq m \|A\|_\infty \|B\|_\infty$. For any $n \in \mathbb{N}$ and $i \in [n]$, $\mathbf{u}_{n,i} \in \mathbb{Z}_q^n$ is an unit vector whose i-th entry is 1 and the others are 0.

Let $\mathbb{Z}_q$ denote the ring of integers modulo q. For an integer $x \in \mathbb{Z}_q$, define $\mathsf{bits}(x) := (x_1, \ldots, x_{\lceil \log_2 q\rceil}) \in \{0,1\}^{1\times\lceil \log_2 q\rceil}$ as the vector of bits corresponding to the binary representation of x in little-endian form—i.e., $x = \Sigma_{i\in[\lceil \log_2 q\rceil]} x_i 2^{i-1}$. For a matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, $\mathsf{bits}(\mathbf{A}) \in \{0,1\}^{nm\lceil \log_2 q\rceil}$ is similarly defined as below:

$$\mathsf{bits}(\mathbf{A}) := (\mathsf{bits}(a_{1,1}), \ldots, \mathsf{bits}(a_{n,1}), \ldots, \mathsf{bits}(a_{1,m}), \ldots, \mathsf{bits}(a_{n,m}))^T$$

For any distribution $\mathcal{X}$, $x \leftarrow\$\ \mathcal{X}$ indicates that x is sampled from $\mathcal{X}$. Let $x \leftarrow\$\ \mathbb{Z}_q$ denote that x is sampled from a unifrom distribution over $\mathbb{Z}_q$.

Let λ be a security parameter. A function $\mathsf{negl}(\lambda) : \mathbb{N} \to \mathbb{R}$ is called negligible, if for all constants $c > 0$, there exists $n \in \mathbb{N}$ such that $\mathsf{negl}(\lambda) < \lambda^{-c}$ holds for all $\lambda > n$. For any two distributions $\mathcal{X}$ and $\mathcal{Y}$, we write $\mathcal{X} \overset{c}{\approx} \mathcal{Y}$ (resp., $\mathcal{X} \overset{s}{\approx} \mathcal{Y}$) to denote that $\mathcal{X}$ and $\mathcal{Y}$ are computationally (resp., statistically) indistinguishable for all probabilistic polynomial-time (PPT) adversaries.

3.2 Lattices and Gaussians Preliminaries

We recall some facts on lattices and gaussian distributions.

Lattices. Fix some integers $n, m, q \in \mathbb{N}$. For a full-rank integer matrix $\mathbf{B} \in \mathbb{Z}^{n\times m}$, an integer lattice is defined by $\Lambda(\mathbf{B}) := \{\mathbf{Bx} \mid \mathbf{x} \in \mathbb{Z}^m\}$. For any matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$, a q-ary orthogonal lattice is defined by $\Lambda^\perp(\mathbf{A}) := \{\mathbf{x} \in \mathbb{Z}^m \mid \mathbf{Ax} = \mathbf{0} \mod q\}$. For any vector $\mathbf{u} \in \mathbb{Z}_q^n$, its coset is defined by $\Lambda_{\mathbf{u}}^\perp(\mathbf{A}) := \{\mathbf{x} \in \mathbb{Z}^m \mid \mathbf{Ax} = \mathbf{u} \mod q\}$.

We adopt the leftover hash lemma from [2].

Lemma 1 (Leftover Hash Lemma). Fix some $n, m, q \in \mathbb{N}$. The leftover hash lemma states that if $m \geq 2n \log q$, then for $\mathbf{A} \leftarrow \mathbb{Z}_q^{n\times m}$, $\mathbf{x} \leftarrow \{0,1\}^m$ and $\mathbf{y} \leftarrow \mathbb{Z}_q^n$, the statistical distance between $(\mathbf{A}, \mathbf{Ax})$ and $(\mathbf{A}, \mathbf{y})$ is negligible. More concretely, it is bounded by $q^n\sqrt{2^{1-m}}$.

Gadget vector and matrix. A gadget vector $\mathbf{g}$ and a gadget matrix $\mathbf{G}_n$ are defined as follows, respectively:

$$\mathbf{g}^T := (2^0, 2^1, \ldots, 2^{\lceil \log_2 q\rceil - 1}) \in \mathbb{Z}_q^{\lceil \log_2 q\rceil}$$
$$\mathbf{G}_n := \mathbf{I}_n \otimes \mathbf{g}^T \in \mathbb{Z}_q^{n\times n\lceil \log_2 q\rceil}$$

For any vector $\mathbf{a} \in \mathbb{Z}_q^n$, $\mathbf{G}_n^{-1}(\mathbf{a}) \in \Lambda_{\mathbf{a}}^{\perp}(\mathbf{G}_n)$ is defined as below:

$$\mathbf{G}_n^{-1}(\mathbf{a}) := (\mathsf{bits}(a_1), \dots, \mathsf{bits}(a_n))^T \in \mathbb{Z}_q^{n\lceil \log_2 q \rceil}$$

This is straightforwardly extended to a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ as $\mathbf{G}_n^{-1}(\mathbf{A}) := (\mathbf{G}_n^{-1}(\mathbf{a}_1), \dots, \mathbf{G}_n^{-1}(\mathbf{a}_m)) \in \mathbb{Z}_q^{n\lceil \log_2 q \rceil \times m}$, which satisfies $\mathbf{G}_n \mathbf{G}_n^{-1}(\mathbf{A}) = \mathbf{A}$.

Gaussian distributions. We adopt the definition of the discrete Gaussian distribution from [28, 19]. For any lattice $\Lambda \subseteq \mathbb{R}^b$ and any Gaussian parameter $\sigma > 0$, the discrete Gaussian distribution $\mathcal{D}_{\Lambda,\sigma}$ is defined by the discrete probability distribution via the probability mass function

$$\Pr[\mathbf{e} = \mathbf{x}] \ := \begin{cases} \frac{\rho_\sigma(\mathbf{x})}{\rho_\sigma(\Lambda)} & \text{if } \mathbf{x} \in \Lambda, \\ 0 & \text{otherwise.} \end{cases}$$

for an $\mathbf{e} \sim \mathcal{D}_{\Lambda,\sigma}$, where $\rho_\sigma : \mathbb{R}^n \to \mathbb{R}_{\leq 0}$ is the Gaussian function defined by $\rho_\sigma(\mathbf{x}) := \exp(-\pi \left\|\mathbf{x}\right\|_2^2 / \sigma^2)$. Here, $\rho_\sigma(\Lambda) := \Sigma_{\mathbf{y} \in \Lambda} \ \rho_\sigma(\mathbf{y})$ is finite, ensuring that $\Pr[\mathbf{e} = \mathbf{x}]$ is well-defined as a probability mass function.

The following lemma bounds the tail probability of the discrete Gaussian distribution [44, 19].

Lemma 2 (Tail Bound of the Discrete Gaussian Distribution [44, 19]). For any Gaussian parameter $\sigma > 0$, $\lambda \in \mathbb{N}$, and $e \sim \mathcal{D}_{\mathbb{Z},\sigma}$, it holds that

$$\Pr\Big[e \geq \sqrt{\lambda}\sigma\Big] \leq 2\exp(-\pi\lambda) \leq \widetilde{O}\big(2^{-\lambda}\big)$$

We adopt the smudging lemma from [51, 2].

Lemma 3 (Smudging Lemma [51, 2]). Let λ be a security parameter. Take any $a \in \mathbb{Z}$ where $|a| \leq B$. Suppose $\sigma \geq B\lambda^{\omega(1)}$. Then the statistical distance between the distributions $\{e \mid e \leftarrow\!\!\$\ \mathcal{D}_{\mathbb{Z},\sigma}\}$ and $\{e + a \mid e \leftarrow\!\!\$\ \mathcal{D}_{\mathbb{Z},\sigma}\}$ is $\mathsf{negl}(\lambda)$.

Trapdoor and preimage sampling. We adopt the definitions and syntax of trapdoor and preimage sampling algorithms from [2]. For integers $n, q \in \mathbb{N}$, let $m := \mathcal{O}(n \log q)$ and $\gamma := \omega(\sqrt{n \log q \log m})$. Previous works [28, 44, 26] propose an algorithm $\mathsf{TrapGen}(1^n, 1^m, q)$ that outputs a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ which is 2^{-n} close to unifrom, and a γ-trapdoor for $\mathbf{A}$ denoted by $\mathbf{A}_\gamma^{-1}$.

The trapdoor allows efficient sampling of a preimage for a target vector $\mathbf{u} \in \mathbb{Z}_q^n$ from the discrete Gaussian distribution over $\Lambda_{\mathbf{u}}^{\perp}(\mathbf{A})$, namely $\mathcal{D}_{\Lambda_{\mathbf{u}}^{\perp}(\mathbf{A}),\gamma}$. To sample a preimage for a matrix $\mathbf{U} \in \mathbb{Z}_q^{n \times l}$, we sample l preimages, one for each column of $\mathbf{U}$—that is,

$$\mathbf{A}_\gamma^{-1}(\mathbf{U}) := (\mathbf{A}_\gamma^{-1}(\mathbf{u}_1), \dots, \mathbf{A}_\gamma^{-1}(\mathbf{u}_l)),$$

where $\mathbf{u}_i$ denotes the i-th column of $\mathbf{U}$.

3.3 Hardness Assumptions

We rely on the following hardness assumptions on lattices. The subexponential hardness of the learning with errors (LWE) assumption defined in Assumption 1 is adopted from Assumption A.5 of [3] with necessarily modifications.

Assumption 1 (Subexponential Hardness of (Binary) Learning With Errors (LWE) [47, 3]). Let $n = n(\lambda), m = m(\lambda), q = q(\lambda) > 2$ be integers, and $\chi = \chi(\lambda)$ be a distribution over $\mathbb{Z}_q$. The LWE assumption is said to be subexponentially hard if there exists some constant $\delta \in (0,1)$ such that for every non-uniform adversary $\mathcal{A}$ of size at most 2^{n^δ}, it holds that

$$\left|\Pr\left[\mathcal{A}(\mathbf{A}, \mathbf{s}^T\mathbf{A} + \mathbf{e}^T) = 1\right] \;-\; \Pr\left[\mathcal{A}(\mathbf{A}, \mathbf{u}^T) = 1\right]\right| \;\leq\; 2^{-n^\delta},$$

where $\mathbf{A} \leftarrow\!\$\ \mathbb{Z}_q^{n\times m}$, $\mathbf{s} \leftarrow\!\$\ \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow\!\$\ \chi^m$, $\mathbf{u} \leftarrow\!\$\ \mathbb{Z}_q^m$, and the probability is taken over the randomness of $\mathcal{A}$. When the same holds with the secret $\mathbf{s}$ sampled uniformly from bits (i.e., $\mathbf{s} \leftarrow\!\$\ \{0,1\}^n$), we say that the binary LWE assumption is subexponentially hard for non-uniform adversaries.

We adopt a variant of the evasive LWE assumption defined in [3], called non-uniform κ-evasive LWE assumption [3]. However, unlike their definition, our definition assumes that an adversary can access to a public matrix $\mathbf{P}$. This is because Brzuska et al. [20] show that the adversary who can see only another public matrix $\mathbf{B}$ can break the security of the evasive LWE with a non-negligible advantage in the private-coin setting. To fix that vulnerability, they introduced a definition, called private-coin binding evasive LWE [20], which includes $\mathbf{P}$ in the adversary's view.

Assumption 2 (Private-coin Binding Non-Uniform κ-Evasive LWE [3, 20]). Let $n, m, t, m', q \in \mathbb{N}$ be parameters and λ be a security parameter. Let χ and χ' be Gaussian parameters. Let Samp be a PPT algorithm that outputs

$$\mathbf{S} \in \mathbb{Z}_q^{m'\times n}, \mathbf{P} \in \mathbb{Z}_q^{n\times t}, \mathsf{aux} \in \{0,1\}^*$$

on input 1^λ. For non-uniform adversaries $\mathcal{A}_0 = \{\mathcal{A}_{0,\lambda}\}_\lambda$, $\mathcal{A}_1 = \{\mathcal{A}_{1,\lambda}\}_\lambda$, we define the following advantage functions:

$$\begin{aligned}
\mathrm{Adv}^{\mathrm{PRE}}_{\mathcal{A}_0}(\lambda) &:= \Pr\left[\mathcal{A}_0\left(\mathbf{B}, \mathbf{P}, \mathbf{SB} + \mathbf{E}, \mathbf{SP} + \mathbf{E}', \mathsf{aux}\right) = 1\right] \\
&\quad - \Pr\left[\mathcal{A}_0\left(\mathbf{B}, \mathbf{P}, \mathbf{C}_0, \mathbf{C}', \mathsf{aux}\right) = 1\right], \\
\mathrm{Adv}^{\mathrm{POST}}_{\mathcal{A}_1}(\lambda) &:= \Pr\left[\mathcal{A}_1\left(\mathbf{B}, \mathbf{P}, \mathbf{SB} + \mathbf{E}, \mathbf{K}, \mathsf{aux}\right) = 1\right] \\
&\quad - \Pr\left[\mathcal{A}_1\left(\mathbf{B}, \mathbf{P}, \mathbf{C}_0, \mathbf{K}, \mathsf{aux}\right) = 1\right],
\end{aligned}$$

where

$$\begin{aligned}
&(\mathbf{S}, \mathbf{P}, \mathsf{aux}) \leftarrow\!\$\ \mathsf{Samp}(1^\lambda), \mathbf{B} \leftarrow\!\$\ \mathbf{Z}_q^{n\times m}, \\
&\mathbf{C}_0 \leftarrow\!\$\ \mathbf{Z}_q^{m'\times m}, \mathbf{C}' \leftarrow\!\$\ \mathbf{Z}_q^{m'\times t}, \\
&\mathbf{E} \leftarrow\!\$\ \mathcal{D}_{\mathbb{Z},\chi}^{m'\times m}, \mathbf{E}' \leftarrow\!\$\ \mathcal{D}_{\mathbb{Z},\chi'}^{m'\times t}, \\
&\mathbf{K} \leftarrow\!\$\ \mathbf{B}_\gamma^{-1}(\mathbf{P}),
\end{aligned}$$

and $\gamma = \mathcal{O}(\sqrt{m \log q})$.

For a function $\kappa = \kappa(\lambda)$ of the security parameter λ, the private-coin binding non-uniform κ-evasive LWE assumption is said to hold, if for every non-uniform sampler Samp and every non-uniform adversary $\mathcal{A}_1$ such that $\mathrm{Size}(\mathsf{Samp}) \leq \mathsf{poly}(\lambda')$ and $\mathrm{Size}(\mathcal{A}_1) \leq \mathsf{poly}(\kappa)$ for $\lambda'(\lambda) \leq \kappa(\lambda)$, there exists another non-uniform adversary $\mathcal{A}_0$ and a polynomial $Q(\cdot)$ such that the following relations hold:

$$\begin{aligned} \mathrm{Adv}^{\mathrm{PRE}}_{\mathcal{A}_0}(\lambda) \ &\geq \ \mathrm{Adv}^{\mathrm{POST}}_{\mathcal{A}_1}(\lambda)/Q(\lambda') \ - \ \mathrm{negl}(\kappa), \\ \mathrm{Size}(\mathcal{A}_0) \ &\leq \ Q(\lambda') \ \cdot \ \mathrm{Size}(\mathcal{A}_1). \end{aligned}$$

Rather than directly relying on Assumption 2, we adopt Lemma 2.5 from [3] with necessarily modifications. This distinguishes between the portion of the auxiliary information that dependes on the secret $\mathbf{S}$ and the portion that does not, and claims that if the former portion in the precondition is pseudorandom, the same holds in the postcondition.

Lemma 4. Let $n, m, t, m', q \in \mathbb{N}$ be parameters and λ be a security parameter. Let χ and χ' be Gaussian parameters. Let Samp be a PPT algorithm that takes as input 1^λ and outputs

$$\mathbf{S} \in \mathbb{Z}_q^{m' \times n}, \mathbf{P} \in \mathbb{Z}_q^{n \times t}, \mathsf{aux} = (\mathsf{aux}_1, \mathsf{aux}_2) \in \mathcal{S} \times \{0,1\}^*$$

for some set $\mathcal{S}$. Furthermore, we assume that there exists a public deterministic polynomial-time algorithm $\mathsf{Reconstruct}$ that allows to derive $\mathbf{P}$ from aux_2, i.e., $\mathbf{P} = \mathsf{Reconstruct}(\mathsf{aux}_2)$. For a non-uniform adversary $\mathcal{A}$, we define the following advantage functions:

$$\begin{aligned} \mathrm{Adv}^{\mathrm{PRE}}_{\mathcal{A}}(\lambda) &:= \Pr\big[\mathcal{A}\big(\mathbf{B},\, \mathbf{P},\, \mathbf{SB}+\mathbf{E},\, \mathbf{SP}+\mathbf{E}',\, \mathsf{aux}_1,\, \mathsf{aux}_2\big) = 1\big] \\ &\quad - \ \Pr\big[\mathcal{A}\big(\mathbf{B},\, \mathbf{P},\, \mathbf{C}_0,\, \mathbf{C}',\, \mathbf{c},\, \mathsf{aux}_2\big) = 1\big], \\ \mathrm{Adv}^{\mathrm{POST}}_{\mathcal{A}}(\lambda) &:= \Pr\big[\mathcal{A}\big(\mathbf{B},\, \mathbf{P},\, \mathbf{SB}+\mathbf{E},\, \mathbf{K},\, \mathsf{aux}_1,\, \mathsf{aux}_2\big) = 1\big] \\ &\quad - \ \Pr\big[\mathcal{A}\big(\mathbf{B},\, \mathbf{P},\, \mathbf{C}_0,\, \mathbf{K},\, \mathbf{c},\, \mathsf{aux}_2\big) = 1\big], \end{aligned}$$

where

$$\begin{aligned} &(\mathbf{S}, \mathbf{P}, \mathsf{aux} = (\mathsf{aux}_1, \mathsf{aux}_2)) \leftarrow\!\!\$\ \mathsf{Samp}(1^\lambda), \mathbf{B} \leftarrow\!\!\$\ \mathbf{Z}_q^{n \times m}, \\ &\mathbf{C}_0 \leftarrow\!\!\$\ \mathbf{Z}_q^{m' \times m}, \mathbf{C}' \leftarrow\!\!\$\ \mathbf{Z}_q^{m' \times t}, \mathbf{c} \leftarrow\!\!\$\ \mathcal{S}, \\ &\mathbf{E} \leftarrow\!\!\$\ \mathcal{D}_{\mathbb{Z},\chi}^{m' \times m}, \mathbf{E}' \leftarrow\!\!\$\ \mathcal{D}_{\mathbb{Z},\chi'}^{m' \times t}, \mathbf{K} \leftarrow\!\!\$\ \mathbf{B}_\gamma^{-1}(\mathbf{P}), \end{aligned}$$

and $\gamma = \mathcal{O}(\sqrt{m \log q})$.

Then, for a function $\kappa := \kappa(\lambda)$ of the security parameter λ, under the private-coin binding non-uniform κ-evaisve LWE assumption (Assumption 2), if $\mathrm{Size}(\mathsf{Samp}) \leq \mathsf{poly}(\lambda')$ and $\mathrm{Size}(\mathcal{A}_1) \leq \mathsf{poly}(\kappa)$ for $\lambda' \leq \kappa(\lambda)$, there exists another non-uniform adversary $\mathcal{A}_0$ and a polynomial $Q(\cdot)$ such that the following relations hold:

$$\begin{aligned} \mathrm{Adv}^{\mathrm{PRE}}_{\mathcal{A}_0}(\lambda) \ &\geq \ \mathrm{Adv}^{\mathrm{POST}}_{\mathcal{A}_1}(\lambda)/Q(\lambda') \ - \ \mathrm{negl}(\kappa), \\ \mathrm{Size}(\mathcal{A}_0) \ &\leq \ Q(\lambda') \ \cdot \ \mathrm{Size}(\mathcal{A}_1). \end{aligned}$$

Proof. The only difference between our lemma and the original lemma in [3] is that $\mathbf{P}$ is added to the adversaries' views. Since the proof of the original lemma does not depend on the presence of $\mathbf{P}$, their proof can be directly applied to the proof of our lemma. $\square$

Additionally, we employ a private-coin hiding evasive LWE assumption introduced in [20]. In this variant, the adversary is not given the public matrices $\mathbf{B}$ and $\mathbf{P}$. This assumption requires $\mathbf{P}$ to be hidden from the adversary. Formally, for a parameter ℓ, the auxiliary information aux should not help the adversary distinguish $\mathbf{P}$ from $\mathbf{P}+\mathbf{R}$, where $\mathbf{R}$ is a random matrix whose entries are sampled uniformly from $\{0,1,\ldots,\ell\}$. We fix ℓ to its maximum value q because the assumption becomes weaker as ℓ grows. As with the previous assumption, we assume non-uniform adversaries.

Assumption 3 (Private-coin Hiding Non-Uniform κ-Evasive LWE [20]). Let $n, m, t, m', q \in \mathbb{N}$ be parameters and λ be a security parameter. Let χ and χ' be Gaussian parameters. Let Samp be a PPT algorithm that outputs

$$\mathbf{S} \in \mathbb{Z}_q^{m' \times n}, \mathbf{P} \in \mathbb{Z}_q^{n \times t}, \mathsf{aux} \in \{0,1\}^*$$

on input 1^λ. For non-uniform adversaries $\mathcal{A}_0 = \{\mathcal{A}_{0,\lambda}\}_\lambda$, $\mathcal{A}_1 = \{\mathcal{A}_{1,\lambda}\}_\lambda$, we define the following advantage functions:

$$\begin{aligned}
\mathrm{Adv}^{\mathrm{PRE1}}_{\mathcal{A}_0}(\lambda) &:= \Pr\big[\mathcal{A}_0\big(\mathbf{SB}+\mathbf{E},\, \mathbf{SP}+\mathbf{E}',\, \mathsf{aux}\big)=1\big] \\
&\quad - \Pr\big[\mathcal{A}_0\big(\mathbf{C}_0,\, \mathbf{C}',\, \mathsf{aux}\big)=1\big], \\
\mathrm{Adv}^{\mathrm{PRE2}}_{\mathcal{A}_0}(\lambda) &:= \Pr\big[\mathcal{A}_0\big(\mathbf{P},\, \mathsf{aux}\big)=1\big] \\
&\quad - \Pr\big[\mathcal{A}_0\big(\mathbf{P}+\mathbf{R},\, \mathsf{aux}\big)=1\big], \\
\mathrm{Adv}^{\mathrm{POST}}_{\mathcal{A}_1}(\lambda) &:= \Pr\big[\mathcal{A}_1\big(\mathbf{SB}+\mathbf{E},\, \mathbf{K},\, \mathsf{aux}\big)=1\big] \\
&\quad - \Pr\big[\mathcal{A}_1\big(\mathbf{C}_0,\, \mathbf{K},\, \mathsf{aux}\big)=1\big],
\end{aligned}$$

where

$$\begin{aligned}
&(\mathbf{S}, \mathbf{P}, \mathsf{aux}) \leftarrow\!\$\, \mathsf{Samp}(1^\lambda), \mathbf{B} \leftarrow\!\$\, \mathbf{Z}_q^{n \times m}, \\
&\mathbf{C}_0 \leftarrow\!\$\, \mathbf{Z}_q^{m' \times m}, \mathbf{C}' \leftarrow\!\$\, \mathbf{Z}_q^{m' \times t}, \\
&\mathbf{E} \leftarrow\!\$\, \mathcal{D}_{\mathbb{Z},\chi}^{m' \times m}, \mathbf{E}' \leftarrow\!\$\, \mathcal{D}_{\mathbb{Z},\chi'}^{m' \times t}, \\
&\mathbf{R} \leftarrow\!\$\, \mathbf{Z}_q^{n \times t}, \mathbf{K} \leftarrow\!\$\, \mathbf{B}_\gamma^{-1}(\mathbf{P}),
\end{aligned}$$

and $\gamma = \mathcal{O}(\sqrt{m \log q})$.

For a function $\kappa = \kappa(\lambda)$ of the security parameter λ, the private-coin binding non-uniform κ-evasive LWE assumption is said to hold, if for every non-uniform sampler Samp and every non-uniform adversary $\mathcal{A}_1$ such that $\mathrm{Size}(\mathsf{Samp}) \leq \mathsf{poly}(\lambda')$ and $\mathrm{Size}(\mathcal{A}_1) \leq \mathsf{poly}(\kappa)$ for $\lambda'(\lambda) \leq \kappa(\lambda)$, there exists another non-uniform adversary $\mathcal{A}_0$ and a polynomial $Q(\cdot)$ such that the following relations hold:

$$\begin{aligned}
\mathrm{Adv}^{\mathrm{PRE1}}_{\mathcal{A}_0}(\lambda) + \mathrm{Adv}^{\mathrm{PRE2}}_{\mathcal{A}_0}(\lambda) \;&\geq\; \mathrm{Adv}^{\mathrm{POST}}_{\mathcal{A}_1}(\lambda)/Q(\lambda') \;-\; \mathrm{negl}(\kappa), \\
\mathrm{Size}(\mathcal{A}_0) \;&\leq\; Q(\lambda') \,\cdot\, \mathrm{Size}(\mathcal{A}_1).
\end{aligned}$$

3.4 Fully Homomorphic Encryption

We adopt a fully homomorphic encryption (FHE) scheme with the following syntax:

- $\mathsf{KeyGen}(1^\lambda) \rightarrow (\mathsf{sk}, \mathsf{pk})$: it takes as input the security parameter 1^λ and outputs a secret key sk and a public key pk.
- $\mathsf{Enc}(\mathsf{pk}, \mathsf{msg}) \rightarrow \mathsf{ct}$: it takes as input the public key pk and a message msg and outputs a ciphertext ct.
- $\mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}) \rightarrow \mathsf{ct}'$: it takes as input the public key pk, a circuit C, and a ciphertext ct, and outputs a ciphertext ct'.
- $\mathsf{Dec}(\mathsf{sk}, \mathsf{ct}) \rightarrow \mathsf{msg}$: it takes as input the secret key sk and a ciphertext ct and outputs a message msg.

For simplicity, the $\mathsf{FHE.Eval}$ algorithm takes a single encrypted message as input. The semantic security of the FHE scheme is defined as follows.

Definition 1 ((Subexponential) Semantic Security of FHE [16]). A FHE scheme is said to be semantically secure if for all polynomial-sized adversaries $\mathcal{A}$ and pairs of messages $(\mathsf{msg}_0, \mathsf{msg}_1)$, it holds that

$$|\Pr[\mathcal{A}(\mathsf{pk}, \mathsf{Enc}(\mathsf{pk}, \mathsf{msg}_0)) = 1] - \Pr[\mathcal{A}(\mathsf{pk}, \mathsf{Enc}(\mathsf{pk}, \mathsf{msg}_1)) = 1]| \leq \mathsf{negl}(\lambda),$$

where $(\mathsf{sk}, \mathsf{pk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$. The FHE scheme is said to satisfy the subexponential semantic security if there exists some constant $\delta \in (0, 1)$ such that for all non-uniform adversaries of size at most 2^{λ^δ}, the advantage defined above is bounded by $2^{-\lambda^\delta}$ for sufficiently large λ.

In addition to the subexponential semantic security, we require that (1) a ciphertext can be decrypted by multiplying it with the secret key vector sampled over bits, and (2) the evaluation algorithm is shallow. Lemma 5 shows that there exists such a FHE scheme.

Lemma 5. There exists a FHE scheme parameterized by $q, n, m, L > 1, \ell, \mathsf{dep} \in \mathbb{N}$ such that in addition to the subexponential semantic security (Definition 1), the following properties hold:

- **Correctness with Linear Decryption.** The FHE scheme supports evaluation of a vector-valued circuit $C : \{0,1\}^L \rightarrow \mathbb{Z}_q^{1\times\ell}$ of a depth at most dep, in which the secret key $\mathbf{t}$ and the evaluated ciphertext $\mathbf{Y}$ are defined, respectively, over $\{0,1\}^n$ and $\mathbb{Z}_q^{n\times\ell}$, respectively. For all circuits C and inputs $\mathsf{msg} \in \{0,1\}^L$, there exists a small-norm vector $\mathbf{e} \in \mathbb{Z}_q^\ell$ such that the following holds:

$$\mathbf{t}^T\mathbf{Y} = C(\mathsf{msg}) + \mathbf{e}^T,\ \|\mathbf{e}\|_\infty \leq \mathsf{poly}(\lambda) m^{\mathcal{O}(\mathsf{poly}(\mathsf{dep}, \log q))},$$

where $(\mathbf{t}, \mathsf{pk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$, $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, \mathsf{msg})$, $\mathbf{Y} \leftarrow \mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct})$.

– **Shallowness of the FHE Evaluation Algorithm.** For all public keys pk and circuits $C : \{0,1\}^L \to \mathbb{Z}_q^{1\times\ell}$ of depth at most dep, let $C_{\mathsf{Eval}(\mathsf{pk},C,\cdot)}$ be a circuit that on input a ciphertext ct outputs

$$\mathbf{Y} = \mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}).$$

The depth of $C_{\mathsf{Eval}(\mathsf{pk},C,\cdot)}$ is bounded by $\mathsf{poly}(\mathsf{dep}, \log n, \log m, \log q)$.

Proof. We show that a variant of the Gentry, Sahai, and Waters (GSW) leveled FHE scheme [29] satisfies the claimed properties. Let σ_E be a Gaussian parameter such that the LWE assumption with the parameters (n, m, q, σ_E) is subexponentially hard. We adopt syntax and properties from [35] with necessary modifications.

– $\mathsf{KeyGen}(1^\lambda) \to (\mathbf{t}^T, \mathbf{A}_{\mathsf{fhe}})$:

$$\text{(secret)} \quad \mathbf{t}^T := \left(\bar{\mathbf{t}}^T, -1\right) \in \mathbb{Z}_q^{n+1},$$
$$\text{(public)} \quad \mathbf{A}_{\mathsf{fhe}} := \begin{pmatrix} \bar{\mathbf{A}}_{\mathsf{fhe}} \\ \bar{\mathbf{t}}^T \bar{\mathbf{A}}_{\mathsf{fhe}} + \mathbf{e}_{\mathsf{fhe}}^T \end{pmatrix} \in \mathbb{Z}_q^{(n+1)\times m},$$

where $\bar{\mathbf{t}} \leftarrow\$ \{0,1\}^n$, $\bar{\mathbf{A}}_{\mathsf{fhe}} \leftarrow\$ \mathbb{Z}_q^{n\times m}$, $\bar{\mathbf{e}}_{\mathsf{fhe}} \leftarrow\$ \mathcal{D}_{\mathbb{Z},\sigma_E}^m$.

– $\mathsf{Enc}(\mathbf{A}_{\mathsf{fhe}}, \mathbf{x}) \to \mathbf{X}$:

$$\mathbf{X} := \mathbf{A}_{\mathsf{fhe}}\mathbf{R} - \mathbf{x}^T \otimes \mathbf{G}_{n+1} \in \mathbb{Z}_q^{(n+1)\times Lm},$$

where $\mathbf{x} \in \{0,1\}^L$ and $\mathbf{R} \leftarrow\$ \{0,1\}^{m\times Lm}$ is the encryption randomness.

– $\mathsf{Eval}(\mathbf{A}_{\mathsf{fhe}}, C, \mathbf{X}) \to \mathbf{Y}$ where C is a vector-valued circuit $C : \{0,1\}^L \to \mathbb{Z}_q^{1\times m'}$.

– $\mathsf{Dec}(\mathbf{t}^T, \mathbf{X}) \to \mathbf{x}$

$$\mathbf{t}^T\mathbf{X} = -\mathbf{e}_{\mathsf{fhe}}^T\mathbf{R} - \mathbf{x} \otimes \mathbf{t}^T\mathbf{G} \in \mathbb{Z}_q^m,$$

which can be used to extract x via multiplication by $\mathbf{G}^{-1}(\lceil q/2 \rceil \mathbf{u}_{n+1,n+1})$.

We now prove the claimed properties.

Subexponential semantic security. We show that the public key $\mathbf{A}_{\mathsf{fhe}}$ and the ciphertext $\mathbf{X}$ are indistinguishable from random matrices, respectively. Specifically, it holds that

$$\begin{aligned}
&\left(\bar{\mathbf{A}} := \begin{pmatrix} \bar{\mathbf{A}}_{\mathsf{fhe}} \\ \bar{\mathbf{b}}^T := \bar{\mathbf{t}}^T\bar{\mathbf{A}}_{\mathsf{fhe}} + \mathbf{e}_{\mathsf{fhe}}^T \end{pmatrix}, \mathbf{X} := \mathbf{A}_{\mathsf{fhe}}\mathbf{R} - \mathbf{x}^T \otimes \mathbf{G}_{n+1}\right) \\
\overset{c}{\approx} &\left(\bar{\mathbf{A}} := \begin{pmatrix} \bar{\mathbf{A}}_{\mathsf{fhe}} \\ \bar{\mathbf{b}}^T \leftarrow\$ \mathbb{Z}_q^m \end{pmatrix}, \mathbf{X} := \mathbf{A}_{\mathsf{fhe}}\mathbf{R} - \mathbf{x}^T \otimes \mathbf{G}_{n+1}\right) \\
\overset{s}{\approx} &\left(\bar{\mathbf{A}}_{\mathsf{fhe}} := \begin{pmatrix} \bar{\mathbf{A}}_{\mathsf{fhe}} \\ \bar{\mathbf{b}}^T \leftarrow\$ \mathbb{Z}_q^m \end{pmatrix}, \mathbf{X}, \leftarrow\$ \mathbb{Z}_q^{(n+1)\times Lm}\right).
\end{aligned}$$

Here, the first computationally indistinguishability follows from the subexponential hardness of the LWE assumption (Assumption 1). Since $L > 1$—namely $Lm \geq 2n\lceil \log_2 q \rceil$—the leftover hash lemma (Lemma 1) implies that the second statistical indistinguishability also holds. Hence, this completes the proof of the claimed security.

Correctness. For an input ciphertext $\mathbf{X}_\ell := \mathbf{A}_{\mathsf{fhe}}\mathbf{R}_\ell - \mathbf{x} \otimes \mathbf{G}$ and a vector-valued circuit $C : \{0,1\}^L \to \mathbb{Z}_q^{1\times m'}$, Lemma 5 in [35] shows that there exists a matrix $\mathbf{R}_C$ such that the following holds:

$$\mathbf{Y} \; := \; \mathbf{A}_{\mathsf{fhe}}\mathbf{R}_C \; - \; \begin{pmatrix} \mathbf{0}_{n\times m'} \\ C(\mathbf{x}) \end{pmatrix} \; \in \; \mathbb{Z}_q^{(n+1)\times m'},$$

where $\|\mathbf{R}_C\|_\infty \leq (m+2)^{\mathsf{dep}}\lceil \log_2 q \rceil \max_{\ell\in[L]} \left\|\mathbf{R}_\ell^T\right\|_\infty$. We can obtain

$$\mathbf{t}^T\mathbf{Y} \; = \; -\mathbf{e}_{\mathsf{fhe}}^T\mathbf{R}_C + C(\mathbf{x}).$$

As shown in Lemma 2, it holds that $\left\|\mathbf{e}_{\mathsf{fhe}}^T\right\|_\infty \leq \sqrt{\lambda}\sigma_E$ with overwhelming probability. Additionally, we adopt $\|\mathbf{R}_C\|_\infty \leq m(m+2)^{\mathsf{dep}}\lceil \log_2 q \rceil$ from Lemma 4 in [35]. Substituting these bounds we obtain:

$$\begin{aligned} \|\mathbf{e}\|_\infty &\leq \sqrt{\lambda}\sigma_E m(m+2)^{\mathsf{dep}}\lceil \log_2 q \rceil \\ &\leq \mathsf{poly}(\lambda) m^{\mathcal{O}(\mathsf{poly}(n,\lceil \log_2 q \rceil))}, \end{aligned}$$

which satisfies the claimed bound.

Shallowness. As shown in Lemma 5 in [35], the depth of $C_{\mathsf{Eval}(\mathsf{pk},C,\cdot)}$ is bounded by $\left(\mathsf{dep}\,\mathcal{O}(\log m \log\log q) + \mathcal{O}(\log^2 \log q)\right)$ for a circuit C of depth dep. Therefore, it is bounded by $\mathsf{poly}(\mathsf{dep}, \log n, \log m, \log q)$. $\square$

3.5 Homomorphic Evaluation of BGG+ Encodings

We adopt the syntax of homomorphic evaluation algorithms used for BGG+ encodings [13] from [52]. For parameters $q, n, L, \ell \in \mathbb{N}$ and $m = n\lceil \log_2 q \rceil$, a public matrix for the input encoding is denoted by $\mathbf{A}_{\mathsf{att}} \in \mathbb{Z}_q^{n\times(L+1)m}$.

The following two deterministic algorithms allows homomorphically evaluating a boolean circuit $C : \{0,1\}^L \to \{0,1\}^\ell$ with bits $\mathbf{x} \in \{0,1\}^L$ on the input encoding [13].

- $\mathsf{EvalC}(\mathbf{A}_{\mathsf{att}}, C) \to \mathbf{A}_C$: it takes as input the public matrix $\mathbf{A}_{\mathsf{att}}$ and the circuit C and outputs a matrix $\mathbf{A}_C \in \mathbb{Z}_q^{n\times \ell m}$.
- $\mathsf{EvalCX}(\mathbf{A}_{\mathsf{att}}, C, \mathbf{x}) \to \mathbf{H}_{C,x}$: it takes as input the public matrix $\mathbf{A}_{\mathsf{att}}$, the circuit C, and the input $\mathbf{x}$, and outputs a matrix $\mathbf{H}_{C,x} \in \mathbb{Z}_q^{(L+1)m\times \ell m}$.

The outputs of these algorithms satisfies the following properties:

- For a circuit C of a depth dep, $\left\|\mathbf{H}_{C,x}^T\right\|_\infty \leq m^{\mathcal{O}(\mathsf{dep})}$ holds.

– For any $\mathbf{A}_{\mathsf{att}}$, C, and $\mathbf{x}$, $(\mathbf{A}_{\mathsf{att}} - (1, \mathbf{x}^T) \otimes \mathbf{G}_n)\mathbf{H}_{C,x} = \mathbf{A}_C - C(\mathbf{x}) \otimes \mathbf{G}_n$ holds, where $\mathbf{A}_C \leftarrow \mathsf{EvalC}(\mathbf{A}_{\mathsf{att}}, C)$ and $\mathbf{H}_{C,x} \leftarrow \mathsf{EvalCX}(\mathbf{A}_{\mathsf{att}}, C, \mathbf{x})$.

Additionally, we adopt evaluation algorithms for an inner product proposed in [31]. The original algorithms are defined for an inner product bettween a vector $\mathbf{t} \in \mathbb{Z}_q^{\ell}$ and the output of the boolean circuit $C(\mathbf{x}) \in \{0,1\}^{\ell}$; ours in Lemma 6 extends the output of the circuit to a matrix $\mathbf{Y} \in \mathbb{Z}_q^{\ell \times \ell'}$ by combining a technique introduced in Section 4.1 of [18].

Lemma 6 (Evaluation of BGG+ encodings for a vector-matrix multiplication). For every parameters $q, n, L, \ell, \ell' \in \mathbb{N}$, and $m = n\lceil \log_2 q \rceil$, the following two deterministic algorithms are defined:

– $\mathsf{VMEvalC}((\mathbf{A}_{\mathsf{att}}, \mathbf{A}_t), F) \to \mathbf{A}_F$: it takes as input the public matrixes $\mathbf{A}_{\mathsf{att}} \in \mathbb{Z}_q^{n \times (L+1)m}$ and $\mathbf{A}_t \in \mathbb{Z}_q^{n \times \ell m}$, and the matrix-valued circuit $F : \{0,1\}^L \to \mathbb{Z}_q^{\ell \times \ell'}$, and outputs a matrix $\mathbf{A}_F \in \mathbb{Z}_q^{n \times \ell' m}$.
– $\mathsf{VMEvalCX}((\mathbf{A}_{\mathsf{att}}, \mathbf{A}_t), F, \mathbf{x}) \to \mathbf{H}_{F,x}$: it takes as input the public matrixes $\mathbf{A}_{\mathsf{att}} \in \mathbb{Z}_q^{n \times (L+1)m}$ and $\mathbf{A}_t \in \mathbb{Z}_q^{n \times \ell m}$, the matrix-valued circuit $F : \{0,1\}^L \to \mathbb{Z}_q^{\ell \times \ell'}$, and the input $\mathbf{x} \in \{0,1\}^L$, and outputs a matrix $\mathbf{H}_{F,x} \in \mathbb{Z}_q^{(L+1+\ell)m \times \ell' m}$.

The outputs of these algorithms satisfy the following properties:

– For a circuit F of a depth dep, $\left\|\mathbf{H}_{F,x}\right\|_\infty \leq 2\ell\ell' \lceil \log_2 q \rceil^2 m^{\mathcal{O}(\mathsf{dep})+1}$ holds.
– For any $\mathbf{A}_{\mathsf{att}}$, $\mathbf{A}_t$, F, $\mathbf{x}$, and $\mathbf{t}$, $((\mathbf{A}_{\mathsf{att}} - (1, \mathbf{x}^T) \otimes \mathbf{G}_n), (\mathbf{A}_t - \mathbf{t}^T \otimes \mathbf{G}_n))\mathbf{H}_{F,x} = \mathbf{A}_F - \mathbf{t}^T\mathbf{Y} \otimes \mathbf{G}_n$ holds, where $\mathbf{A}_F \leftarrow \mathsf{VMEvalC}((\mathbf{A}_{\mathsf{att}}, \mathbf{A}_t), F)$, $\mathbf{H}_{F,x} \leftarrow \mathsf{VMEvalCX}((\mathbf{A}_{\mathsf{att}}, \mathbf{A}_t), F, \mathbf{x})$, and $\mathbf{Y} \leftarrow F(\mathbf{x})$.

Proof. Before describing the concrete consturctions of the algorithms, we present how to compute the encoding of $\mathbf{t}^T\mathbf{Y}$. Let F' be a boolean circuit such that the $((k-1)\ell\ell' + (j-1)\ell + i)$-th output bit is the k-th bit of the (i,j)-th element of $\mathbf{Y} = F(\mathbf{x})$, denoted by $y_{i,j,k} \in \{0,1\}$, for every $i \in [\ell]$, $j \in [\ell']$, and $k \in [\lceil \log_2 q \rceil]$. The matrixes $\mathbf{A}_{F'} \leftarrow \mathsf{EvalC}(\mathbf{A}_{\mathsf{att}}, F')$ and $\mathbf{H}_{F',x} \leftarrow \mathsf{EvalCX}(\mathbf{A}_{\mathsf{att}}, F', \mathbf{x})$ satisfies an equation $(\mathbf{A}_{\mathsf{att}} - (1, \mathbf{x}^T) \otimes \mathbf{G}_n)\mathbf{H}_{F',x} = \mathbf{A}_{F'} - \mathsf{bits}(\mathbf{Y}) \otimes \mathbf{G}_n$.

We observe that the multiplication $\mathbf{t}^T\mathbf{Y}$ is decomposed as follows:

$$\mathbf{t}^T\mathbf{Y} = \sum_{k \in [\lceil \log_2 q \rceil]} 2^{k-1} \sum_{j \in [\ell']} (\sum_{i \in [\ell]} t_i y_{i,j,k}) \mathbf{u}_{\ell',j}$$

, where $\mathbf{u}_{\ell',j} \in \mathbb{Z}_q^{\ell'}$ is a unit vector whose j-th element is 1 and the other elements are 0. For every $j \in [\ell']$ and $k \in [\lceil \log_2 q \rceil]$, the encoding of $\sum_{i \in [\ell]} t_i y_{i,j,k}$ is computed as below:

$$
\begin{aligned}
&\Sigma_{i\in[\ell]}\left((\mathbf{A}_{\mathsf{att}}-(1,\mathbf{x}^T)\otimes\mathbf{G}_n),(\mathbf{A}_t-\mathbf{t}^T\otimes\mathbf{G}_n)\right)\begin{pmatrix}\mathbf{H}_{F',x,y_{i,j,k}} & \mathbf{0}_m\\ \mathbf{0}_m & \mathbf{H}_{t,i}\end{pmatrix}\begin{pmatrix}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\\ y_{i,j,k}\mathbf{I}_m\end{pmatrix}\\
&=\Sigma_{i\in[\ell]}\left((\mathbf{A}_{F',y_{i,j,k}}-y_{i,j,k}\mathbf{G}_n),(\mathbf{A}_t\mathbf{H}_{t,i}-t_i\mathbf{G}_n)\right)\begin{pmatrix}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\\ y_{i,j,k}\mathbf{I}_m\end{pmatrix}\\
&=\left(\Sigma_{i\in[\ell]}\mathbf{A}_{F',y_{i,j,k}}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\right)-\left(\sum_{i\in[\ell]}t_iy_{i,j,k}\right)\mathbf{G}_n
\end{aligned}
$$

, where $\mathbf{A}_{F',y_{i,j,k}},\mathbf{H}_{F',x,y_{i,j,k}}\in\mathbb{Z}_q^{(L+1)m\times m}$ are the submatrix of $\mathbf{A}_{F'}$ and $\mathbf{H}_{F',x}$ from the $(((k-1)\ell\ell'+(j-1)\ell+i-1)m+1)$-th column to $(((k-1)\ell\ell'+(j-1)\ell+i)m)$-th column, respectively, and $\mathbf{H}_{t,i}\in\mathbb{Z}_q^{\ell m\times m}$ is a matrix such that a submatrix from the $((i-1)m+1)$-th row to (im)-th row is an identity matrix $\mathbf{I}_m$ and the other elements are 0.

We next compute a linear combination of the above encodings to obtain the encoding of $\mathbf{t}^T\mathbf{y}_j$, where $\mathbf{y}_j$ is the j-th row of $\mathbf{Y}$ for every $j\in[\ell']$, as follows:

$$
\begin{aligned}
&\Sigma_{k\in[\lceil\log_2 q\rceil]}\left(\left(\Sigma_{i\in[\ell]}\mathbf{A}_{F',y_{i,j,k}}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\right)-\left(\sum_{i\in[\ell]}t_iy_{i,j,k}\right)\mathbf{G}_n\right)\mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n)\\
&=\Sigma_{k\in[\lceil\log_2 q\rceil]}\left(\Sigma_{i\in[\ell]}\mathbf{A}_{F',y_{i,j,k}}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\right)\mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n)-\mathbf{t}^T\mathbf{y}_j\mathbf{G}_n
\end{aligned}
$$

Hence, the encoding of $\mathbf{t}^T\mathbf{Y}$ can be constructed by concatenating the above encodings along a row:

$$
\begin{aligned}
&\Sigma_{j\in[\ell']}\mathbf{u}_{\ell',j}^T\otimes\left(\Sigma_{k\in[\lceil\log_2 q\rceil]}\left(\Sigma_{i\in[\ell]}\mathbf{A}_{F',y_{i,j,k}}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\right)\mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n)-\mathbf{t}^T\mathbf{y}_j\mathbf{G}_n\right)\\
&=\Sigma_{j\in[\ell']}\mathbf{u}_{\ell',j}^T\otimes\left(\Sigma_{k\in[\lceil\log_2 q\rceil]}\left(\Sigma_{i\in[\ell]}\mathbf{A}_{F',y_{i,j,k}}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\right)\mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n)\right)\\
&-\Sigma_{j\in[\ell']}\mathbf{u}_{\ell',j}^T\otimes\mathbf{t}^T\mathbf{y}_j\mathbf{G}_n\\
&=\Sigma_{j\in[\ell']}\left(\Sigma_{k\in[\lceil\log_2 q\rceil]}\left(\Sigma_{i\in[\ell]}\mathbf{A}_{F',y_{i,j,k}}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\right)\mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n)\right)(\mathbf{u}_{\ell',j}^T\otimes\mathbf{I}_m)\\
&-\mathbf{t}^T\mathbf{Y}\otimes\mathbf{G}_n
\end{aligned}
$$

The above analysis indicates that the output of the $\mathsf{VMEvalC}$ and $\mathsf{VMEvalCX}$ algorithms are defined as follows, respectively:

$$
\begin{aligned}
\mathbf{A}_F&:=\Sigma_{j\in[\ell']}\left(\Sigma_{k\in[\lceil\log_2 q\rceil]}\left(\Sigma_{i\in[\ell]}\mathbf{A}_{F',y_{i,j,k}}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\right)\mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n)\right)(\mathbf{u}_{\ell',j}^T\otimes\mathbf{I}_m)\\
\mathbf{H}_{F,x}&:=\Sigma_{j\in[\ell']}\left(\Sigma_{k\in[\lceil\log_2 q\rceil]}\Sigma_{i\in[\ell]}\begin{pmatrix}\mathbf{H}_{F',x,y_{i,j,k}} & \mathbf{0}_m\\ \mathbf{0}_m & \mathbf{H}_{t,i}\end{pmatrix}\begin{pmatrix}\mathbf{G}_n^{-1}(\mathbf{A}_t\mathbf{H}_{t,i})\\ y_{i,j,k}\mathbf{I}_m\end{pmatrix}\mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n)\right)(\mathbf{u}_{\ell',j}^T\otimes\mathbf{I}_m)
\end{aligned}
$$

They satisfy the claimed equation $((\mathbf{A}_{\mathsf{att}}-(1,\mathbf{x}^T)\otimes\mathbf{G}_n),(\mathbf{A}_t-\mathbf{t}^T\otimes\mathbf{G}_n))\mathbf{H}_{F,x}=\mathbf{A}_F-\mathbf{t}^T\mathbf{Y}\otimes\mathbf{G}_n$.

We finally confirm the upper bounds of $\left\|\mathbf{H}_{F,x}^{T}\right\|_{\infty}$. Since $\left\|\mathbf{H}_{F',x}\right\|_{\infty} \leq m^{\mathcal{O}(\mathsf{dep})}$, these norms are bounded as follows:

$$
\begin{aligned}
&\left\|\mathbf{H}_{F,x}\right\|_{\infty} \\
&\leq \left\| \Sigma_{j\in[\ell']} \left(\Sigma_{k\in[\lceil \log_2 q\rceil]} \Sigma_{i\in[\ell]} \begin{pmatrix} \mathbf{H}_{F',x,y_{i,j,k}} & \mathbf{0}_m \\ \mathbf{0}_m & \mathbf{H}_{t,i} \end{pmatrix} \begin{pmatrix} \mathbf{G}_n^{-1}(\mathbf{A}_t \mathbf{H}_{t,i}) \\ y_{i,j,k}\mathbf{I}_m \end{pmatrix} \mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n) \right) (\mathbf{u}_{\ell',j}^T \otimes \mathbf{I}_m) \right\|_{\infty} \\
&\leq \ell' \left\| \Sigma_{k\in[\lceil \log_2 q\rceil]} \Sigma_{i\in[\ell]} \begin{pmatrix} \mathbf{H}_{F',x,y_{i,j,k}} & \mathbf{0}_m \\ \mathbf{0}_m & \mathbf{H}_{t,i} \end{pmatrix} \begin{pmatrix} \mathbf{G}_n^{-1}(\mathbf{A}_t \mathbf{H}_{t,i}) \\ y_{i,j,k}\mathbf{I}_m \end{pmatrix} \mathbf{G}_n^{-1}(2^{k-1}\mathbf{G}_n) \right\|_{\infty} \\
&\leq \ell' \lceil \log_2 q\rceil^2 \Sigma_{i\in[\ell]} \left\| \begin{pmatrix} \mathbf{H}_{F',x,y_{i,j,k}} & \mathbf{0}_m \\ \mathbf{0}_m & \mathbf{H}_{t,i} \end{pmatrix} \right\|_{\infty} \left\| \begin{pmatrix} \mathbf{G}_n^{-1}(\mathbf{A}_t \mathbf{H}_{t,i}) \\ y_{i,j,k}\mathbf{I}_m \end{pmatrix} \right\|_{\infty} \\
&\leq 2\ell\ell' \lceil \log_2 q\rceil^2 m^{\mathcal{O}(\mathsf{dep})+1}
\end{aligned}
$$

This completes the proof. $\square$

3.6 Pseudorandom Oracle Model

We recall the definition of pseudorandom oracle model (PROM) in [38].

Definition 2 (Pseudorandom Oracle Model [38]). Let PRF be a pseudorandom function with a key domain $\{0,1\}^\lambda$. The pseudorandom oracle model for PRF is the model in which all algorithms, including adversaries, can access to the oracle O. This oracle internally uses a random permutation $\mathsf{hMap} : \{0,1\}^\lambda \to \{0,1\}^\lambda$ and responds to the following two types of queries:

- $\mathsf{O}(\mathsf{hGen}, K) = \mathsf{hMap}(K)$,
- $\mathsf{O}(\mathsf{hEval}, h, \mathbf{x}) = \mathsf{PRF}(\mathsf{hMap}^{-1}(h), \mathbf{x})$.

3.7 Indistinguishability Obfuscation and Pseudorandom Obfuscation

We recall the definition of iO in [9].

Definition 3 (Indistinguishability Obfuscation [9]). For input and output sizes $L, \ell \in \mathbb{N}$, an indistinguishability obfuscator (iO) scheme is defined by two PPT algorithms $(\mathsf{Obf}, \mathsf{Eval})$ with the following syntax.

- $\mathsf{Obf}(1^\lambda, C) \to \widetilde{C}$: it takes as input a security parameter 1^λ and a circuit $C : \{0,1\}^L \to \{0,1\}^\ell$ and outputs an obfuscated circuit $\widetilde{C}$.
- $\mathsf{Eval}(\widetilde{C}, \mathbf{x}) \to \mathbf{y}$: it takes as input an obfuscated circuit $\widetilde{C}$ and an input $\mathbf{x} \in \{0,1\}^L$ and outputs the evaluation result $\mathbf{y} \in \{0,1\}^\ell$.

These algorithms satisfy the following properties.

- **Correctness.** For all $\lambda \in \mathbb{N}$, circuits $C : \{0,1\}^L \to \{0,1\}^\ell$, and inputs $\mathbf{x} \in \{0,1\}^L$, it holds that

$$\Pr\left[\mathsf{Eval}(\mathsf{Obf}(1^\lambda, C), \mathbf{x}) = C(\mathbf{x})\right] \geq 1 - \mathsf{negl}(\lambda),$$

where the probability is taken over the random coins of the Obf algorithm.

– **Polynomial Slowdown.** For all $\lambda \in \mathbb{N}$, circuits $C : \{0,1\}^L \to \{0,1\}^\ell$, it holds that

$$|\mathsf{Obf}(1^\lambda, C)| \le \mathsf{poly}(|C|).$$

– **Indistinguishability for Equivalent Functionalities.** For all $\lambda \in \mathbb{N}$, pairs of two circuits $C_0, C_1 : \{0,1\}^L \to \{0,1\}^\ell$ that computes the same function and are of the same size, it holds that

$$\{C_0, C_1, \mathsf{Obf}(1^\lambda, C_0)\} \overset{c}{\approx} \{C_0, C_1, \mathsf{Obf}(1^\lambda, C_1)\}$$

A weaker variant of iO that supports only pseudorandom functionalities—called pseudorandom obfuscation (PRO)—is introduced in [19, 3]. We adopt its security definition from Definition 5.3 in [3] with necessarily modifications as follows.

Definition 4. For the security parameter $\lambda \in \mathbb{N}$, and the input and output sizes $L, \ell \in \mathbb{N}$, let $\mathsf{PROSamp}$ be a PPT algorithm that on input 1^λ, outputs

$$(C_0, C_1, \mathsf{aux}_C \in \{0,1\}^\star),$$

where $C_0 : \{0,1\}^L \to \{0,1\}^\ell$ and $C_1 : \{0,1\}^L \to \{0,1\}^\ell$ have the same circuit size.

For the input and output sizes $L, \ell \in \mathbb{N}$, a pseudorandomness obfuscation (PRO) is defined by two PPT algorithms $(\mathsf{Obf}, \mathsf{Eval})$ with the same syntax as iO in Definition 3. In addition to the correctness and polynomial slowdown properties of iO, the PRO scheme satisfies the indistinguishability for pseudorandom functionalities with respect to the sampler class $\mathcal{SC}$: for all samplers $\mathsf{PROSamp} \in \mathcal{SC}$ the following holds.

$$\begin{aligned}
&\left\{1^\kappa, \{C_0(\mathbf{x})\}_{\mathbf{x}\in\{0,1\}^L}, \mathsf{aux}_C\right\} \overset{c}{\approx} \left\{1^\kappa, \{\Delta_{\mathbf{x}} \leftarrow\!\$\ \{0,1\}^\ell\}_{\mathbf{x}\in\{0,1\}^L}, \mathsf{aux}_C\right\} \\
&\qquad\qquad\qquad\qquad\qquad\qquad \overset{c}{\approx} \left\{1^\kappa, \{C_1(\mathbf{x})\}_{\mathbf{x}\in\{0,1\}^L}, \mathsf{aux}_C\right\} \\
&\Rightarrow \left\{\mathsf{Obf}(1^\lambda, C_0), \mathsf{aux}_C\right\} \overset{c}{\approx} \left\{\mathsf{Obf}(1^\lambda, C_1), \mathsf{aux}_C\right\},
\end{aligned}$$

where $\kappa \ge 2^n$ and $(C_0, C_1, \mathsf{aux}_K) \leftarrow\!\$\ \mathsf{PROSamp}(1^\lambda)$.

4 Diamond iO: Main Constructions

We first present a PRO scheme in Subsection 4.1 and then upgrade it to an iO scheme in Subsection 4.2, incurring only minor additional complexity.

4.1 Construction of PRO

We straightforwardly construct a PRO scheme for a circuit $C : \{0,1\}^L \to \{0,1\}^\ell$ based on the ideas shown in Section 2.

Parameters. The construction depends on the following parameters:

- A security parameter: $\lambda \in \mathbb{N}$.
- Input and output sizes of the circuit: $L, \ell \in \mathbb{N}$.
- A circuit depth $\mathsf{dep} \in \mathbb{N}$.
- A size of the bit representation of the circuit C: $\mathsf{size}_C \in \mathbb{N}$.
- A modulus $q \in \mathbb{N}$.
- An integer $M \in \mathbb{N}$ such that M divides q.
- Dimensions of binary LWE: $n \in \mathbb{N}$ and $m := (n+1)\lceil \log_2 q \rceil$.
- Gaussian parameters σ_E and σ_S.
- A dimension of a FHE secret key: $n_t \in \mathbb{N}$.
- A total bit size of the FHE encryptions of $\mathsf{size}_C + \lambda$ bits: $s' \in \mathbb{N}$.
- A total size of the bits encoded into BGG+ encodings: $L' = 1 + s' + L + n_t$.
- Parameters for trapdoors and preimages:

$$\begin{aligned} &n_B = L'(n+1), m_B = \mathcal{O}(n_B \lceil \log_2 q \rceil), \\ &\sigma_B = \omega(\sqrt{n_B \lceil \log_2 q \rceil \log m_B}). \end{aligned}$$

- A bound $B \in \mathbb{N}$, which is exponentially smaller than $\frac{q}{4}$. Specifically, it is defined as follows:

$$B := B_F + m_B B_p \sigma_B \sqrt{\lambda}$$

  where $B_p, B_F \in \mathbb{N}$ are defined in Equations 3 and 7, respectively.

Construction. With the black-box use of the FHE scheme satisfying the properties in Lemma 5 and a pseudorandom function $\mathsf{PRF}_B : \{0,1\}^\lambda \times \{0,1\}^\lambda \to [-q/4 + B, q/4 - B]^{1 \times \ell}$ with the depth $\mathsf{dep}_{\mathsf{PRF}_B} = \mathsf{poly}(\lambda)$, the constructions of the $\mathsf{PRO.Obf}$ and $\mathsf{PRO.Eval}$ algorithms are, respectively, defined as follows.

- $\mathsf{PRO.Obf}(1^\lambda, C) \to \widetilde{C}$:
  1. Sample $\bar{\mathbf{s}}_\epsilon \leftarrow\!\!\$\ \{0,1\}^n$ and set $\mathbf{s}_\epsilon^T := (\bar{\mathbf{s}}_\epsilon^T, -1)$.
  2. For every $i \in [L]$ and $b \in \{0,1\}$, sample $\bar{\mathbf{S}}_{i,b} \leftarrow\!\!\$\ \{0,1\}^{n \times n}$ and set $\mathbf{S}_{i,b} := \begin{pmatrix} \bar{\mathbf{S}}_{i,b} & \mathbf{0}_{n \times 1} \\ \mathbf{0}_{1 \times n} & 1 \end{pmatrix}$.
  3. Execute $(\mathbf{t}, \mathsf{pk}) \leftarrow \mathsf{FHE.KeyGen}(1^\lambda)$, where $\mathbf{t} \in \{0,1\}^{n_t}$.
  4. Sample $K_B \leftarrow\!\!\$\ \{0,1\}^\lambda$.
  5. Execute $\mathsf{ct} \leftarrow \mathsf{FHE.Enc}(\mathsf{pk}, (\mathsf{bits}(C), K_B))$.
  6. Compute $\xi^T := \mathsf{bits}(\mathsf{ct}) \in \{0,1\}^{s'}$.
  7. For every $i \in [0, L]$, execute $(\mathbf{B}_i, \mathbf{B}_{i,\gamma_B}^{-1}) \leftarrow \mathsf{TrapGen}(1^{n_B}, 1^{m_B}, q)$.
  8. Compute $\mathbf{p}_\epsilon^T := \left((1, \xi^T, \mathbf{0}_{1 \times L}^T, \mathbf{t}^T) \otimes \mathbf{s}_\epsilon^T\right) \mathbf{B}_0 + \mathbf{e}_{p_\epsilon}^T$, where $\mathbf{e}_{p_\epsilon} \leftarrow\!\!\$\ \mathcal{D}_{\mathbb{Z},\sigma_E}^{m_B}$.
  9. For every $i \in [L]$ and $b \in \{0,1\}$, compute $\mathbf{U}_{i,b} := (u_{i',j'}^{i,b})_{i',j'=1}^{L'}$, $u_{i',j'}^{i,b} := \begin{cases} b & \text{if } i' = 1 \wedge j' = 1 + s' + i \\ 1 & \text{if } i' = j' \\ 0 & \text{otherwise} \end{cases}$.
  10. For every $i \in [L]$ and $b \in \{0,1\}$, sample $\mathbf{K}_{i,b} \leftarrow\!\!\$\ \mathbf{B}_{i-1,\gamma_B}^{-1}((\mathbf{U}_{i,b} \otimes \mathbf{S}_{i,b})\mathbf{B}_i + \mathbf{E}_{K_{i,b}})$, where $\mathbf{E}_{K_{i,b}} \leftarrow\!\!\$\ \mathcal{D}_{\mathbb{Z},\sigma_S}^{n_B \times m_B}$.

11. Sample $\mathbf{A}_{\mathsf{att}} \leftarrow\!\!\$\ \mathbb{Z}_q^{(n+1)\times L'm}$,
12. Compute $\mathbf{P}_{\mathsf{att}} := \mathbf{u}_{1,L'} \otimes \mathbf{A}_{\mathsf{att}} - \mathbf{I}_{L'} \otimes \mathbf{G}_{n+1}$.
13. Sample $\mathbf{K}_{\mathsf{att}} \leftarrow\!\!\$\ \mathbf{B}_{L,\gamma_B}^{-1}(\mathbf{P}_{\mathsf{att}})$.
14. Let $f[\mathbf{x}_L]$ be a circuit that on input C and a PRF key K_B outputs $\frac{q}{2}C(\mathbf{x}_L) + \mathsf{PRF}_B(K_B, \mathbf{x}_L)$.
15. Let $f_{\mathsf{high}}[\mathbf{x}_L]$ and $f_{\mathsf{low}}[\mathbf{x}_L]$ are circuits that satisfy $f[\mathbf{x}_L](C, K_B) = M \cdot f_{\mathsf{high}}[\mathbf{x}_L](C, K_B) + f_{\mathsf{low}}[\mathbf{x}_L](C, K_B)$, where $f_{\mathsf{high}}[\mathbf{x}_L](C, K_B) \in [0, q/M]^{1\times\ell}$ and $f_{\mathsf{low}}[\mathbf{x}_L](C, K_B) \in [0, M-1]^{1\times\ell}$. In other words, $f_{\mathsf{high}}[\mathbf{x}_L](C, K_B)$ and $f_{\mathsf{low}}[\mathbf{x}_L](C, K_B)$ represent the quotient and remainder of the division of $f[\mathbf{x}_L](C, K_B)$ by M, respectively.
16. Let $F_{\mathsf{high}}[\mathsf{pk}]$ be a circuit that on input $(1, \xi^T, \mathbf{x}_L^T)$ outputs
$$\mathbf{Y} \leftarrow \mathsf{FHE.Eval}(\mathsf{pk}, M \cdot f_{\mathsf{high}}[\mathbf{x}_L], \mathsf{ct}) \in \mathbb{Z}_q^{n_t\times\ell},$$
where the ciphertext ct is constructed by composing the bits ξ.
17. Let $F_{\mathsf{low}}[\mathsf{pk}]$ be a circuit that on input $(1, \xi^T, \mathbf{x}_L^T)$ outputs
$$\mathbf{Y} \leftarrow \mathsf{FHE.Eval}(\mathsf{pk}, M \cdot f_{\mathsf{low}}[\mathbf{x}_L], \mathsf{ct}) \in \mathbb{Z}_q^{n_t\times\ell},$$
where the ciphertext ct is constructed by composing the bits ξ.
18. Parse $\mathbf{A}_{\mathsf{att}}$ as $(\bar{\mathbf{A}}_{\mathsf{att}}, \bar{\mathbf{A}}_t) \in \mathbb{Z}_q^{(n+1)\times(1+s'+L)m} \times \mathbb{Z}_q^{(n+1)\times n_t m}$.
19. Compute $\mathbf{A}_{F_{\mathsf{high}}} \leftarrow \mathsf{VMEvalC}((\bar{\mathbf{A}}_{\mathsf{att}}, \bar{\mathbf{A}}_t), F_{\mathsf{high}}[\mathsf{pk}])$.
20. Compute $\mathbf{A}_{F_{\mathsf{low}}} \leftarrow \mathsf{VMEvalC}((\bar{\mathbf{A}}_{\mathsf{att}}, \bar{\mathbf{A}}_t), F_{\mathsf{low}}[\mathsf{pk}])$.
21. Compute $\mathbf{J} := (\mathbf{I}_\ell \otimes \mathbf{G}_{n+1}^{-1}(\mathbf{u}_{n+1,n+1}))$.
22. Compute $\mathbf{A}_F := M \cdot \left\lfloor \frac{\mathbf{A}_{F_{\mathsf{high}}}\mathbf{J}}{M} \right\rfloor + \left\lfloor \frac{\mathbf{A}_{F_{\mathsf{low}}}\mathbf{J}}{M} \right\rfloor$.
23. Compute $\mathbf{P}_F := \mathbf{u}_{1,L'} \otimes \mathbf{A}_F$.
24. Sample $\mathbf{K}_F \leftarrow\!\!\$\ \mathbf{B}_{L,\gamma_B}^{-1}(\mathbf{P}_F)$.
25. Output
$$\widetilde{C} := \left(\mathsf{pk}, \xi, \mathbf{p}_\epsilon, \{\mathbf{K}_{i,b}\}_{i\in[L],b\in\{0,1\}}, \mathbf{A}_{\mathsf{att}}, \mathbf{K}_{\mathsf{att}}, \mathbf{K}_F\right).$$

– $\mathsf{PRO.Eval}(\widetilde{C}, \mathbf{x}_L) \to y$:
1. Parse $\widetilde{C}$ as
$$\widetilde{C} := \left(\mathsf{pk}, \xi, \mathbf{p}_\epsilon, \{\mathbf{K}_{i,b}\}_{i\in[L],b\in\{0,1\}}, \mathbf{A}_{\mathsf{att}}, \mathbf{K}_{\mathsf{att}}, \mathbf{K}_F\right).$$
2. For every $i \in [L]$, compute $\mathbf{p}_{\mathbf{x}_i}^T := \mathbf{p}_{\mathbf{x}_{i-1}}^T \mathbf{K}_{i,x_i}$.
3. Compute $\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T := \mathbf{p}_{\mathbf{x}_L}^T \mathbf{K}_{\mathsf{att}}$.
4. Define $F_{\mathsf{high}}[\mathsf{pk}]$ and $F_{\mathsf{low}}[\mathsf{pk}]$ in the same manner as Step 16 and 17 in the $\mathsf{PRO.Obf}$ algorithm.
5. Parse $\mathbf{A}_{\mathsf{att}}$ as $(\bar{\mathbf{A}}_{\mathsf{att}}, \bar{\mathbf{A}}_t) \in \mathbb{Z}_q^{(n+1)\times(1+s'+L)m} \times \mathbb{Z}_q^{(n+1)\times n_t m}$.
6. Compute $\mathbf{H}_{F_{\mathsf{high}},x} \leftarrow \mathsf{VMEvalCX}((\bar{\mathbf{A}}_{\mathsf{att}}, \bar{\mathbf{A}}_t), F_{\mathsf{high}}[\mathsf{pk}], (1, \xi, \mathbf{x}_L))$.
7. Compute $\mathbf{H}_{F_{\mathsf{low}},x} \leftarrow \mathsf{VMEvalCX}((\bar{\mathbf{A}}_{\mathsf{att}}, \bar{\mathbf{A}}_t), F_{\mathsf{low}}[\mathsf{pk}], (1, \xi, \mathbf{x}_L))$.

8. Compute $\mathbf{c}^T_{F_{\mathsf{high}},\mathbf{x}_L} := \mathbf{c}^T_{\mathsf{att},\mathbf{x}_L}\mathbf{H}_{F_{\mathsf{high}},x}\mathbf{J}$, where $\mathbf{J}$ is defined in the same manner as the $\mathsf{PRO.Obf}$ algorithm.
9. Compute $\mathbf{c}^T_{F_{\mathsf{low}},\mathbf{x}_L} := \mathbf{c}^T_{\mathsf{att},\mathbf{x}_L}\mathbf{H}_{F_{\mathsf{low}},x}\mathbf{J}$.
10. Compute $\mathbf{c}^T_{F,\mathbf{x}_L} := M \cdot \left\lfloor \frac{\mathbf{c}^T_{F_{\mathsf{high}},\mathbf{x}_L}}{M} \right\rfloor + \left\lfloor \frac{\mathbf{c}^T_{F_{\mathsf{low}},\mathbf{x}_L}}{M} \right\rfloor$.
11. Compute $\mathbf{v}^T_{\mathbf{x}_L} := \mathbf{p}^T_{\mathbf{x}_L}\mathbf{K}_F$.
12. Compute $\mathbf{z}^T_{\mathbf{x}_L} := \mathbf{c}^T_{F,\mathbf{x}_L} - \mathbf{v}^T_{\mathbf{x}_L}$.
13. For every $i \in [\ell]$, let $y_i = 0$ if $z_i \in [-\frac{q}{4}, \frac{q}{4})$ and $y_i = 1$ otherwise.
14. Output $\mathbf{y}$.

The above construction trivially satisfies the polynomial slowdown property of the PRO scheme (Definition 4), as the size of the obfuscation $\widetilde{C}$ is bounded by $\mathsf{poly}(|C|)$.

Correctness. We confirm that the construction is correct as defined in Definition 4.

Lemma 7 (Correctness of the PRO construction). The PRO construction in Subsection 4.1 satisfies correctness (Definition 4).

Proof. We recall that for a vector $\mathbf{e} \leftarrow\!\!\$\ \mathcal{D}^m_{\mathbb{Z},\sigma_E}$, it holds that $\|\mathbf{e}\|_\infty \leq \sigma_E\sqrt{\lambda}$ except with negligible probability as shown in Lemma 2. Assuming $\|\mathbf{e}\|_\infty \leq \sigma_E\sqrt{\lambda}$, we prove that $\mathsf{PRO.Eval}(\widetilde{C}, \mathbf{x}_L) = C(\mathbf{x}_L)$ holds for any $\mathbf{x}_L \in \{0,1\}^L$.

In the following, we define $\mathbf{s}^T_{\mathbf{x}_i} := \mathbf{s}^T_\epsilon \prod_{i\in[i]} \mathbf{S}_{i,x_i}$ for every $i \in [L]$. This satisfies $\mathbf{s}^T_{\mathbf{x}_i} = \mathbf{s}^T_{\mathbf{x}_{i-1}}\mathbf{S}_{i,x_i}$.

For every $i \in [L]$, it follows from the definition of the matrix $\mathbf{U}_{i,x_i}$ that $(1, \xi^T, \mathbf{x}^T_{i-1}, \mathbf{0}^T_{L-i+1}, \mathbf{t}^T)\mathbf{U}_{i,x_i} = (1, \xi^T, \mathbf{x}^T_i, \mathbf{0}^T_{L-i}, \mathbf{t}^T)$. Therefore, it holds that

$$\begin{aligned}
\mathbf{p}^T_{\mathbf{x}_{i-1}} &= \mathbf{p}^T_{\mathbf{x}_{i-1}}\mathbf{K}_{i,x_i} \\
&= \left(\left((1, \xi^T, \mathbf{x}^T_{i-1}, \mathbf{0}^T_{L-i+1}, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}^T_{i-1}}\right)\mathbf{B}_{i-1} + \mathbf{e}^T_{p_{\mathbf{x}_{i-1}}}\right)\mathbf{K}_{i,x_i} \\
&= \left((1, \xi^T, \mathbf{x}^T_{i-1}, \mathbf{0}^T_{L-i+1}, \mathbf{t}^T) \otimes \mathbf{s}^T_{\mathbf{x}_{i-1}}\right)\left((\mathbf{U}_{i,x_i} \otimes \mathbf{S}_{i,x_i})\mathbf{B}_i + \mathbf{E}_{K_{i,x_i}}\right) \\
&\quad + \mathbf{e}^T_{p_{\mathbf{x}_{i-1}}}\mathbf{K}_{i,x_i} \\
&= \left((1, \xi^T, \mathbf{x}^T_i, \mathbf{0}^T_{L-i}, \mathbf{t}^T) \otimes \mathbf{s}^T_{\mathbf{x}_i}\right)\mathbf{B}_i + \mathbf{e}^T_{p_{\mathbf{x}_i}},
\end{aligned}$$

where $\mathbf{e}^T_{p_{\mathbf{x}_i}} := \left((1, \xi^T, \mathbf{x}^T_{i-1}, \mathbf{0}^T_{L-i+1}, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_{i-1}}\right)\mathbf{E}_{K_{i,x_i}} + \mathbf{e}^T_{p_{\mathbf{x}_{i-1}}}\mathbf{K}_{i,x_i}$. This implies that $\mathbf{p}^T_{\mathbf{x}_L} := \left((1, \xi^T, \mathbf{x}^T_L, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_L}\right)\mathbf{B}_L + \mathbf{e}^T_{p_{\mathbf{x}_L}}$.

We can analyze the upper bound of $\left\|\mathbf{e}^T_{p_{\mathbf{x}_i}}\right\|_\infty$ as follows:

$$\begin{aligned}
\left\|\mathbf{e}^T_{p_{\mathbf{x}_i}}\right\|_\infty &\leq \left\|\left((1, \xi^T, \mathbf{x}^T_{i-1}, \mathbf{0}^T_{L-i+1}, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_{i-1}}\right)\mathbf{E}_{K_{i,x_i}}\right\|_\infty + \left\|\mathbf{e}^T_{p_{\mathbf{x}_{i-1}}}\mathbf{K}_{i,x_i}\right\|_\infty \\
&\leq n_B(n+1)^{i-1}\sigma_S\sqrt{\lambda} + m_B\left\|\mathbf{e}^T_{p_{\mathbf{x}_{i-1}}}\right\|_\infty \sigma_B\sqrt{\lambda},
\end{aligned}$$

where $\left\|\mathbf{e}_{p_\epsilon}^T\right\|_\infty \leq \sigma_E\sqrt{\lambda}$. By treating the above relation as recurrence relations, we obtain $\left\|\mathbf{e}_{p_{\mathbf{x}_L}}^T\right\|_\infty \leq B_p$, where B_p is defined as follows:

$$B_p := n_B\sigma_S\sqrt{\lambda}\frac{(n+1)^L - (m_B\sigma_B\sqrt{\lambda})^L}{n+1-m_B\sigma_B\sqrt{\lambda}} + (m_B\sigma_B\sqrt{\lambda})^L\sigma_E\sqrt{\lambda}. \tag{3}$$

By the definition of $\mathbf{K}_{\mathsf{att}}$, we obtain

$$\begin{aligned}
\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T &:= \mathbf{p}_{\mathbf{x}_L}^T\mathbf{K}_{\mathsf{att}} \\
&= \Big((1,\xi^T,\mathbf{x}_L^T,\mathbf{t}^T)\otimes\mathbf{s}_{\mathbf{x}_L}\Big)\Big(\mathbf{u}_{1,L'}\otimes\mathbf{A}_{\mathsf{att}} - \mathbf{I}_{L'}\otimes\mathbf{G}_{n+1}\Big) + \mathbf{e}_{p_{\mathbf{x}_L}}^T\mathbf{K}_{\mathsf{att}} \\
&= \mathbf{s}_{\mathbf{x}_L}^T\left(\mathbf{A}_{\mathsf{att}} - (1,\xi^T,\mathbf{x}_L^T,\mathbf{t}^T)\otimes\mathbf{G}_{n+1}\right) + \mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}}^T,
\end{aligned}$$

where $\mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}}^T := \mathbf{e}_{p_{\mathbf{x}_L}}^T\mathbf{K}_{\mathsf{att}}$. It holds that $\left\|\mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}}^T\right\|_\infty \leq B_c := m_B B_p\sigma_B\sqrt{\lambda}$.

From Lemma 6 and the correctness of the FHE scheme in Lemma 5, it follows that

$$\begin{aligned}
\mathbf{c}_{F_{\mathsf{high}},\mathbf{x}_L}^T &:= \mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T\mathbf{H}_{F_{\mathsf{high}},x}\mathbf{J} \\
&= \mathbf{s}_{\mathbf{x}_L}^T\Big(\mathbf{A}_{F_{\mathsf{high}}} - (M\cdot f_{\mathsf{high}}[\mathbf{x}_L](C,K_B) + \mathbf{e}_{\mathsf{fhe},\mathsf{high},\mathbf{x}_L}^T)\otimes\mathbf{G}_{n+1}\Big)\mathbf{J} \\
&\quad + \mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}}^T\mathbf{H}_{F_{\mathsf{high}},x}\mathbf{J} \\
&= \mathbf{s}_{\mathbf{x}_L}^T\mathbf{A}_{F_{\mathsf{high}}}\mathbf{J} + M\cdot f_{\mathsf{high}}[\mathbf{x}_L](C,K_B) + \mathbf{e}_{c_{F_{\mathsf{high}},\mathbf{x}_L}}^T,
\end{aligned}$$

where $\mathbf{e}_{c_{F_{\mathsf{high}},\mathbf{x}_L}}^T := \mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}}^T\mathbf{H}_{F_{\mathsf{high}},x}\mathbf{J} + \mathbf{e}_{\mathsf{fhe},\mathsf{high},\mathbf{x}_L}^T$. We note that $\mathbf{s}_{\mathbf{x}_L}^T\mathbf{u}_{n+1,n+1} = -1$ holds since the last row of $\mathbf{s}_{\mathbf{x}_L}^T$ is (-1). Similarly, we have

$$\mathbf{c}_{F_{\mathsf{low}},\mathbf{x}_L}^T = \mathbf{s}_{\mathbf{x}_L}^T\mathbf{A}_{F_{\mathsf{low}}}\mathbf{J} + M\cdot f_{\mathsf{low}}[\mathbf{x}_L](C,K_B) + \mathbf{e}_{c_{F_{\mathsf{low}},\mathbf{x}_L}}^T,$$

where $\mathbf{e}_{c_{F_{\mathsf{low}},\mathbf{x}_L}}^T := \mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}}^T\mathbf{H}_{F_{\mathsf{low}},x}\mathbf{J} + \mathbf{e}_{\mathsf{fhe},\mathsf{low},\mathbf{x}_L}^T$.

By the correctness of the FHE scheme in Lemma 5, it holds that

$$\left\|\mathbf{e}_{\mathsf{fhe},\mathsf{high},\mathbf{x}_L}^T\right\|_\infty, \left\|\mathbf{e}_{\mathsf{fhe},\mathsf{low},\mathbf{x}_L}^T\right\|_\infty \leq B_{\mathsf{fhe}} := \mathsf{poly}(\lambda)m^{\mathcal{O}(\mathsf{poly}(\mathsf{dep},\log q))}. \tag{4}$$

Additionally, the shallowness of the FHE scheme in Lemma 5 ensures that the depth of the circuit $F[\mathsf{pk},C]$ is bounded by $\mathsf{poly}(\mathsf{dep},\log n,\log m,\log q)$; therefore, Lemma 6 implies that $\left\|\mathbf{H}_{F_{\mathsf{high}},x}\right\|_\infty, \left\|\mathbf{H}_{F_{\mathsf{low}},x}\right\|_\infty \leq m^{\mathcal{O}(\mathsf{poly}(\mathsf{dep},\log n,\log m,\log q))}$. Hence, $\left\|\mathbf{e}_{c_{F_{\mathsf{high}},\mathbf{x}_L}}^T\right\|_\infty$ is bounded as follows:

$$\begin{aligned}
\left\|\mathbf{e}_{c_{F_{\mathsf{high}},\mathbf{x}_L}}^T\right\|_\infty &\leq L'mB_c\left\|\mathbf{H}_{F_{\mathsf{high}},x}\mathbf{J}\right\|_\infty + B_{\mathsf{fhe}} \\
&\leq L'B_c\ell m^{\mathcal{O}(\mathsf{poly}(\mathsf{dep},\log n,\log m,\log q))+2}.
\end{aligned}$$

Similarly,

$$\left\|\mathbf{e}^T_{c_{F_{\mathsf{low}},\mathbf{x}_L}}\right\|_\infty \leq L' B_c \ell m^{\mathcal{O}(\mathsf{poly}(\mathsf{dep},\log n,\log m,\log q))+2}$$

holds.

Since $\left\|\mathbf{e}^T_{c_{F_{\mathsf{high}},\mathbf{x}_L}}\right\|_\infty$, we have

$$\left\lfloor \frac{\mathbf{c}^T_{F_{\mathsf{high}},\mathbf{x}_L}}{M} \right\rfloor = \mathbf{s}^T_{\mathbf{x}_L} \left\lfloor \frac{\mathbf{A}_{F_{\mathsf{high}}}\mathbf{J}}{M} \right\rfloor + f_{\mathsf{high}}[\mathbf{x}_L](C, K_B) + \epsilon^T_{\mathsf{high},\mathbf{x}_L} + \mathbf{e}^T_{\mathsf{high},\mathbf{s}_{\mathbf{x}_L}}, \tag{5}$$

where $\epsilon^T_{\mathsf{high},\mathbf{x}_L}$ is a rounding error that is 1 if $\left\|\mathbf{s}^T_{\mathbf{x}_L}\mathbf{A}_{F_{\mathsf{high}}}\mathbf{J} + \mathbf{e}^T_{c_{F_{\mathsf{high}},\mathbf{x}_L}}\right\|_\infty \geq M$ and 0 otherwise, and $\mathbf{e}^T_{\mathsf{high},\mathbf{s}_{\mathbf{x}_L}}$ is another rounding error such that $\left\|\mathbf{e}^T_{\mathsf{high},\mathbf{s}_{\mathbf{x}_L}}\right\|_\infty \leq (n+1)\left\|\mathbf{s}^T_{\mathbf{x}_L}\right\|_\infty < (n+1)^{L+1}$ holds. The same holds for $\mathbf{c}^T_{F_{\mathsf{low}},\mathbf{x}_L}$, namely,

$$\left\lfloor \frac{\mathbf{c}^T_{F_{\mathsf{low}},\mathbf{x}_L}}{M} \right\rfloor = \mathbf{s}^T_{\mathbf{x}_L} \left\lfloor \frac{\mathbf{A}_{F_{\mathsf{low}}}\mathbf{J}}{M} \right\rfloor + f_{\mathsf{low}}[\mathbf{x}_L](C, K_B) + \epsilon^T_{\mathsf{low},\mathbf{x}_L} + \mathbf{e}^T_{\mathsf{low},\mathbf{s}_{\mathbf{x}_L}}. \tag{6}$$

Equations 5 and 6 imply the following equation:

$$\begin{aligned}
\mathbf{c}^T_{F,\mathbf{x}_L} &:= M \cdot \left\lfloor \frac{\mathbf{c}^T_{F_{\mathsf{high}},\mathbf{x}_L}}{M} \right\rfloor + \left\lfloor \frac{\mathbf{c}^T_{F_{\mathsf{low}},\mathbf{x}_L}}{M} \right\rfloor \\
&= \mathbf{s}^T_{\mathbf{x}_L}\left(M \cdot \left\lfloor \frac{\mathbf{A}_{F_{\mathsf{high}}}\mathbf{J}}{M} \right\rfloor + \left\lfloor \frac{\mathbf{A}_{F_{\mathsf{low}}}\mathbf{J}}{M} \right\rfloor\right) + \left(M \cdot f_{\mathsf{high}}[\mathbf{x}_L](C, K_B) + f_{\mathsf{low}}[\mathbf{x}_L](C, K_B)\right) + \mathbf{e}^T_{c_{F,\mathbf{x}_L}} \\
&= \mathbf{s}^T_{\mathbf{x}_L}\mathbf{A}_F + f[\mathbf{x}_L](C, K_B) + \mathbf{e}^T_{c_{F,\mathbf{x}_L}},
\end{aligned}$$

where $\mathbf{e}^T_{c_{F,\mathbf{x}_L}} := M \cdot \left(\epsilon^T_{\mathsf{high},\mathbf{x}_L} + \mathbf{e}^T_{\mathsf{high},\mathbf{s}_{\mathbf{x}_L}}\right) + \epsilon^T_{\mathsf{low},\mathbf{x}_L} + \mathbf{e}^T_{\mathsf{low},\mathbf{s}_{\mathbf{x}_L}}$. This error norm is bounded as follows:

$$\begin{aligned}
\left\|\mathbf{e}^T_{c_{F,\mathbf{x}_L}}\right\|_\infty &\leq B_F \\
&:= M\left((n+1)^{L+1} + 1\right) + \left((n+1)^{L+1} + 1\right) \qquad (7) \\
&\leq (M+1)\left((n+1)^{L+1} + 1\right)
\end{aligned}$$

We also have

$$\begin{aligned}
\mathbf{v}^T_{\mathbf{x}_L} &= \mathbf{p}^T_{\mathbf{x}_L}\mathbf{K}_F \\
&= \mathbf{s}^T_{\mathbf{x}_L}\mathbf{A}_F + \mathbf{e}^T_{p_{\mathbf{x}_L}}\mathbf{K}_F.
\end{aligned}$$

Hence, we obtain

$$\begin{aligned}
\mathbf{z}^T_{\mathbf{x}_L} &= \mathbf{c}^T_{F,\mathbf{x}_L} - \mathbf{v}^T_{\mathbf{x}_L} \\
&= f[\mathbf{x}_L](C, K_B) + \mathbf{e}^T_{c_{F,\mathbf{x}_L}} - \mathbf{e}^T_{p_{\mathbf{x}_L}}\mathbf{K}_F \\
&= \frac{q}{2}C(\mathbf{x}_L) + \mathsf{PRF}_B(K_B, \mathbf{x}_L) + \mathbf{e}^T_{c_{F,\mathbf{x}_L}} - \mathbf{e}^T_{p_{\mathbf{x}_L}}\mathbf{K}_F.
\end{aligned}$$

The last equality follows from the fact that $f[\mathbf{x}_L](C, K_B) := \frac{q}{2}C(\mathbf{x}_L) + \mathsf{PRF}_B(K_B, \mathbf{x}_L)$.

As $y_i = 0$ if and only if $z_i \in [-\frac{q}{4}, \frac{q}{4})$, it suffices to show that

$$\left\| \mathsf{PRF}_B(K_B, \mathbf{x}_L) + \mathbf{e}^T_{c_{F,\mathbf{x}_L}} - \mathbf{e}^T_{p_{\mathbf{x}_L}} \mathbf{K}_F \right\|_\infty \leq \frac{q}{4}.$$

We can confirm that

$$\begin{aligned} \left\| \mathbf{e}^T_{c_{F,\mathbf{x}_L}} - \mathbf{e}^T_{p_{\mathbf{x}_L}} \mathbf{K}_F \right\|_\infty &\leq B_F + m_B B_p \sigma_B \sqrt{\lambda} \\ &\leq B \end{aligned}$$

The output of the PRF is bounded by $\frac{q}{4} - B$, which completes the proof of correctness.

$\square$

Security. Before proving the security of the above PRO construction, we introduce our new lattice assumption called all-product LWE assumption. In a nutshell, we considers the following instances:

$$\begin{aligned} &\{\mathbf{s}^T_{\mathbf{x}_L}(\mathbf{A} - \mathbf{x}^T_L \otimes \mathbf{G}_{n+1}) + \mathbf{e}^T_{A,\mathbf{x}_L}\}_{\mathbf{x}_L \in \{0,1\}^L}, \\ &\{\mathbf{s}^T_{\mathbf{x}_i}\mathbf{B}_i + \mathbf{e}^T_{B,\mathbf{x}_i}\}_{i \in [0,L], \mathbf{x}_i \in \{0,1\}^i}, \end{aligned}$$

where $\mathbf{s}^T_{\mathbf{x}_i} := \left(\bar{\mathbf{s}}^T \prod_{j \in [i]} \bar{\mathbf{S}}_{j,x_{i,j}}, -1\right)$. The assumption claims that even if the adversary can simultaneously see the above instances for every $i \in [0, L]$ and $\mathbf{x}_i \in \{0,1\}^i$ patterns, all of them remain pseudorandom, which is formally defined as follows:

Assumption 4 (All-Product LWE Assumption). Let $n, m_B, q, L \in \mathbb{N}$ be parameters, and $\lambda \in \mathbb{N}$ be a security parameter. Let σ_A, and $\{\sigma_{B,i}\}_{i \in [0,L]}$ be Gaussian parameters. We define the following two distributions:

$$\mathcal{D}_{\mathsf{APLWE},0} := \begin{pmatrix} \mathbf{A}, \{\mathbf{B}_i\}_{i \in [0,L]}, \\ \{\mathbf{c}^T_{A,\mathbf{x}_L} := \mathbf{s}^T_{\mathbf{x}_L}(\mathbf{A} - \mathbf{x}^T_L \otimes \mathbf{G}_{n+1}) + \mathbf{e}^T_{A,\mathbf{x}_L}\}_{\mathbf{x}_L \in \{0,1\}^L} \\ \{\mathbf{c}^T_{B,\mathbf{x}_i} := \mathbf{s}^T_{\mathbf{x}_i}\mathbf{B}_i + \mathbf{e}^T_{B,\mathbf{x}_i}\}_{i \in [0,L], \mathbf{x}_i \in \{0,1\}^i} \end{pmatrix}$$

$$\mathcal{D}_{\mathsf{APLWE},1} := \begin{pmatrix} \mathbf{A}, \{\mathbf{B}_i\}_{i \in [0,L]}, \\ \{\mathbf{c}_{A,\mathbf{x}_L} \leftarrow\!\$\; \mathbb{Z}_q^{1 \times L(n+1)\lceil \log_2 q \rceil}\}_{\mathbf{x}_L \in \{0,1\}^L} \\ \{\mathbf{c}_{B,\mathbf{x}_i} \leftarrow\!\$\; \mathbb{Z}_q^{1 \times m_B}\}_{i \in [0,L], \mathbf{x}_i \in \{0,1\}^i} \end{pmatrix},$$

where

- $\mathbf{A} \leftarrow\!\$\; \mathbb{Z}_q^{(n+1) \times L(n+1)\lceil \log_2 q \rceil}$,
- $\mathbf{B}_i \leftarrow\!\$\; \mathbb{Z}_q^{(n+1) \times m_B}$ for every $i \in [0, L]$,
- $\mathbf{e}_{A,\mathbf{x}_L} \leftarrow\!\$\; \mathcal{D}_{\mathbb{Z},\sigma_A}^{L(n+1)\lceil \log_2 q \rceil}$ for every $\mathbf{x}_L \in \{0,1\}^L$,
- $\mathbf{e}_{B,\mathbf{x}_i} \leftarrow\!\$\; \mathcal{D}_{\mathbb{Z},\sigma_{B,i}}^{m_B}$ for every $i \in [0, L]$ and $\mathbf{x}_i \in \{0,1\}^i$,

– $\bar{\mathbf{s}}_\epsilon \leftarrow\!\$\ \{0,1\}^n$,
– $\bar{\mathbf{S}}_{i,b} \leftarrow\!\$\ \{0,1\}^{n\times n}$ for every $i \in [L]$ and $b \in \{0,1\}$,
– $\mathbf{s}_{\mathbf{x}_i}^T := \left(\bar{\mathbf{s}}_{\mathbf{x}_i}^T \prod_{j\in[i]} \bar{\mathbf{S}}_{i,x_{i,j}}, -1\right)$ for every $i \in [0,L]$ and $\mathbf{x}_i \in \{0,1\}^i$.

For a function $\kappa = \kappa(\lambda)$, the all-product LWE assumption is said to hold, if for every non-uniform adversary $\mathcal{A}$ such that $\mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa)$, it holds that

$$\left|\Pr\left[\mathcal{A}\left(\mathcal{D}_{\mathsf{APLWE},0}\right) = 1\right] - \Pr\left[\mathcal{A}\left(\mathcal{D}_{\mathsf{APLWE},1}\right) = 1\right]\right| \leq \mathsf{negl}(\kappa).$$

We show that the above PRO construction satisfies the indistinguishability required in Definition 4.

Theorem 1 (Indistinguishability of the PRO construction). Let $\mathcal{SC}$ be a sampler class defined for the PRO scheme and, and set $\kappa = \lambda^{L^2 \log\lambda}$. Let Λ be a security parameter such that $2^{\Lambda^\delta} \geq \kappa^{\omega(1)}$ holds for some $\delta \in (0,1)$. Assume that the following hold:

1. the LWE assumption is subexponentially hard (Assumption 1);
2. the private-coin binding and hiding non-uniform κ-evasive LWE assumptions hold (Assumptions 2 and 3);
3. the all-product LWE assumption holds (Assumption 4);
4. the underlying FHE scheme satisfies the subexponential semantic security for the security parameter Λ (Definition 1); and
5. the PRF scheme denoted by $\mathsf{PRF}_B : \{0,1\}^\lambda \times \{0,1\}^\lambda \to [-q/4+B, q/4-B]^{1\times\ell}$ is subexponentially secure for the security parameter Λ.

Then the PRO construction in Subsection 4.1 achieves indistinguishability for pseudorandom functionalities with respect to $\mathcal{SC}$ (Definition 4).

Proof. We recall that $(C_0, C_1, \mathsf{aux}_K) \leftarrow\!\$\ \mathsf{PROSamp}(1^\lambda)$. Assuming that the following indistinguishabilities hold:

$$\left\{1^\kappa, \{C_0(\mathbf{x})\}_{\mathbf{x}\in\{0,1\}^L}, \mathsf{aux}_C\right\} \overset{c}{\approx} \left\{1^\kappa, \{\Delta_\mathbf{x} \leftarrow\!\$\ \{0,1\}^\ell\}_{\mathbf{x}\in\{0,1\}^L}, \mathsf{aux}_C\right\} \tag{8}$$

$$\overset{c}{\approx} \left\{1^\kappa, \{C_1(\mathbf{x})\}_{\mathbf{x}\in\{0,1\}^L}, \mathsf{aux}_C\right\}, \tag{9}$$

it suffices to show

$$\left\{\mathsf{Obf}(1^\lambda, C_0), \mathsf{aux}_C\right\} \overset{c}{\approx} \left\{\mathsf{Obf}(1^\lambda, C_1), \mathsf{aux}_C\right\}. \tag{10}$$

This is proven in two steps:

1. We first show that, for $C = C_0$ (resp., $C = C_1$) the indistinguishabilities of Relations 8 (resp., Relation 9) implies the pseudorandomness of the joint distribution in which all preimages in $\widetilde{C}$ are multiplied by the corresponding LWE instances. Formally, the corresponding advantage function is defined

as follows:

$$
\begin{aligned}
&\mathrm{Adv}_{\mathcal{A}}^{L+2}(\lambda)\\
&:= \Pr\left[\mathcal{A}\begin{pmatrix}\mathsf{pk}, \xi := \mathsf{bits}(\mathsf{ct}), \mathbf{A}_{\mathsf{att}}, \mathbf{B}_L, \mathsf{aux}_C,\\ \{\mathbf{p}_{\mathbf{x}_i}^T := \left((1, \xi^T, \mathbf{x}_i^T, \mathbf{0}_{L-i}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_i}^T\right)\mathbf{B}_i + \mathbf{e}_{p_{\mathbf{x}_i}}^T\}_{i\in[0,L],\mathbf{x}_i\in\{0,1\}^i}\\ \{\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T := \mathbf{s}_{\mathbf{x}_L}^T\left(\mathbf{A}_{\mathsf{att}} - (1, \xi^T, \mathbf{x}_L^T, \mathbf{t}^T) \otimes \mathbf{G}_{n+1}\right) + \mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}}^T\}_{\mathbf{x}_L\in\{0,1\}^L}\\ \{\mathbf{v}_{\mathbf{x}_L}^T := \mathbf{s}_{\mathbf{x}_L}^T\mathbf{A}_F + \mathbf{e}_{v_{\mathbf{x}_L}}^T\}_{\mathbf{x}_L\in\{0,1\}^L}\end{pmatrix} = 1\right]\\
&- \Pr\left[\mathcal{A}\begin{pmatrix}\mathsf{pk}, \xi := \mathsf{bits}(\mathsf{ct}'), \mathbf{A}_{\mathsf{att}}, \mathbf{B}_L, \mathsf{aux}_C,\\ \{\mathbf{p}_{\mathbf{x}_i}^T \leftarrow\!\$\ \mathbb{Z}_q^{1\times m_B}\}_{i\in[0,L],\mathbf{x}_i\in\{0,1\}^i}\\ \{\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T \leftarrow\!\$\ \mathbb{Z}_q^{1\times L'm}\}_{\mathbf{x}_L\in\{0,1\}^L}\\ \{\mathbf{v}_{\mathbf{x}_L}^T \leftarrow\!\$\ \mathbb{Z}_q^{1\times \ell}\}_{\mathbf{x}_L\in\{0,1\}^L}\end{pmatrix} = 1\right].
\end{aligned} \tag{11}
$$

where $\mathbf{s}_{\mathbf{x}_i} := \mathbf{s}_\epsilon^T \prod_{j\in[i]} \mathbf{S}_{j,x_j}$ for every $i \in [L]$ and $\mathbf{x}_i \in \{0,1\}^i$, $\mathsf{ct}' \leftarrow \mathsf{FHE.Enc}(\mathsf{pk}, \mathbf{0}_{\mathsf{size}_C})$, and $\mathbf{s}_\epsilon$, $\{\mathbf{S}_{i,b}\}_{i\in[L],b\in\{0,1\}}$, $\mathbf{t}$, K_B, pk, $\mathbf{A}_{\mathsf{att}}$, $\mathbf{A}_F$, $\{\mathbf{B}_i\}_{i\in[0,L]}$ are defined in the same manner as the construction in Subsection 4.1. Here, in addition to σ_E, σ_S, and σ_B, other Gaussian parameters $\{\chi_{p,i}\}_{i\in[0,L]}$, $\{\chi_{B,i}\}_{i\in[0,L]}$, χ_{att}, and χ_F, are defined so as to satisfy the following constraints:

- $\sigma_E = \chi_{p,0}$.
- For every $i \in [0, L]$, $\chi_{p,i} \geq \left(L'\chi_{B,i}\sqrt{\lambda}\right)\lambda^{\omega(1)}$.
- For every $i \in [L]$, $\chi_{p,i}\sqrt{\lambda} < m_B\chi_{p,i-1}\sigma_B\lambda + n_B(n+1)^{i-1}\sigma_S\sqrt{\lambda}$.
- $\chi_F \geq B_F\lambda^{\omega(1)}$.
- $\chi_{\mathsf{att}}\sqrt{\lambda} < m_B\chi_{p,L}\sigma_B\lambda$ and $\chi_F\sqrt{\lambda} < m_B\chi_{p,L}\sigma_B\lambda$.

In the above distributions, the errors are sampled as follows:

$$\{\mathbf{e}_{p_{\mathbf{x}_i}} \leftarrow\!\$\ \mathcal{D}_{\mathbb{Z},\chi_{p,i}}^{m_B}\}_{i\in[0,L],\mathbf{x}_i\in\{0,1\}^i}, \quad \{\mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}} \leftarrow\!\$\ \mathcal{D}_{\mathbb{Z},\chi_{\mathsf{att}}}^{L'm}\}_{\mathbf{x}_L\in\{0,1\}^L}, \{\mathbf{e}_{v_{\mathbf{x}_L}} \leftarrow\!\$\ \mathcal{D}_{\mathbb{Z},\chi_F}^{\ell}\}_{\mathbf{x}_L\in\{0,1\}^L}.$$

2. We next show that the indistinguishability shown in the first step implies that of Relation 10.

Lemma 8. Let C be either C_0 or C_1. For every non-uniform adversaries $\mathcal{A} = \{\mathcal{A}_\lambda\}_\lambda$ such that $\mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa)$, $\mathrm{Adv}_{\mathcal{A}}^{L+2}(\lambda)$ is negligible in κ.

Proof. We define the following sequence of hybrids.

Hyb_0. This is same as the first distribution in the RHS of Equation 11.

Hyb_1. This is same as Hyb_0 except that, for every $\mathbf{x}_L \in \{0,1\}^L$, $\mathbf{v}_{\mathbf{x}_L}$ is replaced with

$$\mathbf{v}_{\mathbf{x}_L}^T := \mathbf{c}_{F,\mathbf{x}_L}^T - \mathbf{r}_{\mathbf{x}_L}^T - \mathsf{PRF}_B(K_B, \mathbf{x}_L) + \mathbf{e}_{v_{\mathbf{x}_L}}^T$$

where $\mathbf{c}_{F,\mathbf{x}_L}^T := \mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T\mathbf{H}_{F,x}(\mathbf{I}_\ell \otimes \mathbf{G}_{n+1}^{-1}(\mathbf{u}_{n+1,n+1}))$ and $\mathbf{r}_{\mathbf{x}_L}^T \leftarrow \mathsf{PRF}_B(K_B, \mathbf{x}_L)$.

Hyb_2. This is same as Hyb_2 except that we set

$$\{\mathbf{p}^T_{\mathbf{x}_i} \leftarrow\$ \mathbb{Z}_q^{1\times m_B}\}_{i\in[0,L],\mathbf{x}_i\in\{0,1\}^i}, \{\mathbf{c}^T_{\mathsf{att},\mathbf{x}_L} \leftarrow\$ \mathbb{Z}_q^{1\times L'm}\}_{\mathbf{x}_L\in\{0,1\}^L}.$$

Hyb_3. This is same as Hyb_2 except that ξ^T is replaced with $\xi^T := \mathsf{bits}(\mathsf{ct}')$, where $\mathsf{ct}' \leftarrow \mathsf{FHE.Enc}(\mathsf{pk}, \mathbf{0}_{\mathsf{size}_C})$.

Hyb_4. This is same as Hyb_3 except that, for every $\mathbf{x}_L \in \{0,1\}^L$, $\mathbf{r}^T_{\mathbf{x}_L}$ is sampled from $[-\frac{q}{4}+B, \frac{q}{4}-B]^{1\times\ell}$.

Hyb_5. This is same as Hyb_4 except that, for every $\mathbf{x}_L \in \{0,1\}^L$, $\mathbf{r}^T_{\mathbf{x}_L}$ is sampled from $[-\frac{q}{4}, \frac{q}{4}]^{1\times\ell}$.

Hyb_6. This is same as Hyb_5 except that, for every $\mathbf{x}_L \in \{0,1\}^L$, $\mathbf{v}_{\mathbf{x}_L}$ is replaced with $\mathbf{v}_{\mathbf{x}_L} \leftarrow\$ \mathbb{Z}_q^\ell$.

We note that the last hybrid is identical to the second distribution in the RHS of Equation 11. Therefore, it suffices to show that each consecutive hybrids are indistinguishable.

Indistinguishability between Hyb_0 and Hyb_1. We prove their indistinguishability for every $\mathbf{x}_L \in \{0,1\}^L$. In Hyb_1, from the definition of $\mathbf{v}_{\mathbf{x}_L}$, it follows that

$$\begin{aligned}\mathbf{v}^T_{\mathbf{x}_L} &:= \mathbf{c}^T_{F,\mathbf{x}_L} - \frac{q}{2}C(\mathbf{x}_L) - \mathbf{r}^T_{\mathbf{x}_L} + \mathbf{e}^T_{v_{\mathbf{x}_L}} \\ &= \mathbf{s}^T_{\mathbf{x}_L}\mathbf{A}_F + \mathbf{e}^T_{c_{F,\mathbf{x}_L}} + \mathbf{e}^T_{v_{\mathbf{x}_L}},\end{aligned}$$

where $\mathbf{e}^T_{c_{F,\mathbf{x}_L}} := B_F$ as shown in Equation 7. By the definition of the parameter χ_F, noise flooding (Definition 3) implies that $\mathbf{e}^T_{c_{F,\mathbf{x}_L}} + \mathbf{e}^T_{v_{\mathbf{x}_L}} \overset{s}{\approx} \mathbf{e}^T_{v_{\mathbf{x}_L}}$ holds. Therefore, these two hybrids are statistically indistinguishable.

Indistinguishability between Hyb_1 and Hyb_2. We prove that if there is an adversary $\mathcal{A}$ that can distinguish between Hyb_1 and Hyb_2, then we can construct an adversary $\mathcal{B}$ that can break the security of the all-product LWE assumption (Definition 4) with non-negligible advantage. The parameters for the assumption are set to $(n, L'm_B + (1+s'+n_t)m, q, L, \chi_{\mathsf{att}}, \{\chi_{B,i}\}_{i\in[0,L]})$. In the following, we let $b_1 = 0$ if $\mathcal{B}$ is in $\mathcal{D}_{\mathsf{APLWE},0}$ and $b_1 = 1$ otherwise. $\mathcal{A}$ outputs $b_2 = 0$ if it guesses that it is in Hyb_1 and $b_2 = 1$ otherwise. $\mathcal{B}$ internally invokes $\mathcal{A}$ as follows.

1. For the circuit C, $\mathcal{B}$ computes pk, K_B, ξ, and $\mathbf{t}$, in the same manner as the $\mathsf{PRO.Obf}$ algorithm.
2. $\mathcal{B}$ receives the following instances from the challenger of the security game for the all-product LWE assumption.

$$\begin{pmatrix}\mathbf{A}, \{\mathbf{B}'_i\}_{i\in[0,L]}, \\ \{\mathbf{c}^T_{A,\mathbf{x}_L}\}_{\mathbf{x}_L\in\{0,1\}^L} \\ \{\mathbf{c}^T_{B,\mathbf{x}_i}\}_{i\in[0,L],\mathbf{x}_i\in\{0,1\}^i}\end{pmatrix}$$

3. $\mathcal{B}$ define the following matrices and vectors:
   (a) Let $\mathbf{A}_{\mathsf{att,dyn}} := \mathbf{A}$.
   (b) For every $i \in [0, L-1]$, parse $\mathbf{B}'_i \in \mathbb{Z}_q^{(n+1)\times(L'm_B+(1+s'+n_t)m)}$ as
   $$\left(\mathbf{B}_{i,1} \in \mathbb{Z}_q^{(n+1)\times m_B}, \ldots, \mathbf{B}_{i,L'} \in \mathbb{Z}_q^{(n+1)\times m_B}, \mathbf{B}_{i,\mathsf{other}} \in \mathbb{Z}_q^{(n+1)\times(1+s'+n_t)m}\right).$$
   (c) Parse $\mathbf{B}'_L \in \mathbb{Z}_q^{(n+1)\times(L'm_B+(1+s'+n_t)m)}$ as
   $$\left(\mathbf{B}_{L,1} \in \mathbb{Z}_q^{(n+1)\times m_B}, \ldots, \mathbf{B}_{L,L'} \in \mathbb{Z}_q^{(n+1)\times m_B}, \bar{\mathbf{A}}_{\mathsf{att,fix1}} \in \mathbb{Z}_q^{(n+1)\times(1+s')m}, \bar{\mathbf{A}}_{\mathsf{att,fix2}} \in \mathbb{Z}_q^{(n+1)\times n_t m}\right).$$
   (d) Let $\mathbf{A}_{\mathsf{att,fix1}} := \bar{\mathbf{A}}_{\mathsf{att,fix1}} + (1, \xi^T) \otimes \mathbf{G}_{n+1}$ and $\mathbf{A}_{\mathsf{att,fix2}} := \bar{\mathbf{A}}_{\mathsf{att,fix2}} + \mathbf{t}^T \otimes \mathbf{G}_{n+1}$.
   (e) Let $\mathbf{A}_{\mathsf{att}} := (\mathbf{A}_{\mathsf{att,fix1}}, \mathbf{A}_{\mathsf{att,dyn}}, \mathbf{A}_{\mathsf{att,fix2}})$.
   (f) For every $i \in [0, L]$, let $\mathbf{B}_i := \begin{pmatrix} \mathbf{B}_{i,1} \\ \vdots \\ \mathbf{B}_{i,L'} \end{pmatrix}$.
   (g) For every $i \in [0, L-1]$ and $\mathbf{x}_o \in \{0,1\}^i$, parse $\mathbf{c}_{B,\mathbf{x}_i}^T$ as
   $$\left(\mathbf{b}_{\mathbf{x}_i,1}^T \in \mathbb{Z}_q^{1\times m_B}, \ldots, \mathbf{b}_{\mathbf{x}_i,L'}^T \in \mathbb{Z}_q^{1\times m_B}, \mathbf{b}_{\mathbf{x}_i,\mathsf{other}}^T \in \mathbb{Z}_q^{1\times(1+s'+n_t)m}\right).$$
   (h) For every $\mathbf{x}_L \in \{0,1\}^L$, parse $\mathbf{c}_{B,\mathbf{x}_L}^T$ as
   $$\left(\mathbf{b}_{\mathbf{x}_L,1}^T \in \mathbb{Z}_q^{1\times m_B}, \ldots, \mathbf{b}_{\mathbf{x}_L,L'}^T \in \mathbb{Z}_q^{1\times m_B}, \mathbf{c}_{\mathbf{x}_L,\mathsf{fix1}}^T \in \mathbb{Z}_q^{1\times(1+s')m}, \mathbf{c}_{\mathbf{x}_L,\mathsf{fix2}}^T \in \mathbb{Z}_q^{1\times n_t m}\right).$$
   (i) For every $i \in [0, L]$ and $\mathbf{x}_i \in \{0,1\}^i$, let $\mathbf{p}_{\mathbf{x}_i}^T := \Sigma_{j\in L'} \eta_{\mathbf{x}_i,j} \mathbf{b}_{\mathbf{x}_i,j}^T + \mathbf{e}_{p_{\mathbf{x}_i}}^T$, where $\eta_{\mathbf{x}_i} := (1, \xi_T, \mathbf{x}_i^T, \mathbf{0}_{L-i}^T, \mathbf{t}^T)$.
   (j) For every $\mathbf{x}_L \in \{0,1\}^L$, let $\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T := \left(\mathbf{c}_{\mathbf{x}_L,\mathsf{fix1}}^T + \mathbf{e'}_{c_{\mathbf{x}_L,\mathsf{fix1}}}^T, \mathbf{c}_{A,\mathbf{x}_L}^T, \mathbf{c}_{\mathbf{x}_L,\mathsf{fix2}}^T + \mathbf{e'}_{c_{\mathbf{x}_L,\mathsf{fix2}}}^T\right)$, where $\mathbf{e'}_{c_{\mathbf{x}_L,\mathsf{fix1}}}^T$ and $\mathbf{e'}_{c_{\mathbf{x}_L,\mathsf{fix2}}}^T$ are injected, respectively, so that the total errors follow $\mathcal{D}_{\mathbb{Z},\chi_{\mathsf{att}}}^{L'm}$.
   (k) For every $\mathbf{x}_L \in \{0,1\}^L$, let $\mathbf{v}_{\mathbf{x}_L}^T := \mathbf{c}_{F,\mathbf{x}_L}^T - \frac{q}{2}C(\mathbf{x}_L) - \mathsf{PRF}_B(K_B, \mathbf{x}_L) + \mathbf{e}_{v_{\mathbf{x}_L}}^T$.
4. $\mathcal{B}$ provides $\mathcal{A}$ with
$$\left(\begin{matrix}\mathsf{pk}, \xi, \mathbf{A}_{\mathsf{att}}, \mathbf{B}_L, \mathsf{aux}_C, \\ \{\mathbf{p}_{\mathbf{x}_i}^T\}_{i\in[0,L],\mathbf{x}_i\in\{0,1\}^i}, \{\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T\}_{\mathbf{x}_L\in\{0,1\}^L}, \{\mathbf{v}_{\mathbf{x}_L}^T\}_{\mathbf{x}_L\in\{0,1\}^L}\end{matrix}\right).$$
5. $\mathcal{A}$ forwards the $\mathcal{A}$'s output b_2.

We confirm that the above Distribution of the instances provided for $\mathcal{A}$ simulates Hyb_2 if $b_1 = 0$ and Hyb_3 otherwise. We focus on the distributions of $\mathbf{p}_{\mathbf{x}_i}$, $\mathbf{c}_{\mathsf{att},\mathbf{x}_L}$, and $\mathbf{v}_{\mathbf{x}_L}$ since there is no difference of the other distributions between Hyb_2 and Hyb_3.

When $b_1 = 0$, $\mathbf{p}_{\mathbf{x}_i}$ is parsed as follows:

$$\begin{aligned}
\mathbf{p}_{\mathbf{x}_i}^T &:= \Sigma_{j\in L'} \eta_{\mathbf{x}_i,j} \mathbf{b}_{\mathbf{x}_i,j}^T + \mathbf{e}_{p_{\mathbf{x}_i}}^T \\
&= \Sigma_{j\in L'} \eta_{\mathbf{x}_i,j} \left(\mathbf{s}_{\mathbf{x}_i}^T \mathbf{B}_{i,j} + \mathbf{e}_{b_{\mathbf{x}_i,j}}^T \right) + \mathbf{e}_{p_{\mathbf{x}_i}}^T \\
&= \left(\eta_{\mathbf{x}_i} \otimes \mathbf{s}_{\mathbf{x}_i}^T \right) \mathbf{B}_i + \Sigma_{j\in L'} \eta_{\mathbf{x}_i,j} \mathbf{e}_{b_{\mathbf{x}_i,j}}^T + \mathbf{e}_{p_{\mathbf{x}_i}}^T \\
&\overset{s}{\approx} \left((1, \xi^T, \mathbf{x}_i^T, \mathbf{0}_{L-i}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_i}^T \right) \mathbf{B}_i + \mathbf{e}_{p_{\mathbf{x}_i}}^T,
\end{aligned}$$

where the last statistical indistinguishability follows from the noise flooding. Similarly, we have

$$\begin{aligned}
\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T &:= \left(\mathbf{c}_{\mathbf{x}_L,\mathsf{fix1}}^T + \mathbf{e}'^T_{c_{\mathbf{x}_L,\mathsf{fix1}}}, \mathbf{c}_{A,\mathbf{x}_L}^T, \mathbf{c}_{\mathbf{x}_L,\mathsf{fix2}}^T + \mathbf{e}'^T_{c_{\mathbf{x}_L,\mathsf{fix2}}} \right) \\
&= \mathbf{s}_{\mathbf{x}_L}^T \left(\bar{\mathbf{A}}_{\mathsf{att,fix1}}, \mathbf{A}_{\mathsf{att,dyn}} - \mathbf{x}_L \otimes \mathbf{G}_{n+1}, \bar{\mathbf{A}}_{\mathsf{att,fix2}} \right) + \left(\mathbf{e}_{c_{\mathbf{x}_L,\mathsf{fix1}}}^T + \mathbf{e}'^T_{c_{\mathbf{x}_L,\mathsf{fix1}}}, \mathbf{e}_{c_{A,\mathbf{x}_L}}^T, \mathbf{e}_{c_{\mathbf{x}_L,\mathsf{fix2}}}^T + \mathbf{e}'^T_{c_{\mathbf{x}_L,\mathsf{fix2}}} \right) \\
&\overset{s}{\approx} \mathbf{s}_{\mathbf{x}_L}^T \left(\mathbf{A}_{\mathsf{att}} - (1, \xi_T, \mathbf{x}_L, \mathbf{t}_T) \otimes \mathbf{G}_{n+1} \right) + \mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}}^T,
\end{aligned}$$

where the last statistical indistinguishability follows because the distribution of $\left(\mathbf{e}_{c_{\mathbf{x}_L,\mathsf{fix1}}}^T + \mathbf{e}'^T_{c_{\mathbf{x}_L,\mathsf{fix1}}}, \mathbf{e}_{c_{A,\mathbf{x}_L}}^T, \mathbf{e}_{c_{\mathbf{x}_L,\mathsf{fix2}}}^T + \mathbf{e}'^T_{c_{\mathbf{x}_L,\mathsf{fix2}}} \right)$ is identitcal to $\mathbf{e}_{c_{\mathsf{att},\mathbf{x}_L}}^T$. Clearly, the distribution of $\mathbf{v}_{\mathbf{x}_L}$ provided by $\mathcal{B}$ is statistically indistinguishable from the one in Hyb_2. Therefore, we can confirm that for $b_1 = 0$, $\mathcal{B}$ simulates the view of $\mathcal{A}$ in Hyb_1.

When $b_1 = 1$, $\mathbf{p}_{\mathbf{x}_i}^T$, $\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T$, and $\mathbf{v}_{\mathbf{x}_L}^T$ are trivially indistinguishable from random vectors, since the provided vectors $\mathbf{b}_{\mathbf{x}_i,j}^T$, $\mathbf{c}_{\mathbf{x}_L,\mathsf{fix1}}^T$, $\mathbf{c}_{\mathbf{x}_L,\mathsf{fix2}}^T$, $\mathbf{c}_{A,\mathbf{x}_L}^T$, and $\mathbf{c}_{R,\mathbf{x}_L}^T$ are also random vectors. Hence, we can conclude that if $\mathcal{A}$ can distinguishe between Hyb_1 and Hyb_2, then $\mathcal{B}$ can break the all-product LWE assumption with non-negligible advantage. This is a contradiction; thus, Hyb_1 and Hyb_2 are indistinguishable.

Indistinguishability between Hyb_2 and Hyb_3. The only difference between Hyb_2 and Hyb_3 is the distribution of ξ^T—namely, the FHE ciphertexts. In both hybrids, there is no information about $\mathbf{t}$ in the adversary's view. Therefore, from the subexponential semantic security of the FHE scheme, it immidiately follows that these hybrids are indistinguishable.

Indistinguishability between Hyb_3 and Hyb_4. The only difference between Hyb_3 and Hyb_4 is whether $\mathbf{r}_{\mathbf{x}_L}$ is computed as the output of the PRF or sampled from the uniform distribution. Since there is no information about the PRF key K_B, which is removed from ξ^T in Hyb_3, in the adversary's view, the subexponential security of the PRF scheme implies their indistinguishability.

Indistinguishability between Hyb_4 and Hyb_5. The only difference between Hyb_4 and Hyb_5 is the range of $\mathbf{r}_{\mathbf{x}_L}$ for every $\mathbf{x}_L \in \{0,1\}^L$, which is

$$\begin{aligned}
\mathcal{U}_1 =& [-\frac{q}{4} + B, \frac{q}{4} - B] \text{ in } \mathsf{Hyb}_3 \\
\mathcal{U}_2 =& [-\frac{q}{4}, \frac{q}{4}] \text{ in } \mathsf{Hyb}_4.
\end{aligned}$$

Since B is set to be exponentially smaller than $\frac{q}{4}$, the statistical distance between $\mathcal{U}_1$ and $\mathcal{U}_2$ is negligible. Therefore, these hybrids are statistically indistinguishable.

Indistinguishability between Hyb_5 and Hyb_6. We claim that Hyb_5 is identical to Hyb_6. Since there is no information about C in the adversary's view, when $C = C_0$ (resp., $C = C_1$), the LHS of Relations 8 (resp., the RHS of Relation 9) corresponds to Hyb_5, and the other side corresponds to Hyb_6. Therefore, the indistinguishability of Relations 8 and 9 implies that, for every $\mathbf{x}_L \in \{0,1\}^L$, the highest-order bits of $\mathbf{v}_{\mathbf{x}_L}$ in Hyb_5 are indistinguishable from random bits. The other bits of $\mathbf{v}_{\mathbf{x}_L}$ are also uniformly random over $[-\frac{q}{4}, \frac{q}{4}]^{1\times\ell}$ in both hybrids. Hence, $\mathbf{v}_{\mathbf{x}_L}$ in Hyb_5 is indistinguishable from $\mathbf{v}_{\mathbf{x}_L} \leftarrow\$ \mathbb{Z}_q^{\ell}$ in Hyb_6.

This completes the proof of Lemma 8. □

We now move on to Step 2. The proof is based on Theorem 3.9 of [3].

For a non-uniform adversary $\mathcal{A}$, the following advantage function is defined:

$$
\begin{aligned}
&\mathrm{Adv}_{\mathcal{A}}^{L+1}(\lambda)\\
&:= \Pr\left[\mathcal{A}\begin{pmatrix}\mathsf{pk}, \xi := \mathsf{bits}(\mathsf{ct}), \mathbf{A}_{\mathsf{att}}, \mathbf{B}_L, \mathsf{aux}_C,\\ \{\mathbf{p}_{\mathbf{x}_i}^T := \left((1, \xi^T, \mathbf{x}_i^T, \mathbf{0}_{L-i}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_i}^T\right)\mathbf{B}_i + \mathbf{e}_{p_{\mathbf{x}_i}}^T\}_{i\in[0,L], \mathbf{x}_i\in\{0,1\}^i}\\ \mathbf{K}_{\mathsf{att}}, \mathbf{K}_F\end{pmatrix} = 1\right]\\
&\ -\ \Pr\left[\mathcal{A}\begin{pmatrix}\mathsf{pk}, \xi := \mathsf{bits}(\mathsf{ct}'), \mathbf{A}_{\mathsf{att}}, \mathbf{B}_L, \mathsf{aux}_C,\\ \{\mathbf{p}_{\mathbf{x}_i}^T \leftarrow\$ \mathbb{Z}_q^{1\times m_B}\}_{i\in[0,L], \mathbf{x}_i\in\{0,1\}^i}\\ \mathbf{K}_{\mathsf{att}}, \mathbf{K}_F\end{pmatrix} = 1\right].
\end{aligned} \tag{12}
$$

Additionally, for every $h \in [L]$, we define

$$
\begin{aligned}
&\mathrm{Adv}_{\mathcal{A}}^{h}(\lambda)\\
&:= \Pr\left[\mathcal{A}\begin{pmatrix}\mathsf{pk}, \xi := \mathsf{bits}(\mathsf{ct}), \mathbf{A}_{\mathsf{att}}, \mathbf{B}_L, \mathsf{aux}_C,\\ \{\mathbf{p}_{\mathbf{x}_i}^T := \left((1, \xi^T, \mathbf{x}_i^T, \mathbf{0}_{L-i}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_i}^T\right)\mathbf{B}_i + \mathbf{e}_{p_{\mathbf{x}_i}}^T\}_{i\in[0,h-1], \mathbf{x}_i\in\{0,1\}^i}\\ \{\mathbf{K}_{i,b}\}_{i\in[h,L], b\in\{0,1\}}, \mathbf{K}_{\mathsf{att}}, \mathbf{K}_F\end{pmatrix} = 1\right]\\
&\ -\ \Pr\left[\mathcal{A}\begin{pmatrix}\mathsf{pk}, \xi := \mathsf{bits}(\mathsf{ct}'), \mathbf{A}_{\mathsf{att}}, \mathbf{B}_L, \mathsf{aux}_C,\\ \{\mathbf{p}_{\mathbf{x}_i}^T \leftarrow\$ \mathbb{Z}_q^{1\times m_B}\}_{i\in[0,h-1], \mathbf{x}_i\in\{0,1\}^i}\\ \{\mathbf{K}_{i,b}\}_{i\in[h,L], b\in\{0,1\}}, \mathbf{K}_{\mathsf{att}}, \mathbf{K}_F\end{pmatrix} = 1\right].
\end{aligned} \tag{13}
$$

Our goal is to prove that if $\mathrm{Adv}_{\mathcal{A}}^{L+1}(\lambda)$ is negligible in κ—implied by Lemma 8—then $\mathrm{Adv}_{\mathcal{A}}^{1}(\lambda)$ is negligible in λ. As the first distributions in Equation 16 for $h = 1$ corresponds to the real distribution of the obfuscation $\widetilde{C}$, except for the public matrix $\mathbf{B}_L$, and the second one is trivially independent of the circuit C, this consequence implies that $\mathsf{PRO.Obf}(1^\lambda, C_0)$ and $\mathsf{PRO.Obf}(1^\lambda, C_1)$ are indistinguishable.

Lemma 9. For every non-uniform adversaries $\mathcal{A} = \{\mathcal{A}_\lambda\}_\lambda$ such that $\mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa)$, $\mathrm{Adv}_{\mathcal{A}}^{L+1}(\lambda)$ is negligible in κ.

Proof. We prove this by applying Lemma 4 with $\lambda' = \lambda^L$, which is proven based on the private-coin binding non-uniform κ-evasive LWE assumption (Definition 2), to the distributions in the RHS of Equation 12. Specifically, we define the sampler $\mathsf{Samp}(1^\lambda) \to (\mathbf{S}, \mathbf{P}, \mathsf{aux})$, in which C and aux_C are hardcoded, as follows:

1. Compute $\mathbf{s}_\epsilon$, $\{\mathbf{S}_{i,b}\}_{i\in[L],b\in\{0,1\}}$, $\mathbf{A}_{\mathsf{att}}$, $\{\mathbf{B}_i\}_{i\in[0,L-1]}$, K_B, pk, ξ, $\mathbf{t}$, $\mathbf{P}_{\mathsf{att}}$, and $\mathbf{P}_F$ in the same manner as the $\mathsf{PRO.Obf}$ algorithm.
2. For every $i \in [0, L-1]$ and $\mathbf{x}_i \in \{0,1\}^i$, compute $\mathbf{p}_{\mathbf{x}_i}^T := \left((1, \xi^T, \mathbf{x}_i^T, \mathbf{0}_{L-i}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_i}^T\right) \mathbf{B}_i + \mathbf{e}_{p_{\mathbf{x}_i}}^T$.
3. Let
$$\mathbf{S} := \begin{pmatrix} (1, \xi^T, \mathbf{0}_L^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{0}_L}^T \\ \vdots \\ (1, \xi^T, \mathbf{1}_L^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{1}_L}^T \end{pmatrix}, \mathbf{P} := \left(\mathbf{P}_{\mathsf{att}}, \mathbf{P}_F\right)$$
, where $\mathbf{s}_{\mathbf{x}_L}^T := \mathbf{s}_\epsilon \prod_{j\in[L]} \mathbf{S}_{j,x_j}$ for every $\mathbf{x}_L \in \{0,1\}^L$.
4. Let
$$\mathsf{aux}_1 := \left(\xi^T, \{\mathbf{p}_{\mathbf{x}_i}^T\}_{i\in[0,L],\mathbf{x}_i\in\{0,1\}^i},\right), \mathsf{aux}_2 := \left(\mathsf{pk}, \mathbf{A}_{\mathsf{att}}, \mathsf{aux}_C\right),$$
5. Output $(\mathbf{S}, \mathbf{P}, \mathsf{aux} = (\mathsf{aux}_1, \mathsf{aux}_2))$.

It holds that $\mathrm{Size}(\mathsf{Samp}) \leq \mathsf{poly}(\lambda') = \mathsf{poly}(\lambda^L)$ since $2^L = \mathsf{poly}(\lambda^L)$.

The distributions in the precondition of Lemma 4 for this sanpler is identical to ones in the RHS of Equation 11. This is because when $\mathbf{B}$ provided by the challenger of Lemma 4 is set to $\mathbf{B}_L$, the distributions of $\mathbf{SB} + \mathbf{E}$ and $\mathbf{SP} + \mathbf{E}'$ correspond to $\{\mathbf{p}_{\mathbf{x}_L}^T\}_{\mathbf{x}_L\in\{0,1\}^L}$ and $\{\mathbf{c}_{\mathsf{att},\mathbf{x}_L}^T, \mathbf{v}_{\mathbf{x}_L}^T\}_{\mathbf{x}_L\in\{0,1\}^L}$, respectively[7]. Similarly, the distributions in the postcondition is identical to ones in the RHS of Equation 12. Therefore, Lemma 4 implies the existance of another non-uniform adversary $\mathcal{B}$ and a polynomial $Q(\cdot)$ such that

$$\mathrm{Adv}_{\mathcal{B}}^{L+2}(\lambda) \geq \mathrm{Adv}_{\mathcal{A}}^{L+1}(\lambda)/Q(\lambda') - \mathsf{negl}(\kappa), \mathrm{Size}(\mathcal{B}) \leq \mathsf{poly}(\lambda') \cdot \mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa).$$

From Lemma 8, it follows that $\mathrm{Adv}_{\mathcal{A}}^{L+1}(\lambda) \leq \mathsf{negl}(\kappa)$. This completes the proof of Lemma 9.

$\square$

Lemma 10. For every non-uniform adversaries $\mathcal{A} = \{\mathcal{A}_\lambda\}_\lambda$ such that $\mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa)$, $\mathrm{Adv}_{\mathcal{A}}^{L}(\lambda)$ is negligible in κ.

[7] Although the matrix $\mathbf{P}$ is not explicitly included in distributions in the RHS of Equation 11, it can be publicly computed from aux_2; thus it does not provide any non-negligible advantage to distinguish between these distributions.

Proof. We prove this by applying the private-coin hiding non-uniform κ-evasive LWE (Assumption 3) with $\lambda' = \lambda^L$ to the distributions in the RHS of Equation 13 for $h = L$. Specifically, we define the sampler $\mathsf{Samp}(1^\lambda) \to (\mathbf{S}, \mathbf{P}, \mathsf{aux})$, in which C and aux_C are hardcoded, as follows:

1. Compute $\mathbf{s}_\epsilon$, $\{\mathbf{S}_{i,b}\}_{i\in[L],b\in\{0,1\}}$, $\mathbf{A}_{\mathsf{att}}$, $\mathbf{R}$, $\{\mathbf{B}_i\}_{i\in[0,L]/\{L-1\}}$, pk, ξ, $\mathbf{t}$, $\{\mathbf{U}_{L,b}, \mathbf{E}_{K_{L,b}}\}_{b\in\{0,1\}}$, $\mathbf{K}_{\mathsf{att}}$, and $\mathbf{K}_F$ in the same manner as the $\mathsf{PRO.Obf}$ algorithm.
2. For every $i \in [0, L-2]$ and $\mathbf{x}_i \in \{0,1\}^i$, compute $\mathbf{p}_{\mathbf{x}_i}^T := \left((1, \xi^T, \mathbf{x}_i^T, \mathbf{0}_{L-i}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_i}^T\right)\mathbf{B}_i + \mathbf{e}_{p_{\mathbf{x}_i}}^T$.
3. Let
$$\mathbf{S} := \begin{pmatrix} (1, \xi^T, \mathbf{0}_{L-1}^T, \mathbf{0}_1^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{0}_{L-1}}^T \\ \vdots \\ (1, \xi^T, \mathbf{1}_{L-1}^T, \mathbf{0}_1^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{1}_{L-1}}^T \end{pmatrix},$$
$$\mathbf{P} := \left((\mathbf{U}_{L,0} \otimes \mathbf{S}_{L,0})\mathbf{B}_L + \mathbf{E}_{K_{L,0}}, (\mathbf{U}_{L,1} \otimes \mathbf{S}_{L,1})\mathbf{B}_L + \mathbf{E}_{K_{L,1}}\right)$$
, where $\mathbf{s}_{\mathbf{x}_{L-1}}^T := \mathbf{s}_\epsilon \prod_{j\in[L-1]} \mathbf{S}_{j,x_j}$ for every $\mathbf{x}_L \in \{0,1\}^L$.
4. Let
$$\mathsf{aux} := \begin{pmatrix} \mathsf{pk}, \mathbf{A}_{\mathsf{att}}, \mathbf{R}, \mathbf{B}_L, \mathsf{aux}_C, \mathbf{K}_{\mathsf{att}}, \mathbf{K}_F \\ \xi^T, \{\mathbf{p}_{\mathbf{x}_i}^T\}_{i\in[0,L-2],\mathbf{x}_i\in\{0,1\}^i}, \end{pmatrix},$$
5. Output $(\mathbf{S}, \mathbf{P}, \mathsf{aux})$.

It holds that $\mathrm{Size}(\mathsf{Samp}) \le \mathsf{poly}(\lambda') = \mathsf{poly}(\lambda^L)$.

We recall that the precondition of Assumption 3 requires the following two indistinguishabilities:

$$(\mathbf{SB} + \mathbf{E},\, \mathbf{SP} + \mathbf{E}',\, \mathsf{aux}) \overset{c}{\approx} (\mathbf{C}_0,\, \mathbf{C}',\, \mathsf{aux}) \tag{14}$$
$$(\mathbf{P},\, \mathsf{aux}) \overset{c}{\approx} (\mathbf{P} + \mathbf{R}',\, \mathsf{aux}), \tag{15}$$

where $\mathbf{E} \leftarrow\$ \mathcal{D}_{\mathbb{Z},\chi_{p,L-1}}^{2^{L-1}\times m_B}$, $\mathbf{E}' \leftarrow\$ \mathcal{D}_{\mathbb{Z},\chi_{p,L}}^{2^{L-1}\times 2m_B}$, $\mathbf{C}' \leftarrow\$ \mathbb{Z}_q^{2^{L-1}\times m_B}$, $\mathbf{C}_0 \leftarrow\$ \mathbb{Z}_q^{2^{L-1}\times 2m_B}$, and $\mathbf{R}' \leftarrow\$ \mathbb{Z}_q^{n_B\times 2m_B}$.

The indistinguishability of Relation 14 follows from Lemma 9. This is because the distributions of $\mathbf{SB}+\mathbf{E}$ and $\mathbf{SP}+\mathbf{E}'$ correspond to $\{\mathbf{p}_{\mathbf{x}_{L-1}}^T\}_{\mathbf{x}_{L-1}\in\{0,1\}^{L-1}}$ and $\{\mathbf{p}_{\mathbf{x}_L}^T\}_{\mathbf{x}_L\in\{0,1\}^L}$, respectively. Besides, ξ^T and $\{\mathbf{p}_{\mathbf{x}_i}^T\}_{i\in[0,L-2],\mathbf{x}_i\in\{0,1\}^i}$ included in aux in the second distribution in Equation 12 can be replaced with an encryption of zeros $\xi := \mathsf{bits}(\mathsf{ct}')$ and random vectors $\{\mathbf{p}_{\mathbf{x}_i}^T \leftarrow\$ \mathbb{Z}_q^{1\times m_B}\}_{i\in[0,L-2],\mathbf{x}_i\in\{0,1\}^i}$, respectively, by employing the subexponential semantic security of the FHE scheme (Definition 1) and the all-product LWE assumption (Assumption 4).

The indistinguishability of Relation 15 follows from the LWE assumption (Assumption 1). Since there is no information about $\{\mathbf{S}_{L,b}\}_{b\in\{0,1\}}$ in aux, the LWE instances provided by the challenger of the security game for the LWE assumption, which are $\{\mathbf{S}_{L,b}\mathbf{B}_L + \mathbf{E}_{K_{L,b}}\}_{b\in\{0,1\}}$ in the real case and $\{\mathbf{P}_{L,b} \leftarrow\$$

$\mathbb{Z}_q^{n_B \times 2m_B}\}_{b\in\{0,1\}}$ in the simulated case, can be used as $\mathbf{P}$. Hence, the hardness of the LWE assumption implies that no adversary can distinguish between Relation 15 with non-negligible advantage.

Lemma 4 implies the existance of another non-uniform adversary $\mathcal{B}$ and a polynomial $Q(\cdot)$ such that

$$\begin{aligned}&\mathrm{Adv}_{\mathcal{B}}^{L+1}(\lambda) + \mathrm{Adv}_{\mathcal{B}}^{P_L}(\lambda) \geq \mathrm{Adv}_{\mathcal{A}}^{L}(\lambda)/Q(\lambda') - \mathsf{negl}(\kappa),\\ &\mathrm{Size}(\mathcal{B}) \leq \mathsf{poly}(\lambda') \cdot \mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa),\end{aligned}$$

where $\mathrm{Adv}_{\mathcal{B}}^{P_L}(\lambda)$ denotes the advantage of $\mathcal{B}$ in distinguishing between the distributions in Relation 15. From the above analysis, we can conclude that $\mathrm{Adv}_{\mathcal{A}}^{L}(\lambda) \leq \mathsf{negl}(\kappa)$. This completes the proof of Lemma 10. $\square$

Lemma 11. For every $h \in [L]$ and non-uniform adversary $\mathcal{A} = \{\mathcal{A}_\lambda\}_\lambda$ such that $\mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa)$, there exists another non-uniform adversary $\mathcal{B} := \{\mathcal{B}_\lambda\}_\lambda$ and a polynomial Q such that the following holds:

$$\begin{aligned}&\mathrm{Adv}_{\mathcal{B}}^{h}(\lambda) \geq \mathrm{Adv}_{\mathcal{A}}^{h-1}(\lambda)/Q(\lambda') - \mathsf{negl}(\kappa),\\ &\mathrm{Size}(\mathcal{B}) \leq Q(\lambda') \cdot \mathrm{Size}(\mathcal{A}),\end{aligned}$$

where $\lambda' := \lambda^L$.

Proof. In similar to Lemma 10, Assumption 3 is employed to prove this lemma. We define the sampler $\mathsf{Samp}(1^\lambda) \to (\mathbf{S}, \mathbf{P}, \mathsf{aux})$ for C and aux_C as follows:

1. Compute $\mathbf{s}_\epsilon$, $\{\mathbf{S}_{i,b}\}_{i\in[L],b\in\{0,1\}}$, $\mathbf{A}_{\mathsf{att}}$, $\mathbf{R}$, $\{\mathbf{B}_i\}_{i\in[0,L]/\{h-1\}}$, pk, ξ, $\mathbf{t}$, $\{\mathbf{U}_{h,b}, \mathbf{E}_{K_{h,b}}\}_{b\in\{0,1\}}$, $\{\mathbf{K}_{i,b}\}_{i\in[h,L],b\in\{0,1\}}$, $\mathbf{K}_{\mathsf{att}}$, and $\mathbf{K}_F$ in the same manner as the $\mathsf{PRO.Obf}$ algorithm.
2. If $h > 1$, for every $i \in [0, h-2]$ and $\mathbf{x}_i \in \{0,1\}^i$, compute $\mathbf{p}_{\mathbf{x}_i}^T := \left((1, \xi^T, \mathbf{x}_i^T, \mathbf{0}_{L-i}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{x}_i}^T\right)\mathbf{B}_i + \mathbf{e}_{p_{\mathbf{x}_i}}^T$.
3. Let
$$\begin{aligned}\mathbf{S} &:= \begin{pmatrix}(1, \xi^T, \mathbf{0}_{h-1}^T, \mathbf{0}_{L-h+1}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{0}_{h-1}}^T\\ \vdots \\ (1, \xi^T, \mathbf{1}_{h-1}^T, \mathbf{0}_{L-h+1}^T, \mathbf{t}^T) \otimes \mathbf{s}_{\mathbf{1}_{h-1}}^T\end{pmatrix},\\ \mathbf{P} &:= \left((\mathbf{U}_{h,0} \otimes \mathbf{S}_{h,0})\mathbf{B}_h + \mathbf{E}_{K_{h,0}}, (\mathbf{U}_{h,1} \otimes \mathbf{S}_{h,1})\mathbf{B}_h + \mathbf{E}_{K_{h,1}}\right)\end{aligned}$$
, where $\mathbf{s}_{\mathbf{x}_{h-1}}^T := \mathbf{s}_\epsilon \prod_{j\in[h-1]} \mathbf{S}_{j,x_j}$ for every $\mathbf{x}_L \in \{0,1\}^L$.
4. Let
$$\mathsf{aux} := \begin{pmatrix}\mathsf{pk}, \mathbf{A}_{\mathsf{att}}, \mathbf{R}, \mathbf{B}_L, \mathsf{aux}_C, \{\mathbf{K}_{i,b}\}_{i\in[h,L],b\in\{0,1\}}, \mathbf{K}_{\mathsf{att}}, \mathbf{K}_F\\ \xi^T, \{\mathbf{p}_{\mathbf{x}_i}^T\}_{i\in[0,h-2],\mathbf{x}_i\in\{0,1\}^i}\end{pmatrix},$$
where the last term $\{\mathbf{p}_{\mathbf{x}_i}^T\}_{i\in[0,h-2],\mathbf{x}_i\in\{0,1\}^i}$ is included if $h > 1$.
5. Output $(\mathbf{S}, \mathbf{P}, \mathsf{aux})$.

In the same manner as Lemma 10, Assumption 3 ensures that there is a non-uniform adversary $\mathcal{B}$ and a polynomial $Q(\cdot)$ such that the following holds:

$$\begin{aligned} &\mathrm{Adv}_{\mathcal{B}}^{h}(\lambda) + \mathrm{Adv}_{\mathcal{B}}^{P_h}(\lambda) \geq \mathrm{Adv}_{\mathcal{A}}^{h-1}(\lambda)/Q(\lambda') - \mathsf{negl}(\kappa), \\ &\mathrm{Size}(\mathcal{B}) \leq \mathsf{poly}(\lambda') \cdot \mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa), \end{aligned} \tag{16}$$

where $\mathrm{Adv}_{\mathcal{B}}^{P_h}(\lambda)$ denotes the advantage of $\mathcal{B}$ in distinguishing between $(\mathbf{P}, \mathsf{aux}) \stackrel{c}{\approx} (\mathbf{P} + \mathbf{R}', \mathsf{aux})$ for $\mathbf{R}' \leftarrow\!\!\$\ \mathbb{Z}_q^{n_B \times 2m_B}$. As shown in Lemma 10, assuming the hardness of the LWE assumption, we obtain that $\mathrm{Adv}_{\mathcal{B}}^{P_h}(\lambda) \leq \mathsf{negl}(\kappa)$. Therefore, Inequation 16 implies that $\mathrm{Adv}_{\mathcal{B}}^{h}(\lambda) \geq \mathrm{Adv}_{\mathcal{A}}^{h-1}(\lambda)/Q(\lambda') - \mathsf{negl}(\kappa)$. This completes the proof of Lemma 11. $\square$

We adopt Lemma 3.12 of [3] with necessary modifications.

Lemma 12. Suppose that $\mathrm{Adv}_{\mathcal{A}}^{L}$ is negligible in κ for all non-uniform adversaries $\mathcal{A} := \{\mathcal{A}_\lambda\}_\lambda$ such that $\mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa)$, $\mathrm{Adv}_{\mathcal{B}}^{1}$ is negligible in λ for all non-uniform adversaries $\mathcal{B} := \{\mathcal{B}_\lambda\}_\lambda$ such that $\mathrm{Size}(\mathcal{B}) \leq \mathsf{poly}(\lambda)$.

Proof. We prove this by contradiction. Suppose that there exists a non-uniform adversary $\mathcal{B} := \{\mathcal{B}_\lambda\}_\lambda$ such that $\epsilon_1(\lambda) := \mathrm{Adv}_{\mathcal{B}}^{1}(\lambda)$ is non-negligible in λ, and $t_1(\lambda) := \mathrm{Size}(\mathcal{B})$ is bounded by a polynomial in λ. By the definition of non-negligible functions, there exists an infinite set $\mathcal{L} \in \mathbb{N}$ and some polynomial p such that $\epsilon_1(\lambda) \geq \frac{1}{p(\lambda)}$ and $t_1(\lambda) \leq p(\lambda)$ for all $\lambda \in \mathcal{L}$. For every $i \in [2, L]$, we define $t_i(\lambda) := p(\lambda)\lambda^{L(i-1)\log\lambda}$ and $\epsilon_i(\lambda) := \frac{1}{p(\lambda)\lambda^{L(i-1)\log\lambda}}$. As $\kappa = \lambda^{L^2 \log\lambda}$, it holds that $t_L(\lambda) < \kappa$ and $\epsilon_L(\lambda) > \frac{1}{\kappa}$ for sufficiently large λ. Therefore, there exists $\lambda_0 \in \mathbb{N}$ such that for all $\lambda > \lambda_0$, there is no adversary $\mathcal{A}_\lambda$ such that $\mathrm{Size}(\mathcal{A}_\lambda) \leq t_L(\lambda)$ and $\mathrm{Adv}_{\mathcal{A}_\lambda}^{L} \geq \epsilon_L(\lambda)$[8]. To derive a contradiction, we consider the following statement parameterized by λ and $h \in [L]$:

$\mathsf{Statement}_{\lambda,h}$: There exists an adversary $\mathcal{A}_\lambda$ such that $\mathrm{Size}(\mathcal{A}_\lambda) \leq t_h(\lambda)$ and $\mathrm{Adv}_{\mathcal{A}_\lambda}^{h}(\lambda) \geq \epsilon_h(\lambda)$.

For each $\lambda \in \mathcal{L} \wedge \mathbb{N}_{>\lambda_0}$, there exists $h_\lambda^\star \in [1, h-1]$ such that $\mathsf{Statement}_{\lambda,h_\lambda^\star}$ is true and $\mathsf{Statement}_{\lambda,h_\lambda^\star+1}$ is false, since $\mathsf{Statement}_{\lambda,1}$ is true and $\mathsf{Statement}_{\lambda,L}$ is false. Lemma 11 implies that there exists another non-uniform adversary $\mathcal{A}' := \{\mathcal{A}'_\lambda\}_\lambda$ and a polynomial Q such that

$$\begin{aligned} \mathrm{Adv}_{\mathcal{A}'}^{h_\lambda^\star+1}(\lambda) &\geq \mathrm{Adv}_{\mathcal{A}}^{h_\lambda^\star}(\lambda)/Q(\lambda') - \mathsf{negl}(\kappa) \\ &\geq \epsilon_{h_\lambda^\star}(\lambda)/Q(\lambda') - \mathsf{negl}(\kappa) \\ &\geq \epsilon_{h_\lambda^\star}(\lambda)/2Q(\lambda') \end{aligned}$$

and

$$\begin{aligned} \mathrm{Size}(\mathcal{A}') &\leq Q(\lambda') \cdot \mathrm{Size}(\mathcal{A}) \\ &\leq t_{h_\lambda^\star}(\lambda)Q(\lambda') \end{aligned}$$

[8] We recall that $\mathrm{Adv}_{\mathcal{A}_\lambda}^{L}$ is assumed to be negligible in κ.

hold. For all sufficiently large λ, the fact that $\lambda^{L \log \lambda} > Q(\lambda') = Q(\lambda^L)$ holds implies

$$\mathrm{Adv}_{\mathcal{A}'}^{h^\star_\lambda+1}(\lambda) \geq \epsilon_{h^\star_\lambda+1}(\lambda),\, \mathrm{Size}(\mathcal{A}') \leq t_{h^\star_\lambda+1}(\lambda).$$

This contradicts the assumption that $\mathsf{Statement}_{\lambda,h^\star_\lambda+1}$ is false. Therefore, we conclude that $\mathrm{Adv}_{\mathcal{B}}^{1}(\lambda)$ is negligible in λ for all non-uniform adversaries $\mathcal{B}$ such that $\mathrm{Size}(\mathcal{B}) \leq \mathsf{poly}(\lambda)$. □

Lemmas 10 and 12 imply that $\mathrm{Adv}_{\mathcal{A}}^{1}(\lambda)$ is negligible in λ. Since the first distributions in Equation 16 for $h = 1$ corresponds to the real distribution of the obfuscation $\widetilde{C}$, except for the public matric $\mathbf{B}_L$, and the second one is trivially independent of the circuit C, we can conclude that $\mathsf{PRO.Obf}(1^\lambda, C_0)$ and $\mathsf{PRO.Obf}(1^\lambda, C_1)$ are indistinguishable. This completes the proof of Theorem 1. □

4.2 Bootstrapping from PRO to iO

We adopt the bootstrapping technique that lifts PRO to iO in PROM, which is adopted from Subsection 5.4 of [19]. To keep this exposition self-contained, we provide a complete description of the construction and its security proof.

Using the PRO scheme ($\mathsf{PRO.Obf}, \mathsf{PRO.Eval}$), and the pseudorandom oracle O, which corresponds to PRF, we construct an iO scheme ($\mathsf{iO.Obf}, \mathsf{iO.Eval}$) as follows:

- $\mathsf{iO.Obf}(1^\lambda, C) \to \widetilde{C}$:
  1. Sample a PRF key $K \leftarrow\!\!\$\ \{0,1\}^\lambda$.
  2. Execute $h \leftarrow \mathsf{O}(\mathsf{hGen}, K)$.
  3. Let $C'[K]$ be a circuit that on input $\mathbf{x} \in \{0,1\}^L$ outputs $C(\mathbf{x}) \oplus \mathsf{PRF}(K, \mathbf{x})$.
  4. Execute $\widetilde{C'} \leftarrow \mathsf{PRO.Obf}(1^\lambda, C'[K])$.
  5. Output $\widetilde{C} := \left(\widetilde{C'}, h\right)$
- $\mathsf{iO.Eval}(\widetilde{C}, \mathbf{x}) \to \mathbf{y}$:
  1. Parse $\widetilde{C}$ as $(\widetilde{C'}, h)$.
  2. Execute $\mathbf{y}' \leftarrow \mathsf{PRO.Eval}(\widetilde{C'}, \mathbf{x})$.
  3. Execute $\nu_{\mathbf{x}_L} \leftarrow \mathsf{O}(\mathsf{hEval}, h, \mathbf{x})$.
  4. Compute $\mathbf{y} := \mathbf{y}' \oplus \nu_{\mathbf{x}_L}$.
  5. Output $\mathbf{y}$.

The above construction trivially satisfies the polynomial slowdown property defined in Definition 3.

Correctness. The correctness is confirmed by the following lemma.

Lemma 13 (Correctness of the iO construction). The iO construction in Subsection 4.2 satisfies correctness (Definition 3).

Proof. By the correctness of the PRO scheme (Definition 4), we have

$$\begin{aligned}\mathsf{PRO.Eval}(\widetilde{C}', \mathbf{x}) &= C'[K](\mathbf{x}) \\ &= C(\mathbf{x}) \oplus \mathsf{PRF}(K, \mathbf{x})\end{aligned}$$

with overwhelming probability. Since $\mathsf{PRF}(K, \mathbf{x}) = \nu_{\mathbf{x}_L}$ holds, where $\nu_{\mathbf{x}_L} \leftarrow \mathsf{O}(\mathsf{hEval}, h, \mathbf{x})$, it follows that

$$\begin{aligned}\mathbf{y} &= \mathsf{PRO.Eval}(\widetilde{C}', \mathbf{x}) \oplus \nu_{\mathbf{x}_L} \\ &= (C(\mathbf{x}) \oplus \mathsf{PRF}(K, \mathbf{x})) \oplus \mathsf{PRF}(K, \mathbf{x}) \\ &= C(\mathbf{x}).\end{aligned}$$

Therefore, $\mathsf{iO.Eval}(\widetilde{C}, \mathbf{x}) = C(\mathbf{x})$ holds with overwhelming probability. This completes the proof of Lemma 13. □

Security. The indistinguishability is proven by the following lemma.

Lemma 14 (Indistinguishability of the iO construction). The iO construction in Subsection 4.2 satisfies indistinguishability for equivalent functionalities (Definition 3).

Proof. We prove this by applying the indistinguishability of the PRO scheme (Definition 4) to the iO construction. A PRO sampler $\mathsf{PROSamp}_{C_0,C_1}(1^\lambda)$ is defined as follows:

1. Sample a PRF key $K \leftarrow\!\!\$\ \{0,1\}^\lambda$.
2. Execute $h \leftarrow \mathsf{O}(\mathsf{hGen}, K)$.
3. For $b \in \{0,1\}$, let $C'_b[K]$ be a circuit that on input $\mathbf{x} \in \{0,1\}^L$ outputs $C_b(\mathbf{x}) \oplus \mathsf{PRF}(K, \mathbf{x})$.
4. Let $\mathsf{aux}_{C'} := (C_0, C_1, h)$.
5. Output $(C'_0[K], C'_1[K], \mathsf{aux}_{C'})$.

The distribution of $(C_0, C_1, \mathsf{iO.Obf}(1^\lambda, C_b))$ is identical to that of

$$(\mathsf{PRO.Obf}(1^\lambda, C_b), \mathsf{aux}_{C'}) = (\mathsf{PRO.Obf}(1^\lambda, C_b), (C_0, C_1, h)),$$

, which is provided in the postcondition of the security game for the PRO scheme. Besides, the oracle output $\nu_{\mathbf{x}_L} \leftarrow \mathsf{O}(\mathsf{hEval}, h, \mathbf{x})$ can be simulated without the key K by computing

$$\nu_{\mathbf{x}_L} := \mathsf{PRO.Eval}(\widetilde{C}'_b, \mathbf{x}) \oplus C_b(\mathbf{x}).$$

Therefore, it suffices to show the indistinguishability between $\widetilde{C}'_0$ and $\widetilde{C}'_1$. The indistinguishability of PRO requires that the following precondition holds for $b \in \{0,1\}$:

$$\left\{1^\kappa, \{C'_b(\mathbf{x})\}_{\mathbf{x}\in\{0,1\}^L}, \mathsf{aux}_C\right\} \overset{c}{\approx} \left\{1^\kappa, \{\Delta_{\mathbf{x}} \leftarrow\!\!\$\ \{0,1\}^\ell\}_{\mathbf{x}\in\{0,1\}^L}, \mathsf{aux}_C\right\}$$

Since there is no information about the PRF key K, their indistinguishability immidiately follows from the PRF security. This completes the proof of Lemma 14. □

Combining Lemmas 7, 13, 14, and Theorem 1, we obtain the final theorem in Section 4 as follows:

Theorem 2. Let $\kappa = \lambda^{L^2 \log \lambda}$ and Λ be a security parameter such that $2^{\Lambda^\delta} \geq \kappa^{\omega(1)}$ holds for some $\delta \in (0, 1)$. Assume that the following hold:

1. the LWE assumption is subexponentially hard (Assumption 1);
2. the private-coin binding and hiding non-uniform κ-evasive LWE assumptions hold (Assumptions 2 and 3);
3. the all-product LWE assumption holds (Assumption 4); and
4. the underlying FHE scheme satisfies the subexponential semantic security (Definition 1).
5. the PRF scheme denoted by $\mathsf{PRF}_B : \{0,1\}^\lambda \times \{0,1\}^\lambda \to [-q/4 + B, q/4 - B]^{1\times\ell}$ is subexponentially secure for the security parameter Λ.
6. there exists a pseudorandom oracle (Definition 3.6).

Then the iO construction in Subsection 4.2 instantiated with the PRO construction from Subsection 4.1 satisfies the properties in Definition 3.

5 Cryptanalysis of Nonstandard Assumptions

We analyze the security of nonstandard assumptions used in our construction: the all-product LWE assumption (Assumption 4) and multiple flavors of the private-coin evasive LWE assumptions (Assumptions 2 and 3).

5.1 All-Product LWE Assumption

Although we have not yet established a reduction from the all-product LWE assumption to the standard LWE assumption, we can do that for its variant such that only the LWE-like instances $\mathbf{c}_{B,\mathbf{x}_i}^T := \mathbf{s}_{\mathbf{x}_i}^T \mathbf{B}_i + \mathbf{e}_{B,\mathbf{x}_i}^T$ are provided as shown in Theorem 3.

Theorem 3. Let $n, m_R, m_B, q, p, L \in \mathbb{N}$ be parameters such that $(n+1)^{L+1}p < q$ holds, and $\lambda \in \mathbb{N}$ be a security parameter. Let σ_A, and $\{\sigma_{B,i}\}_{i\in[0,L]}$ be Gaussian parameters. We define a variant of the all-product LWE assumption (Assumption 4) such that an adversary distinguishes between the following two distributions:

$$\mathcal{D}'_{\mathsf{APLWE},0} := \begin{pmatrix} \{\mathbf{B}_i\}_{i\in[0,L]}, \\ \{\mathbf{c}_{B,\mathbf{x}_i}^T := \mathbf{s}_{\mathbf{x}_i}^T \mathbf{B}_i + \mathbf{e}_{B,\mathbf{x}_i}^T\}_{i\in[0,L],\mathbf{x}_i\in\{0,1\}^i} \end{pmatrix}$$

$$\mathcal{D}'_{\mathsf{APLWE},1} := \begin{pmatrix} \{\mathbf{B}_i\}_{i\in[0,L]}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} \leftarrow\$\, \mathbb{Z}_q^{1\times m_B}\}_{i\in[0,L],\mathbf{x}_i\in\{0,1\}^i} \end{pmatrix},$$

where the variables are defined in the same manner for the parameters

$$(n, m_R, m_B, q, p, L, \sigma_A, \{\sigma_{B,i}\}_{i\in[0,L]})$$

as in Assumption 4.

Let σ_E be a Gaussian parameter such that the following holds for every $i \in [L]$:

$$\sigma_{B,i} \geq \left(\sigma_E \sqrt{\lambda}(n+1)^i\right) \lambda^{\omega(1)}. \tag{17}$$

For a function $\kappa := \kappa(\lambda)$ of the security parameter λ, assume that the LWE assumption is subexponentially hard for a security parameter Λ such that $2^{\Lambda^\delta} \geq \kappa^{\omega(1)}$ holds for some $\delta \in (0,1)$ (Assumption 1), along with the sets of the parameters (n, m, q, σ_E) and $\{(n, m, q, \sigma_{B,i})\}_{i \in [0,L]}$. Then, for every non-uniform adversary $\mathcal{A}$ such that $\mathrm{Size}(\mathcal{A}) \leq \mathsf{poly}(\kappa)$, the following holds:

$$\left|\Pr\left[\mathcal{A}\left(\mathcal{D}'_{\mathsf{APLWE},0}\right) = 1\right] - \Pr\left[\mathcal{A}\left(\mathcal{D}'_{\mathsf{APLWE},1}\right) = 1\right]\right| \leq \mathsf{negl}(\kappa).$$

Proof. For every $h \in [0, L]$, we define the distribution $\mathcal{D}_h$ as follows:

$$\mathcal{D}_h := \begin{pmatrix} \{\mathbf{B}_i\}_{i \in [0,L]}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}_{\mathbf{x}_i}^T \mathbf{B}_i + \mathbf{e}_{B,\mathbf{x}_i}^T\}_{i \in [0,h], \mathbf{x}_i \in \{0,1\}^i}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}_{\mathbf{x}_h}^T \mathbf{C}_{\mathbf{0}_h x_{h+1} \dots x_i} + \mathbf{e}_{B,\mathbf{x}_i}^T\}_{i \in [h+1,L], \mathbf{x}_i \in \{0,1\}^i} \end{pmatrix},$$

where $\mathbf{C}_{\mathbf{0}_h x_{h+1} \dots x_i} \leftarrow\!\!\$\; \mathbb{Z}_q^{(n+1) \times m_B}$ for every $i \in [h+1, L]$ and $x_{h+1} \dots x_i \in \{0,1\}^{i-h}$. Notably, $\mathcal{D}_L$ is identical to $\mathcal{D}'_{\mathsf{APLWE},0}$. From the subexponential security of the LWE assumption, it also follows that $\mathcal{D}_0$ is computationally indistinguishable from $\mathcal{D}'_{\mathsf{APLWE},1}$ [9]. Therefore, it suffices to show the indistinguishability between $\mathcal{D}_L$ and $\mathcal{D}_0$.

To prove this, we show that $\mathcal{D}_h$ and $\mathcal{D}_{h-1}$ are indistinguishable for every $h \in \{L, \dots, 1\}$, namely, it holds that

$$\begin{pmatrix} \{\mathbf{B}_i\}_{i \in [0,L]}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}_{\mathbf{x}_i}^T \mathbf{B}_i + \mathbf{e}_{B,\mathbf{x}_i}^T\}_{i \in [0,h-1], \mathbf{x}_i \in \{0,1\}^i}, \\ \{\mathbf{c}_{B,\mathbf{x}_h} := \mathbf{s}_{\mathbf{x}_{h-1}}^T \mathbf{S}_{h,x_h} \mathbf{B}_h + \mathbf{e}_{B,\mathbf{x}_h}^T\}_{\mathbf{x}_h \in \{0,1\}^h}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}_{\mathbf{x}_{h-1}}^T \mathbf{S}_{h,x_h} \mathbf{C}_{\mathbf{0}_h x_{h+1} \dots x_i} + \mathbf{e}_{B,\mathbf{x}_i}^T\}_{i \in [h+1,L], \mathbf{x}_i \in \{0,1\}^i} \end{pmatrix}$$
$$\overset{c}{\approx} \begin{pmatrix} \{\mathbf{B}_i\}_{i \in [0,L]}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}_{\mathbf{x}_i}^T \mathbf{B}_i + \mathbf{e}_{B,\mathbf{x}_i}^T\}_{i \in [0,h-1], \mathbf{x}_i \in \{0,1\}^i}, \\ \{\mathbf{c}_{B,\mathbf{x}_h} := \mathbf{s}_{\mathbf{x}_{h-1}}^T \mathbf{C}_{\mathbf{0}_{h-1} x_h} + \mathbf{e}_{B,\mathbf{x}_h}^T\}_{\mathbf{x}_h \in \{0,1\}^h}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}_{\mathbf{x}_{h-1}}^T \mathbf{C}_{\mathbf{0}_{h-1} x_h \dots x_i} + \mathbf{e}_{B,\mathbf{x}_i}^T\}_{i \in [h+1,L], \mathbf{x}_i \in \{0,1\}^i} \end{pmatrix}.$$

For every $i \in [L-h+1]$ and $\mathbf{x}_i \in \{0,1\}^i$, we define $\mathbf{F}_{\mathbf{x}_i} \leftarrow\!\!\$\; \mathcal{D}_{\mathbb{Z},\sigma_E}^{(n+1) \times m_B}$. By the choice of the parameter σ_E in Inequation 17, the noise flooding (Definition 3)

[9] Specifically, we can apply the LWE assumption by seeing $\mathbf{s}_\epsilon$ as the secret vector, and $\mathbf{B}_0$ and $\{\mathbf{C}_{\mathbf{x}_L}\}_{\mathbf{x}_L \in \{0,1\}^L}$ as public matrices.

implies that the following holds:

$$\begin{pmatrix} \{\mathbf{c}_{B,\mathbf{x}_h} := \mathbf{s}^T_{\mathbf{x}_{h-1}} \mathbf{S}_{h,x_h} \mathbf{B}_h + \mathbf{e}^T_{B,\mathbf{x}_h}\}_{\mathbf{x}_h \in \{0,1\}^h}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}^T_{\mathbf{x}_{h-1}} \mathbf{S}_{h,x_h} \mathbf{C}_{\mathbf{0}_h x_{h+1}\dots x_i} + \mathbf{e}^T_{B,\mathbf{x}_i}\}_{i\in[h+1,L],\mathbf{x}_i\in\{0,1\}^i} \end{pmatrix}$$
$$\overset{s}{\approx} \begin{pmatrix} \{\mathbf{c}_{B,\mathbf{x}_h} := \mathbf{s}^T_{\mathbf{x}_{h-1}} (\mathbf{S}_{h,x_h} \mathbf{B}_h + \mathbf{F}_{x_h}) + \mathbf{e}^T_{B,\mathbf{x}_h}\}_{\mathbf{x}_h \in \{0,1\}^h}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}^T_{\mathbf{x}_{h-1}} (\mathbf{S}_{h,x_h} \mathbf{C}_{\mathbf{0}_h x_{h+1}\dots x_i} + \mathbf{F}_{x_h\dots x_i}) + \mathbf{e}^T_{B,\mathbf{x}_i}\}_{i\in[h+1,L],\mathbf{x}_i\in\{0,1\}^i} \end{pmatrix}. \tag{18}$$

Since each of $\mathbf{B}_h$ and $\{\mathbf{C}_{\mathbf{0}_h x_{h+1}\dots x_i}\}_{i\in[h+1,L],\mathbf{x}_i\in\{0,1\}^i}$ are sampled independently, assuming the LWE assumption, we obtain that

$$\begin{pmatrix} \{\mathbf{c}_{B,\mathbf{x}_h} := \mathbf{s}^T_{\mathbf{x}_{h-1}} (\mathbf{S}_{h,x_h} \mathbf{B}_h + \mathbf{F}_{x_h}) + \mathbf{e}^T_{B,\mathbf{x}_h}\}_{\mathbf{x}_h \in \{0,1\}^h}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}^T_{\mathbf{x}_{h-1}} (\mathbf{S}_{h,x_h} \mathbf{C}_{\mathbf{0}_h x_{h+1}\dots x_i} + \mathbf{F}_{x_h\dots x_i}) + \mathbf{e}^T_{B,\mathbf{x}_i}\}_{i\in[h+1,L],\mathbf{x}_i\in\{0,1\}^i} \end{pmatrix}$$
$$\overset{c}{\approx} \begin{pmatrix} \{\mathbf{c}_{B,\mathbf{x}_h} := \mathbf{s}^T_{\mathbf{x}_{h-1}} \mathbf{C}_{\mathbf{0}_{h-1} x_h} + \mathbf{e}^T_{B,\mathbf{x}_h}\}_{\mathbf{x}_h \in \{0,1\}^h}, \\ \{\mathbf{c}_{B,\mathbf{x}_i} := \mathbf{s}^T_{\mathbf{x}_{h-1}} \mathbf{C}_{\mathbf{0}_{h-1} x_h\dots x_i} + \mathbf{e}^T_{B,\mathbf{x}_i}\}_{i\in[h+1,L],\mathbf{x}_i\in\{0,1\}^i} \end{pmatrix}. \tag{19}$$

The indistinguishabilities of Relations 18 and 19 imply that $\mathcal{D}_h$ and $\mathcal{D}_{h-1}$ are indistinguishable. Hence, we conclude that $\mathcal{D}_L$ and $\mathcal{D}_0$ are indistinguishable. This completes the proof. □

5.2 Evasive LWE Assumption

In this subsection, we review previous works on attacks against the evasive LWE assumption. For each category of attacks, we show that our concrete choice of the parameters and samplers in Section 4 is secure against all known attacks.

According to [33] and [4], the evasive LWE assumption relies on two heuristics:

1. **Heuristics 1: an adversary can use a preimage only in a semi-honest manner.** A preimage $\mathbf{K} \leftarrow_\$ \mathbf{B}^{-1}(\mathbf{P})$ cannot be used for anything other than deriving $\mathbf{SP}+\mathbf{EK}$ from $\mathbf{SB}+\mathbf{E}$; specifically, it does not provides an adversary non-trivial advantage in distinguishing the postcondition.
2. **Heuristics 2: an adversary cannot exploit structured error.** Although the error $\mathbf{EK}$ in the postcondition is structured, $\mathbf{SP}+\mathbf{EK}$ does not provide an adversary non-trivial information about secrets beyond an LWE instance with unstructured error in the precondition—namely, $\mathbf{SP}+\mathbf{E}'$.

These heuristics have been used to build counterexamples against the evasive-LWE assumption, which satisfy the precondition but fail to satisfy the postcondition.

Early attacks (and one recent attack introduced in [36]) exploit the first heuristic [50, 20]. However, each relies on contrived data that fall outside the "natural" distribution of the matrix $\mathbf{P}$ and auxiliary information aux—for example, an obfuscation that hardcodes a trapdoor for $\mathbf{P}$ [50], instance-hiding witness encryption [36], and a trapdoor for a submatrix of $\mathbf{P}$ [20] [10]. Since the evasive LWE samplers employed in the security proof of Theorem 1 does not contain such contrived data, none of these attacks applies to our PRO and iO constructions.

Recent attacks exploit the second heuristic [22, 4, 33]. They are comparatively simpler and require only minor modifications to the natural distributions of P and aux used by existing schemes based on the evasive LWE assumption. Nonetheless, we demonstrate that our construction remains secure against them by grouping the attacks into three categories and showing how each category is thwarted.

Attacks exploiting error correlation in GSW-FHE. The first category comprises the attacks described in Subsections 4.1, 5.1, and 5.2 of [4] and Subsection 7.3 of [33]. In short, these attacks are implemented by maliciously performing the homomorphic evaluation of a PRF within the GSW FHE scheme. Concretely, they employ a contrived circuit for that homomorphic evaluation—originally proposed by Hopkins et al. [32]—so that the PRF output and the FHE decryption error cancel each other out modulo 2. This is problematic because an adversary can then distinguish the postcondition distributions by checking whether certain linear equations over integers are solvable.

For example, with a variant of the sampler adopted in the original construction of the AKY24 FE scheme [2], an adversary can obtain $\mathbf{v}^T := f(\mathbf{x}) + \mathbf{e}_B^T\mathbf{K} + \mathbf{e}_{\mathsf{fhe}}^T - \mathbf{e}_A^T\mathbf{H}_{f,x}$ over $\mathbb{Z}_q$, where $\mathbf{e}_A^T$ and $\mathbf{e}_B^T$ are LWE errors provided by an honest encryptor, $\mathbf{H}_{f,x}$ is the output of the EvalCX algorithm, $\mathbf{K}$ is a preimage, and $f(\mathbf{x})$ is a function that returns pseudorandom outputs by computing PRF [4]. Here, $\left\|\mathbf{e}_B^T\mathbf{K} + \mathbf{e}_{\mathsf{fhe}}^T - \mathbf{e}_A^T\mathbf{H}_{f,x}\right\|_\infty$ is upper bound by some small B. As shown in [4], this allows the adversary to obtain $\mathbf{v}^T$ over integers. Consequently, the contrived circuit enables the adversary to obtain $\mathbf{v}^T \mod 2 = \mathbf{e}_B^T\mathbf{K} - \mathbf{e}_A^T\mathbf{H}_{f,x}$, which can be easily solved about $\mathbf{e}_B^T$ and $\mathbf{e}_A^T$ if and only if $\mathbf{SB} + \mathbf{E}$ is provided. Thereofre, the existence of the solutions tells the adversary whether it is in the real or random case in the postcondition.

Our iO construction prevents this attack in the same manner as the updated version of the AKY24 FE scheme [2]. Specifically, they perform rounding operations on the BGG+ encodings to replace the correlated FHE error with non-correlated rounding errors. Therefore, the adversary no longer has the ability to cancel out the PRF output, maintaining the pseudorandomness of $\mathbf{v'}^T \bmod 2$.

Attacks when the error in the precondition is larger. The second category of attacks is described in Subsection 4.2 of [4]. The attack assumes that the error added to $\mathbf{SP}$ in the precondition is larger than one in the postcondition, i.e., $\|\mathbf{E}'\|_\infty > \|\mathbf{EK}\|_\infty$.

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[10] One might argue that this last counterexample still looks natural. To rule out such a counterexample, Brzuska et al. [20] formalized the private-coin binding and hiding evasive-LWE assumptions. Since our construction is proved under these assumptions, the trapdoor-based attack no longer applies to our construction.

Our construction invalidates this attack simply by avoiding such parameter choices. Specifically, we recall that the Gaussian parameters $\{\chi_{p,i}\}_{i\in[0,L]}$, $\{\chi_{B,i}\}_{i\in[0,L]}$, χ_{att}, and χ_F are defined in Theorem 1 satisfy the following constraints:

- For every $i \in [L]$, $\chi_{p,i}\sqrt{\lambda} < m_B\chi_{p,i-1}\sigma_B\lambda + n_B(n+1)^{i-1}\sigma_S\sqrt{\lambda}$.
- $\chi_{\mathsf{att}}\sqrt{\lambda} < m_B\chi_{p,L}\sigma_B\lambda$ and $\chi_F\sqrt{\lambda} < m_B\chi_{p,L}\sigma_B\lambda$.

In Lemmas 10 and 11, the evasive LWE assumption is applied to $\{\mathbf{B}_i\}_{i\in[0,L]}$. For every $h \in [L]$, the instances $\{\mathbf{p}^T_{\mathbf{x}_{h-1}} := \left((1, \xi^T, \mathbf{x}^T_{h-1}, \mathbf{0}^T_{L-h+1}, \mathbf{t}^T) \otimes \mathbf{s}^T_{\mathbf{x}_{h-1}}\right)\mathbf{B}_{h-1} + \mathbf{e}^T_{p_{\mathbf{x}_{h-1}}}\}_{\mathbf{x}_{h-1}\in\{0,1\}^{h-1}}$ and $\{\mathbf{p}^T_{\mathbf{x}_h} := \left((1, \xi^T, \mathbf{x}^T_h, \mathbf{0}^T_{L-h}, \mathbf{t}^T) \otimes \mathbf{s}^T_{\mathbf{x}_h}\right)\mathbf{B}_h + \mathbf{e}^T_{p_{\mathbf{x}_h}}\}_{\mathbf{x}_h\in\{0,1\}^h}$ are, respectively, used as ones corresponding to $\mathbf{SB} + \mathbf{E}$ and $\mathbf{SP} + \mathbf{E}'$. Therefore, the first condition ensures that $\|\mathbf{E}'\|_\infty$ in the precondition is smaller than $\|\mathbf{EK}\|_\infty$ in the postcondition. Similarly, the second condition ensures the same when $\{\mathbf{p}^T_{\mathbf{x}_L} := \left((1, \xi^T, \mathbf{x}^T_L, \mathbf{t}^T) \otimes \mathbf{s}^T_{\mathbf{x}_L}\right)\mathbf{B}_L + \mathbf{e}^T_{p_{\mathbf{x}_L}}\}_{\mathbf{x}_L\in\{0,1\}^L}$ is used in the precondition. Hence, this attack does not apply to our construction.

Attacks with an overlapping auxiliary instance. The last category is for an attack introduced in [22]. This assumes that aux contains an instance that overlaps with $\mathbf{SP} + \mathbf{E}'$ except for the error, namely $\mathbf{SP} - 2\mathbf{T}$, where $\mathbf{T}$ is a uniformly random matrix whose entries are drawn from $[0, \lfloor \frac{q}{4} \rfloor]$[11]. Our construction is immune to this attack because aux contains no such overlapping instance.

6 Conclusion

Despite recent progress in constructing iO from reasonable cryptographic assumptions, it remains a theoretical rather than a practical cryptographic primitive due to its complexity and inefficiency. We observe that a common bottleneck in recently proposed iO schemes is their reliance on bootstrapping techniques from FE to iO. Specifically, they recursively invoke the FE encryption algorithm for each input bit as a function evaluated during decryption in the FE scheme. Consequently, the size and complexity of the FE encryption algorithm become a lower bound on the size of the functions that must be evaluated by the underlying FE scheme.

To address this bottleneck, we propose diamond iO, a straightforward construction of iO using lattice techniques. Our construction replaces the costly FE recursive encryption process for every input bit required in those bootstrapping techniques with simple matrix operations, by leveraging the FE scheme for pseudorandom functionalities proposed in [2]. We prove the security of our construction under LWE and evasive LWE, along with a new lattice assumption we introduce—called all-product LWE—in the PROM. We analyze the nonstandard assumptions—namely, all-product LWE and evasive LWE—and show that (1) a

[11] In the counterexample of [22], the entries of $\mathbf{T}$ are sampled from $[0, q/2]$. We lower the upper bound to $q/4$ because the original work defines $\mathbb{Z}_q$ as $[0, q)$, whereas we define it as $[-\frac{q}{2}, \frac{q}{2})$.

variant of the former assumption can be reduced to the standard LWE assumption, and (2) existing attacks against the latter assumption are not applicable to our iO construction.

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